#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

Gaseous State

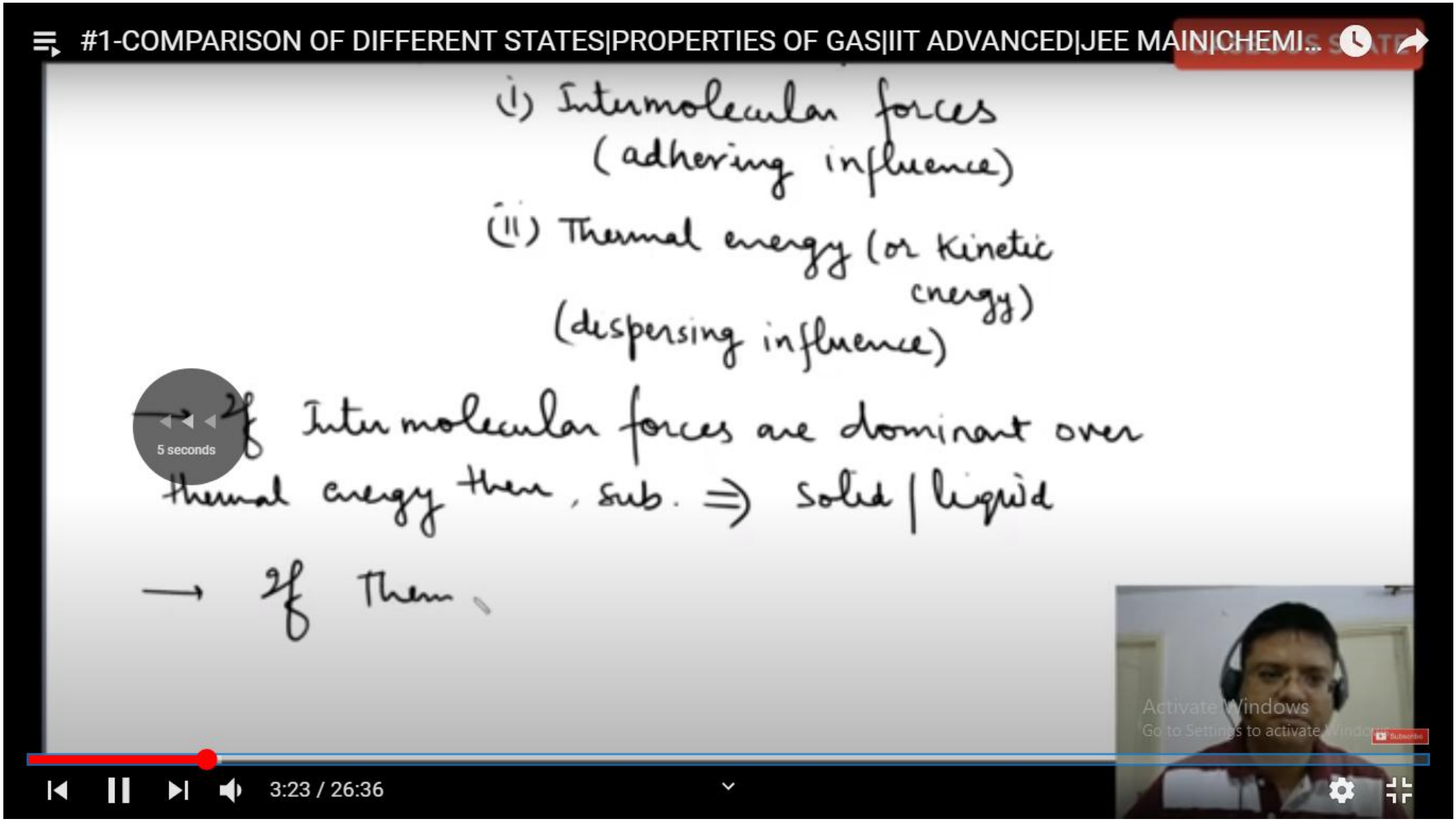
(states of matter)

⇒ state of matter depends on

(i) Intermolecular forces (adhering influence)

(ii) Thermal energy (or Kinetic energy)

2:12 / 26:36


#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...
(i) Intermolecular forces
(adhering influence)
(ii) Thermal energy (or Kinetic energy)
(despersing influence)
5 seconds
If Intermolecular forces are dominant over
thermal energy then, sub. ⇒ solid | liquid
→ If Them
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3:23 / 26:36

energy)

(despersing influence)

→ If Intermolecular forces are dominant over thermal energy then, sub. ⇒ solid | liquid

→ If Thermal energy $(= f(T))$ is dominant, then substance will be in gaseous state.

GASEOUS STATE

Property	solid	liquid	Gas
(1) Intermolecular forces	high	lesser than solid	very low (weak)
(2) Intermolecular space	least	more than solid	very high

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

(2)	Intermolecular space	least	more than solid	very high
(3)	density	very high	slightly lesser than solid	very low.
(4)	Compressibility	negligible	negligible but slightly more than solid	very high
(5)				

7:00 / 26:36

(4) Compressibility	negligible	negligible but slightly more than solid	very high
(5) Diffusibility	negligible	high	very high

		negligible	high	very high
(6)	Shape	Definite	No definite Shape (take the Shape of vessel)	No shape (occupies the entire vol. of vessel)
(7)				

GASEOUS STATE

		negligible	high	very high
(6)	Shape	Definite or fixed	No definite Shape (take the Shape of vessel)	No shape (occupies the entire vol. of vessel)
(7)	Volume	~~fixed~~ or definite	definite	Not fixed (takes the volume of vessel)

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

(7) Volume ~~fixed~~ or definite ≅ definite Not fixed (takes the volume of vessel)

→ To describe the gas, 4 measurable properties are:

(i) Amount: no of moles = $\frac{\text{mass}}{\text{Molar mass}}$

(ii) Pressure: → By barometer (For atmospheric pressure)

11:13 / 26:36

→ To describe the gas, 4 measurable properties are:

(i) **Amount**: no of moles = $\frac{\text{mass}}{\text{Molar mass}}$

(ii) **Pressure**: → By barometer (For atmospheric pressure)

→ By **Manometer** (to measure pr. of gas)

5 seconds

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of gas)

1 atm = 1.01325 × 10^5 Pa

1 bar = 10^5 Pa

⇒ 1 atm = 1.013 bar

1 atm = 760 torr [→ torr unit is on the name of physist who made barometer → Torricelli

= 760 mm of Hg

= 76 cm of Hg

1 Barye = 1 Ba = 0.1 Pa
1 psi = 6.894757 kPa = 6895 Pa
(pound per square inch)

$\rightarrow$ Torricelli

(iii) Volume : $1m^3 = 10^3$ lit

1 lit $= 10^3$ cm^3 or c.c or ml

$\Rightarrow 1m^3 = 10^6$ cm^3.

(iv) Temperature

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$\rightarrow$ Torricelli

(iii) **Volume** : $1m^3 = 10^3$ lit

1 lit $= 10^3$ cm^3 or c.c or ml

$\Rightarrow 1m^3 = 10^6$ cm^3.

(iv) **Temperature** : $K = {}^\circ C + 273.15$

$\Rightarrow K = {}^\circ C + 273$ for solving questions.

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#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

Experimental Gas laws:

① **Boyle's law**: At given temperature, ~~pressure~~ volume of a certain amount of gas is inversely proportional to ~~volume~~ its pressure.

16:23 / 26:36

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

① **Boyle's law**: At given temperature, volume of a certain amount of gas is inversely proportional to its pressure.

$T \rightarrow$ Constant

$n \rightarrow$ Constant

$$V \propto \frac{1}{P}$$

16:53 / 26:36

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

$\Rightarrow V = K \times \frac{1}{P}$

$PV = \text{Constant}$

$P_1 V_1$

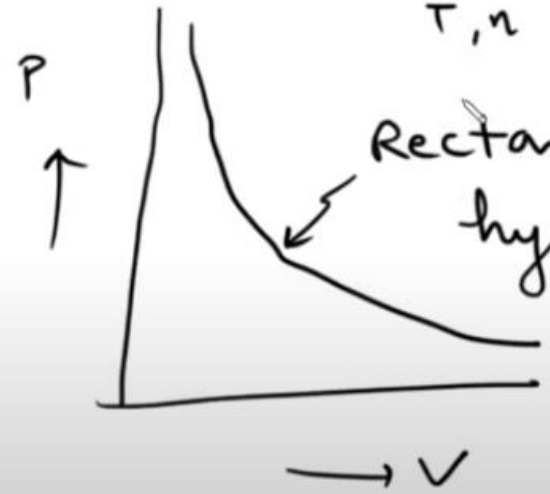
17:16 / 26:36

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

$P_1V_1 = P_2V_2 = \text{Constant}$

⇒ P vs V graph.

T, n → Constant



$PV = K$ (Const)

$P = \frac{K}{V}$

xy = Constant

(Rectangular hyperbola)

19:05 / 26:36

→ V (hyperbola)

Some important Graphs:

① straight line : $y = mx + c$

m = slope = $\tan\theta$

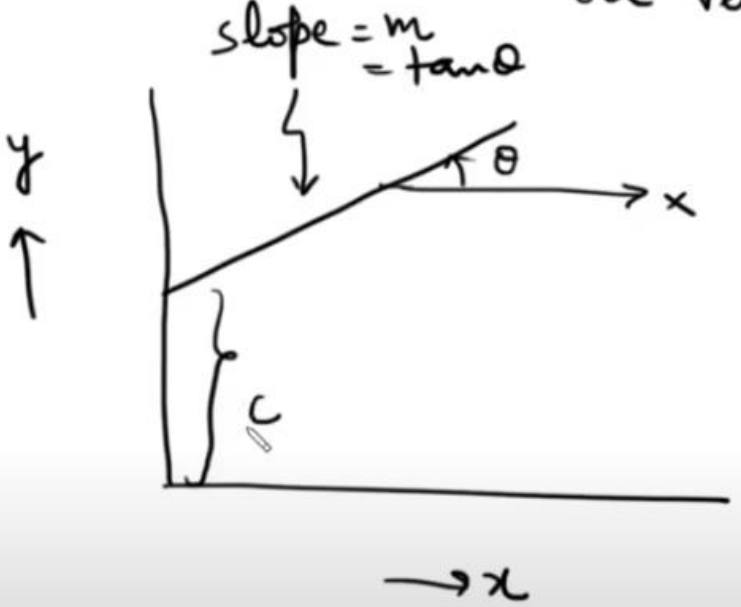
[$\theta \rightarrow$ angle of straight line from x-axis]

c = intercept on vertical (or Y) axis.

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

line from x-axis]

C = intercept on vertical (or y) axis.



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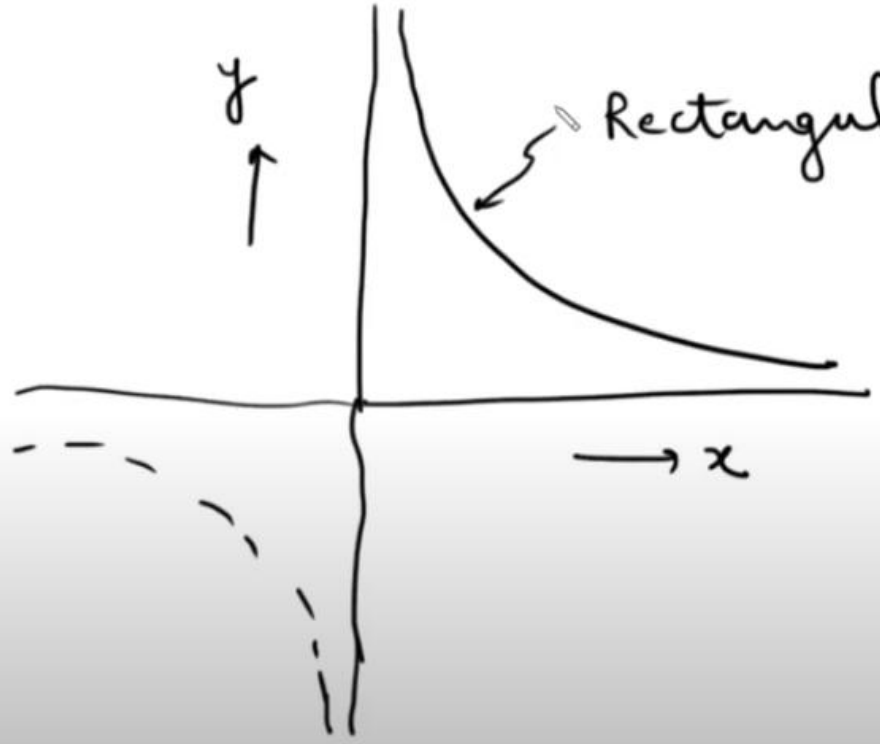
#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

(2) Rectangular hyperbola: $xy = \text{constant}$



In gaseous state,

P, V, n, T

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#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...

(5) Parabola :

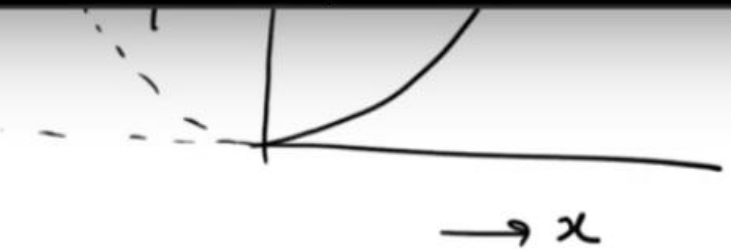
(a) $y = kx^2$ (symmetrical about y axis)

$(k > 0)$

(b) $y^2 = kx$ $(k > 0)$

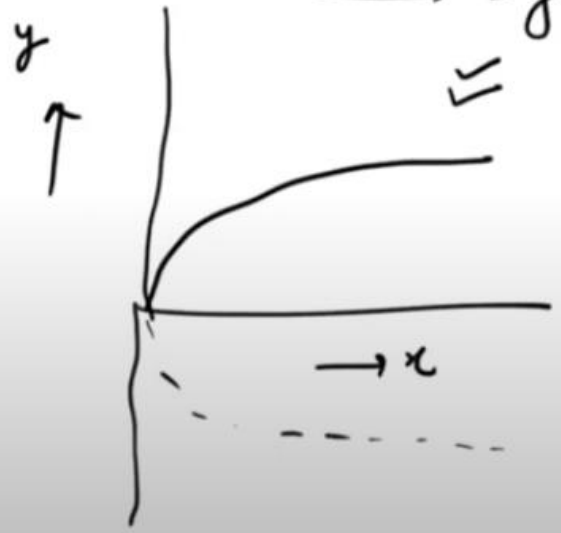
25:50 / 26:36

#1-COMPARISON OF DIFFERENT STATES|PROPERTIES OF GAS|IIT ADVANCED|JEE MAIN|CHEMI...



(b) $y^2 = kx \quad (k > 0)$

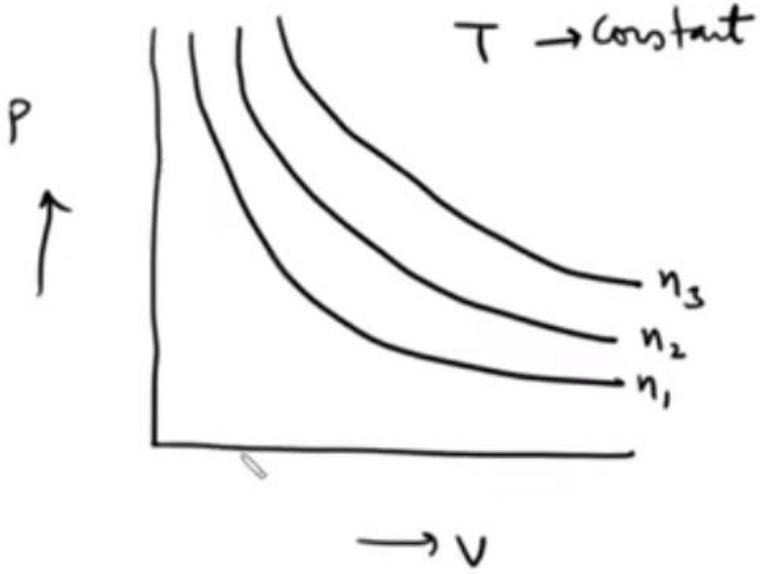
→ symmetrical about x-axis.



26:15 / 26:36

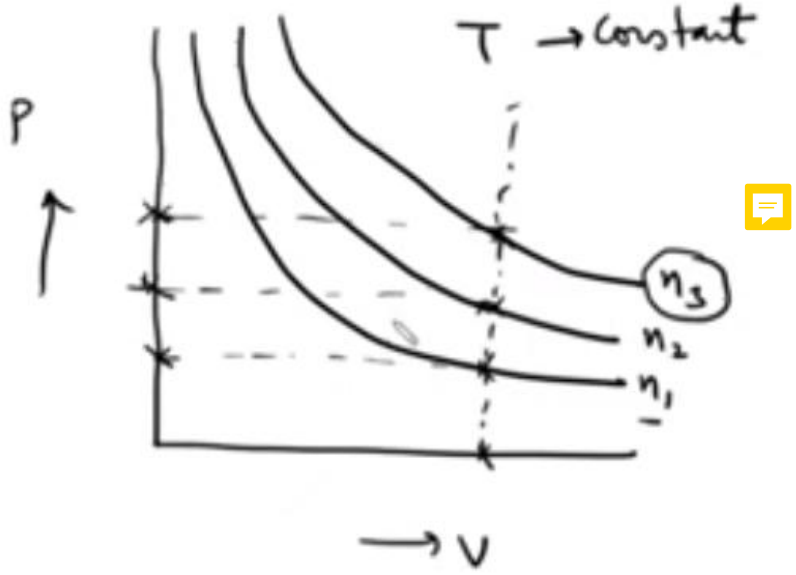
PV = Constant (Boyle's law)

Q →



PV = Constant (Boyle's law)

Q →



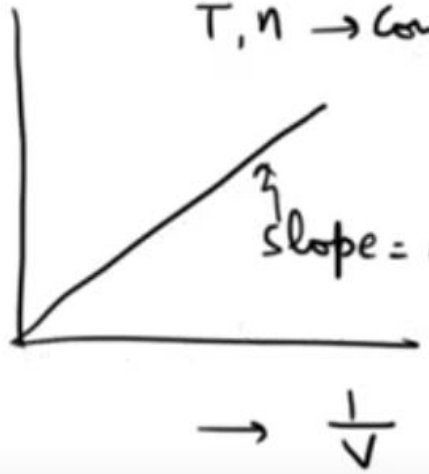
$PV = nRT$

$n_3 > n_2 > n_1$

#2-GASEOUS STATE|GRAPHS OF BOYLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12...

$PV = nRT$ $\quad n_3 > n_2 > n_1$

Q → P



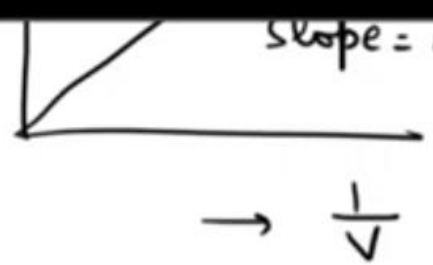
T, n → Constant, PV = Constant = nRT

$P = \frac{K \times 1}{V}$

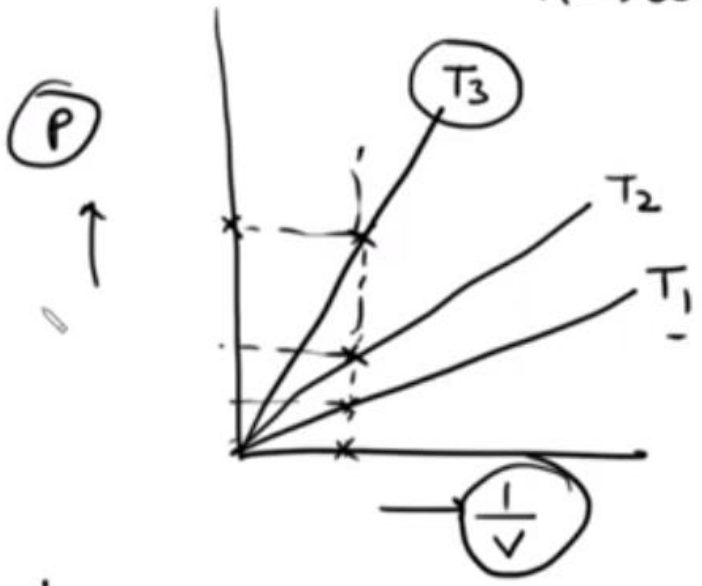
$y = mx$

5:08 / 14:16

GASEOUS STATE



$n \rightarrow$ Constant



Compare T_1, T_2, T_3

Soln : # $PV = nRT$

$P = (nRT) \times \frac{1}{V}$

slope = nRT

$T_3 > T_2 > T_1$

$P = (nRT) \times \frac{1}{V}$

$T_3 > T_2 > T_1$

$T_3 > T_2 > T_1$

Q → Plot $\frac{P}{V}$ vs $\frac{1}{V}$ for an ideal gas at Const. Temp and amount.

Soln: $PV^2 = nRT$

$\frac{P}{V} \times V = nRT$

$\frac{P}{V} = (nRT) \times \left(\frac{1}{V}\right)^2 \Rightarrow y = kx^2$

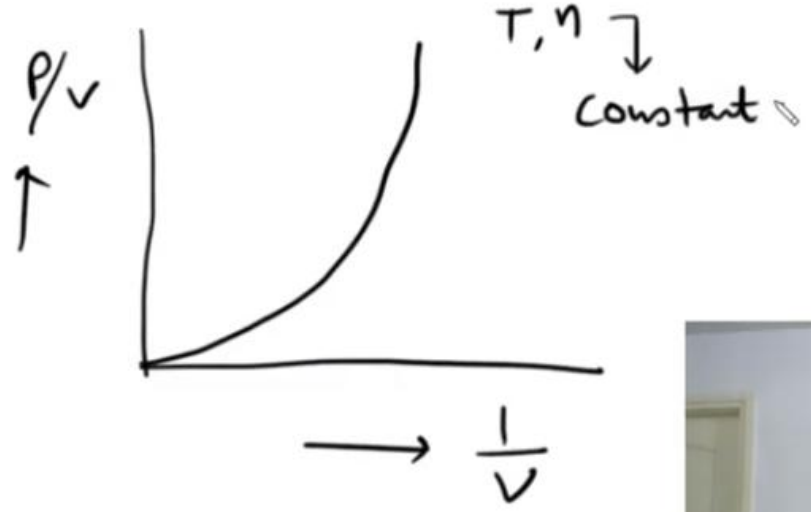
$\frac{P}{V}$ → y, nRT → const., $\frac{1}{V}$ → x

Sol^{n} ;

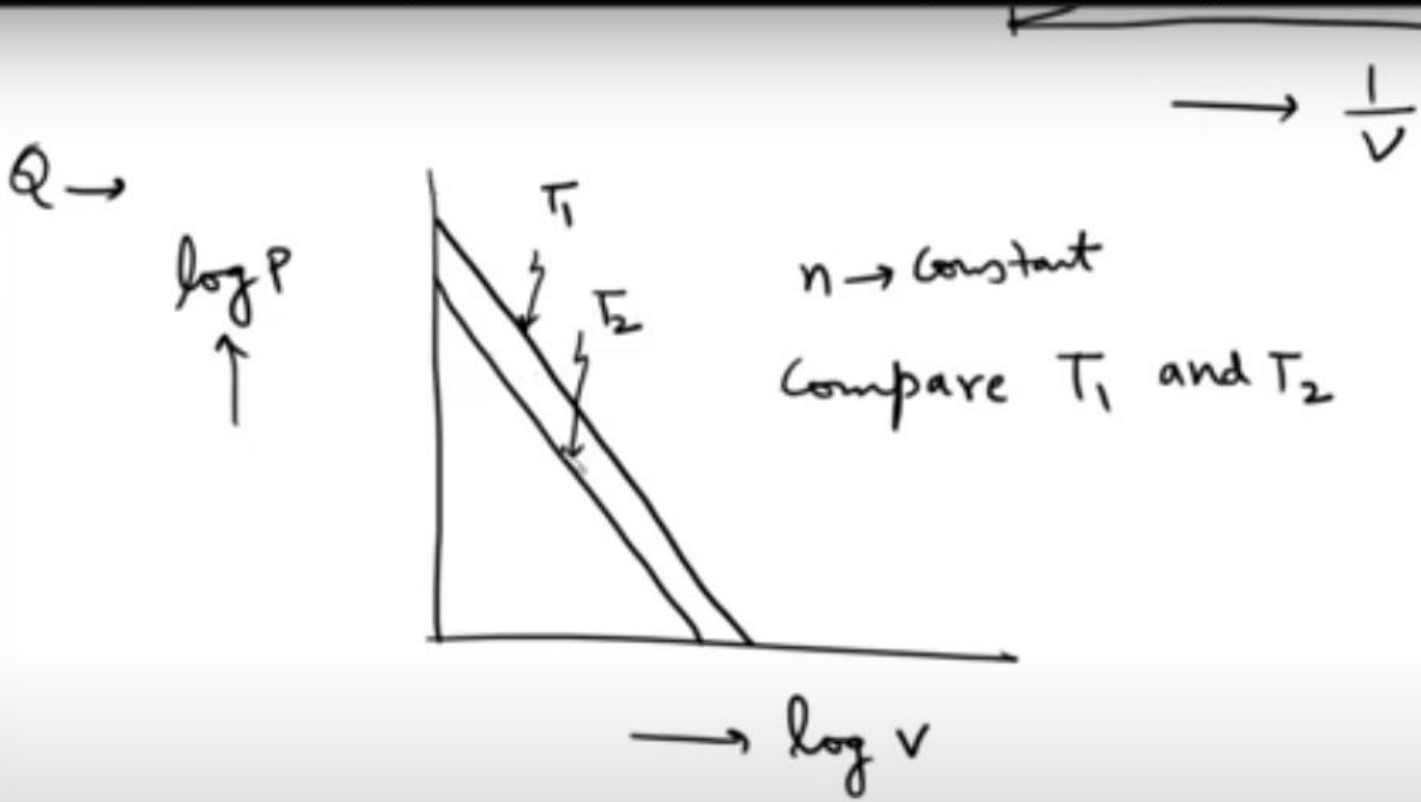
$$PV^2 = nRT$$

$$\frac{P}{V} = (nRT) \times \left(\frac{1}{V}\right)^2 \Rightarrow y = kx^2$$

$\frac{P}{V}$ → y ; (nRT) → const. ; $\frac{1}{V}$ → x



#2-GASEOUS STATE|GRAPHS OF BOYLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12...



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#2-GASEOUS STATE|GRAPHS OF BOYLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12...

$\rightarrow \log V$

$PV = nRT \Rightarrow \log P + \log V = \log(nRT)$

$\log P = (-1)\log V + \log(nRT)$

$y \qquad x$

$y = mx + c$, slope $= -1$

$\theta = 135$

5 seconds

12:02 / 14:16

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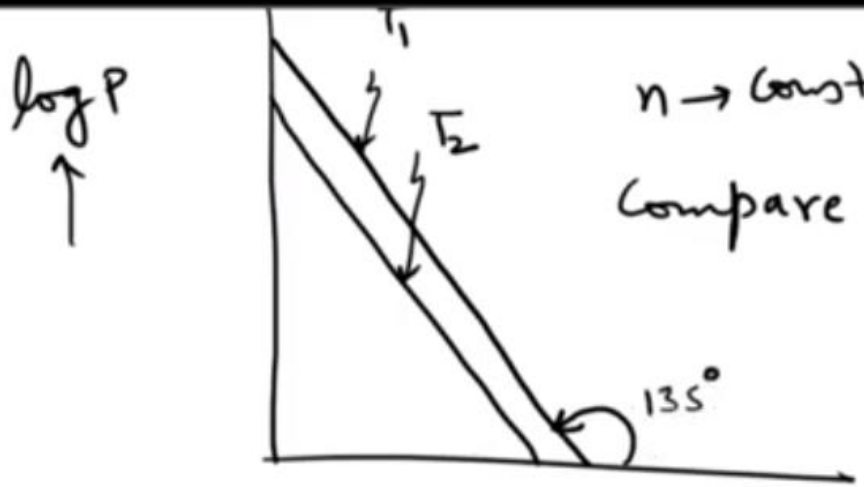


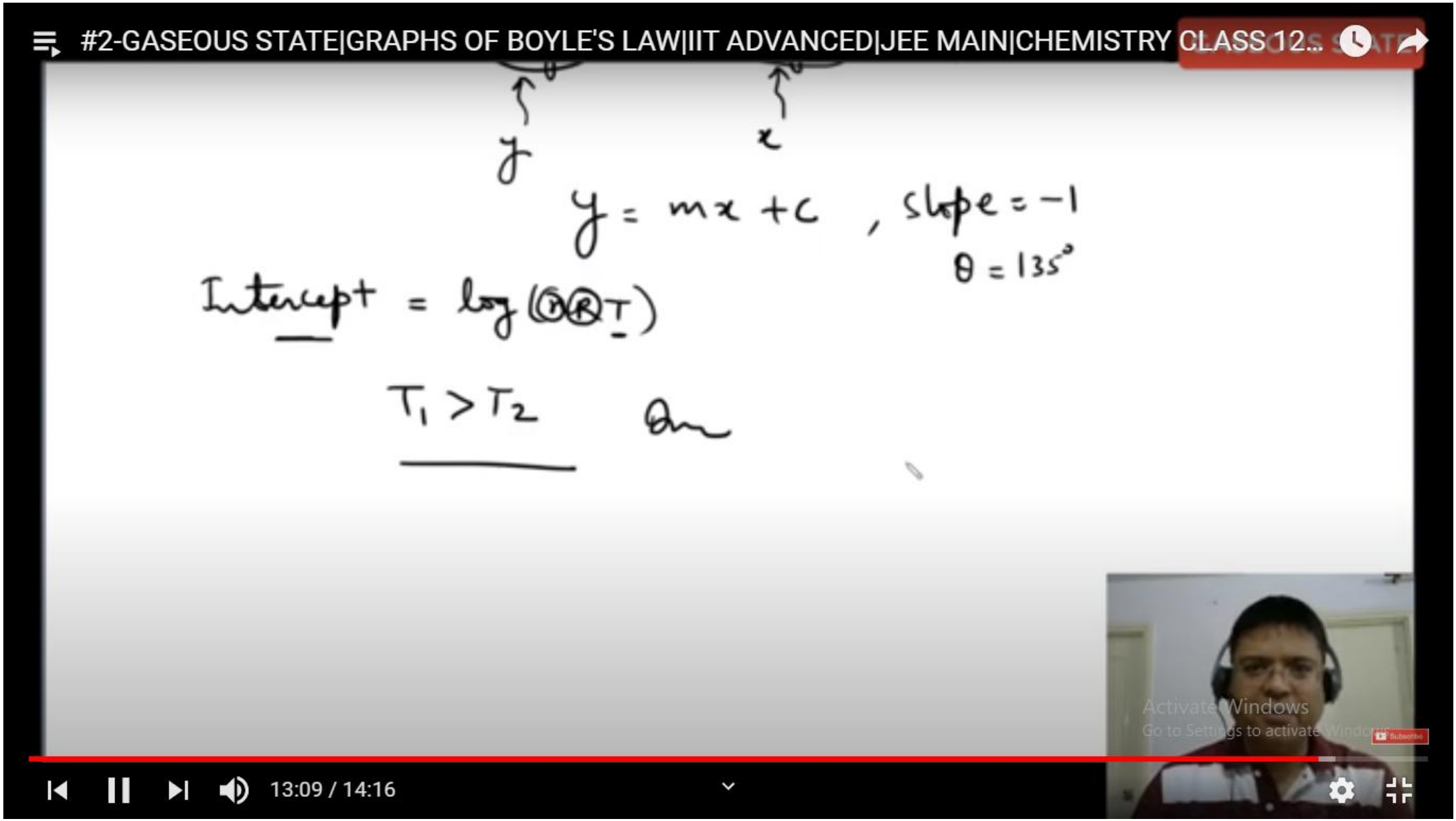
$n \rightarrow$ Constant

Compare T_1 and T_2

$PV = nRT \Rightarrow \log P + \log V = \log(nRT)$

$\log P = (-1)\log V + \log(nRT)$

y x


#2-GASEOUS STATE|GRAPHS OF BOYLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12...
y
x
y = mx + c , slope = −1
θ = 135°
Intercept = log(nRT)
T1 > T2
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13:09 / 14:16

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

(2) Charle's law :

At constant pressure volume of a certain amount of gas is directly proportional to its absolute temp. (temp in Kelvin)

$P, n \rightarrow$ Constant

$V \propto T$

2:02 / 21:37

absolute temp. (temp in Kelvin)

$P, n \rightarrow$ Constant

$V \propto T$

$V = KT$

$\frac{V}{T} = \text{Const}$, $\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2} =}$ Constant

5 seconds

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$\frac{V}{T} = \text{const}$, $\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2} = \text{Constant}}$

OR (originally what charle's said:)

At constant pressure, volume of a certain amount of gas increases or decreases by a factor $\frac{1}{273.15}$ of volume at 0°c for every degree rise or fall in temperature.

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$\frac{1}{273.15}$ of volume at 0c for every degree rise or fall in temperature.

$\Rightarrow$ Volume of gas at 0°c $= V_0$

$$V_{1c} = V_0 + \frac{1}{273.15} \times V_0$$

$$V_2 = V_0 + \frac{1}{273.15} \times$$

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$$V_{1^\circ C} = V_0 + \frac{1}{273.15} \times V_0$$

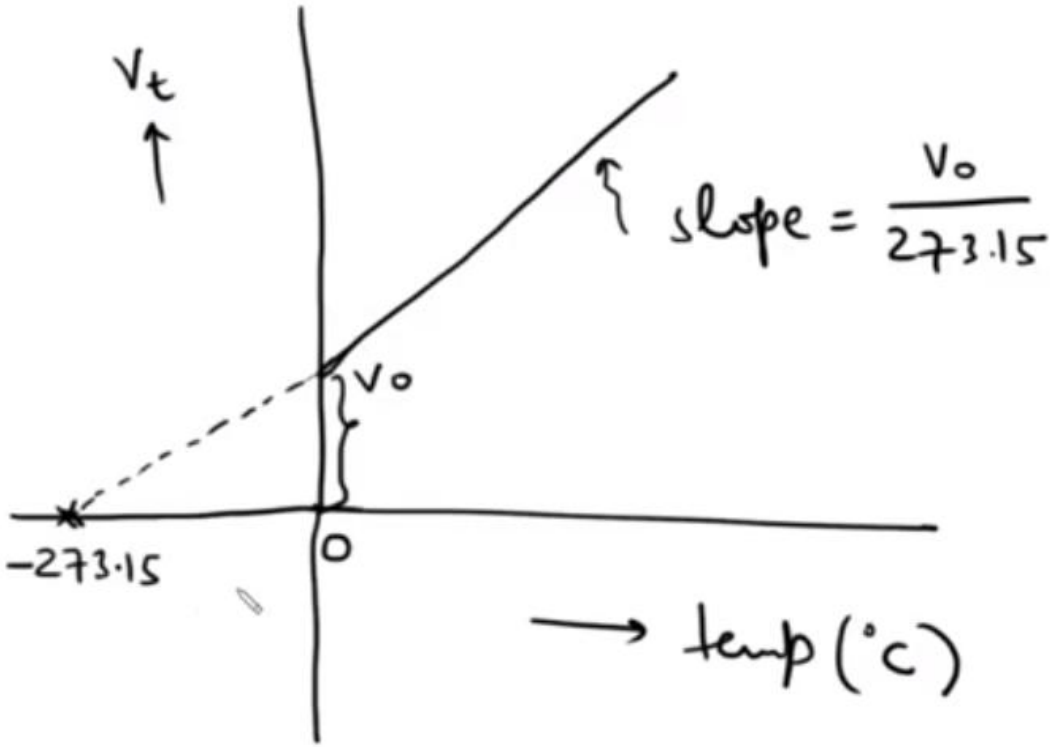
$$V_2 = V_0 + \frac{2}{273.15} \times V_0$$

$$\vdots$$

$$V_t = V_0 + \frac{t}{273.15} \times V_0$$

$$y = mx + c \Rightarrow \text{slope} = \frac{V_0}{273.15}$$

$$\text{intercept} = V_0$$


V_t
slope = $\frac{V_0}{273.15}$
V_0
-273.15
0
temp (°C)

⇒ At temp -273.15°C, volume turns zero

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#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

⇒ At temp −273.15°C, volume thus zero

So −273.15°C is the minimum conceivable temp.

$V_t = V$

8:41 / 21:37

⇒ At temp $-273.15^{\circ}C$, volume thus zero

So $-273.15^{\circ}C$ is the minimum conceivable temp.

$$V_t = V_0 + \frac{t}{273.15} \times V_0$$

$$\Rightarrow V_t = V_0 \left(\frac{t + 273.15}{273.15} \right)$$

$$\Rightarrow \quad V_t = V_0 \left(\frac{t + 273.15}{273.15} \right)$$

$$\frac{V_t}{(t+273.15)} = \frac{V_0}{(273.15+0)} \Rightarrow \frac{V_T}{T} = \frac{V_0}{T_0}$$

$$\Rightarrow \boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$$ Ans

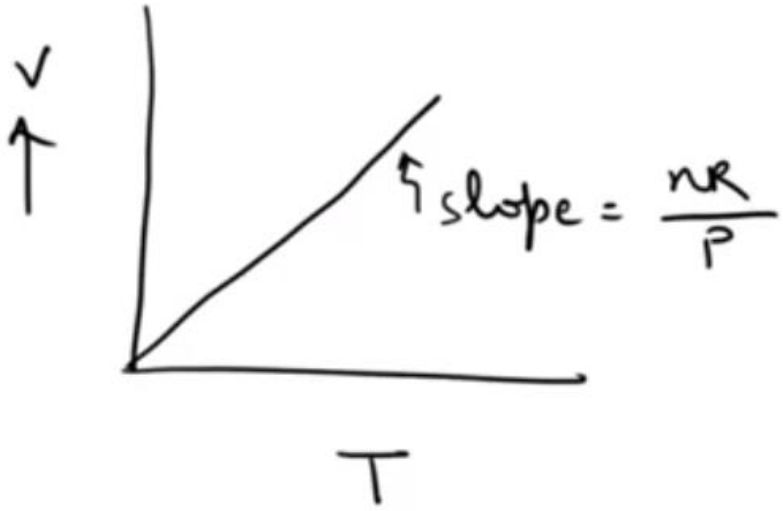
$\Rightarrow T = t + 273.15$

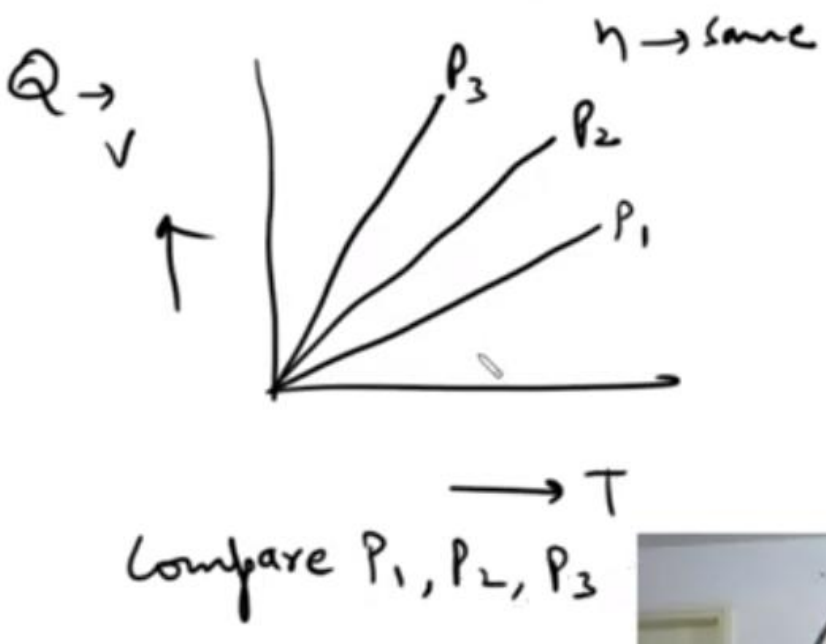
T ↑ temp. in Kelvin

t ↑ in celcius

$\frac{V}{T}$ = constant $\Rightarrow$ $V = KT$ $\Rightarrow$ $PV = nRT$

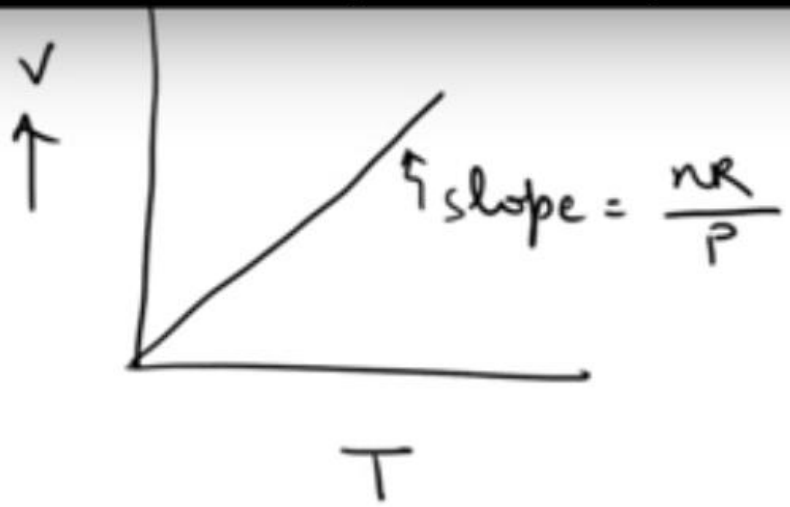
$V = \left(\frac{nR}{P}\right) \times T$

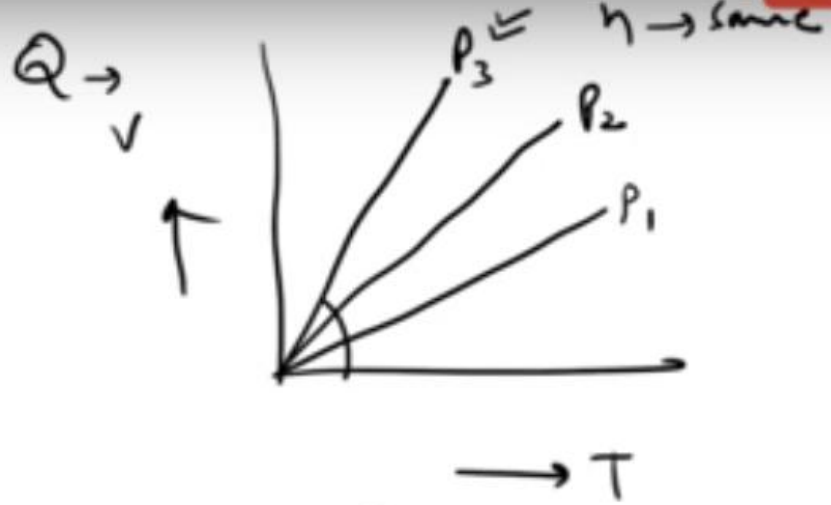




Compare P_1, P_2, P_3

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...





Compare P_1, P_2, P_3

sol^n: $\text{slope} = \frac{nR}{P}$

$P_3 < P_2 < P_1$

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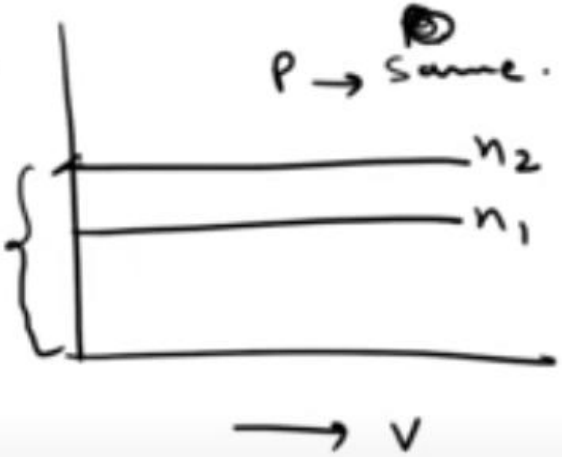
#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

$P_3 < P_2 < P_1$ Ans

Q → $\frac{V}{T}$ ↑

P → same.



$PV = nRT$

$\frac{V}{T} = \frac{nR}{P}$

$n_2 > n_1$ Ans

14:47 / 21:37

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

(3) Gay-Lusac's law of Pressure and temp:

V, n → Cont.

$P \propto T$

$\frac{P}{T} =$

15:46 / 21:37

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

$P \propto T$

$\frac{P}{T} = \text{Const.}$ $\quad \frac{P_1}{T_1} = \frac{P_2}{T_2} = \text{Const}$

(4) Avogadro's law

16:10 / 21:37

$P \propto T$

$\frac{P}{T} = \text{Cont.}$ $\quad \frac{P_1}{T_1} = \frac{P_2}{T_2} = \text{Const}$

(4) Avogadro's law : $P, T \rightarrow$ Constant

$V \propto n$

$\frac{V_1}{n_1} = \frac{V_2}{n_2} = \text{Cont}$

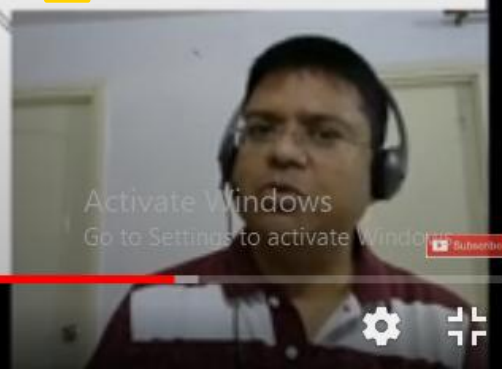
#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

⇒ Equal volume of gases contain equal no. of gas molecules under similar condition of pressure and temperature.

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \Rightarrow \text{if } V_1 = V_2$$

$$\Rightarrow n_1 = n_2$$

⇒ no. of molecules of gas are same.



18:09 / 21:37

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

$\Rightarrow$ no of molecules of gas are same.

~~Combining~~ Combining these gas laws,

$V \propto \frac{1}{P}$ (Boyle's law)

$V \propto T$ (Charle's law)

$V \propto n$ (Avogadro's law)

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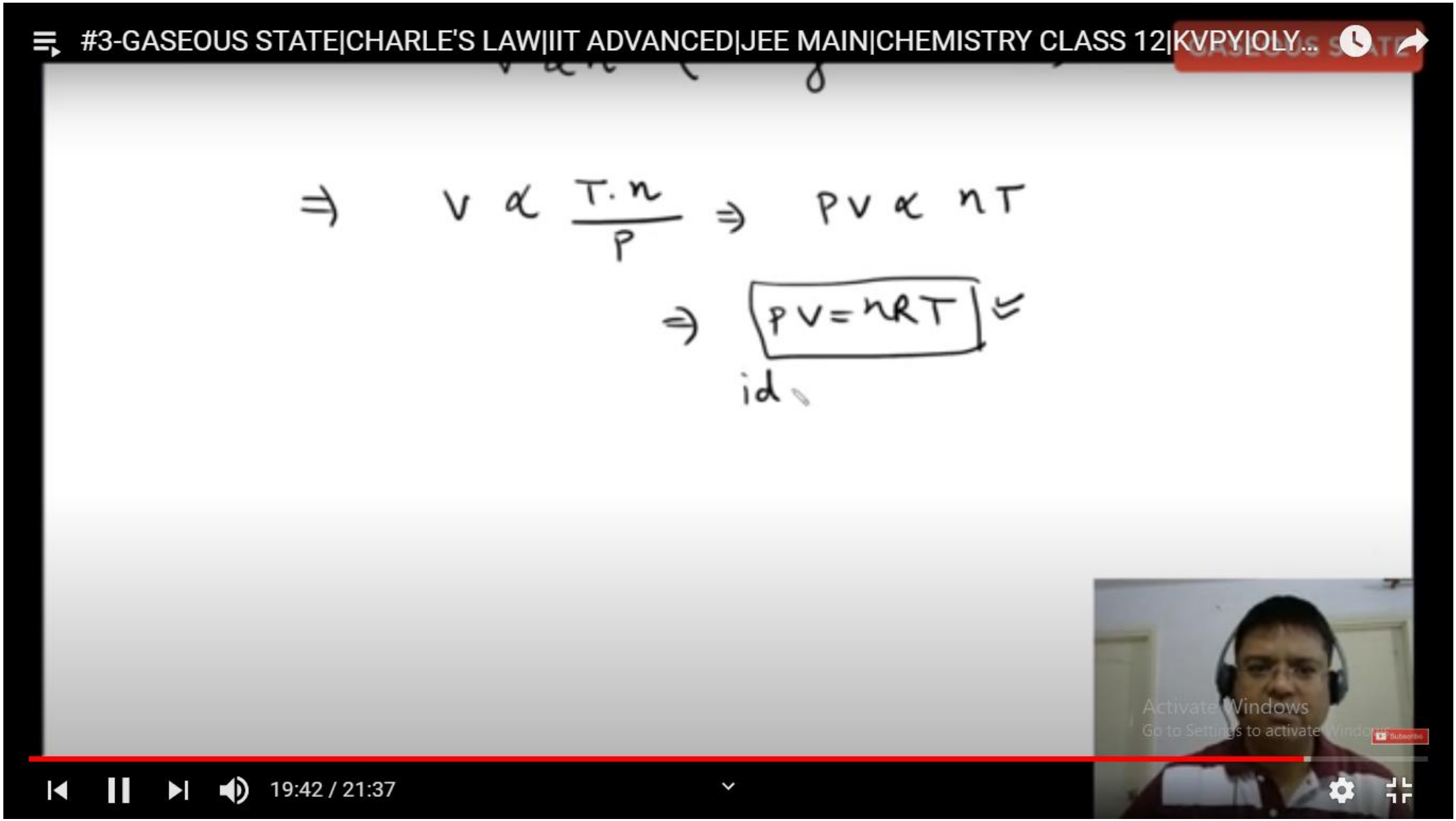
#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...
$\Rightarrow \quad V \propto \frac{T \cdot n}{P} \Rightarrow \quad PV \propto nT$
$\Rightarrow \boxed{PV = nRT}$
id
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19:42 / 21:37

#3-GASEOUS STATE|CHARLE'S LAW|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLY...

$$\Rightarrow \quad V \propto \frac{T \cdot n}{P} \Rightarrow \quad PV \propto nT$$

$$\Rightarrow \boxed{PV = nRT}$$

ideal gas equation

R = Universal Gas Const.

20:03 / 21:37

ideal gas equation.

R = Universal Gas Constant

$= 0.0821$ lit atm $mol^{-1}\ K^{-1} \cong \frac{1}{12}$ lit atm $mol^{-1}\ K^{-1}$

$= 8.314$ Joule $mol^{-1}\ K^{-1} \cong \frac{25}{3}$ J $mol^{-1}\ K^{-1}$.

$\cong$ 2 Cal $mol^{-1}\ K^{-1}$

1 Cal = 4.18 Joule

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#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Q → At constant temperature, pressure of a certain amount of gas is increased by 10%. find % change in volume of gas.

Sol^n : $T, n \rightarrow$ Constant

$PV = nRT$

$P_1V_1 = P_2V_2$

$P \times V = 1.1P \times V_f \Rightarrow V_f = \frac{V}{1.1}$

5 seconds

2:37 / 18:53

$$\% \text{ decrease in volume of gas} = \frac{\Delta V}{V_i} \times 100$$

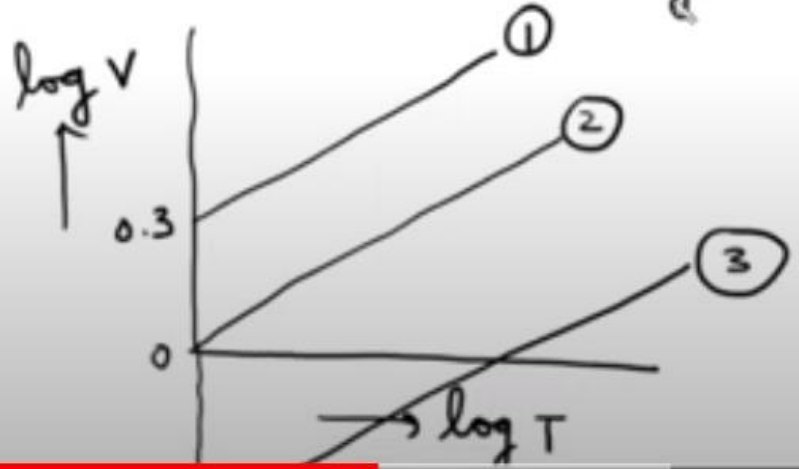
$$= \frac{\left(V - \frac{V}{1.1}\right)}{V} \times 100$$

$$= \frac{0.1}{1.1} \times 100 = \frac{100}{11}\%$$

#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Ip

Q → At constant pressure of 0.0821 atm, log V vs log T are plotted for three samples of an ideal gas as shown. Find no. of moles taken in sample 1, 2, 3. [log 2 = 0.3, log 3 = 0.48]



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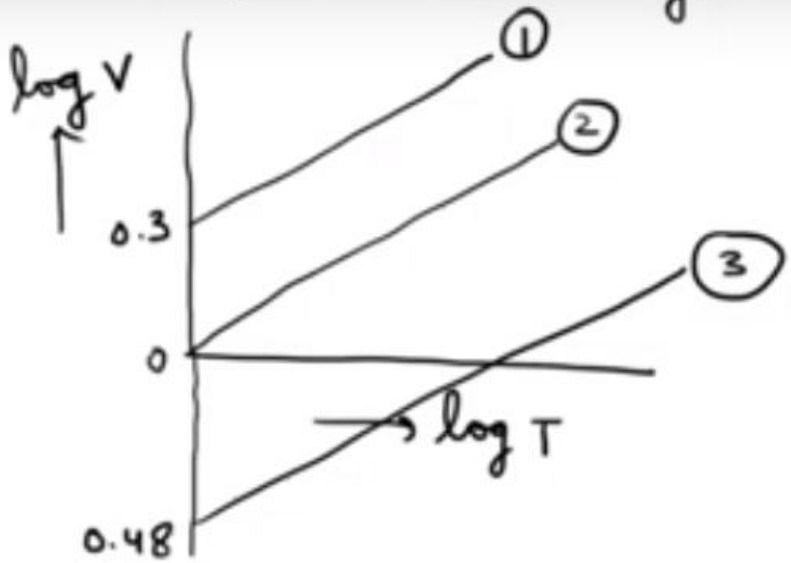
6:17 / 18:53

#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

1, 2, 3 [log 2 = 0.3 , log 3 = 0.48]



Soln ; Pl

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6:25 / 18:53

#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Sol^n : $PV = nRT \Rightarrow V = \frac{nR}{P} \times T$

$\log V = \log T + \log\left(\frac{nR}{P}\right)$

$y \quad\quad x$

$y = mx + c$

Intercept $= \log \frac{nR}{P} = \log \frac{n \times 0.0821}{0.0821} = \log n$

Sample ①, $\log n_1 = 0.3 \Rightarrow n_1 = 2$ mol

Sample ②, $\log n_2 = 0 \Rightarrow n_2 = 1$ mol

Sample ③, $\log n_3 = -0.48 = \log \frac{1}{3}$

$\Rightarrow n_3 = \frac{1}{3}$

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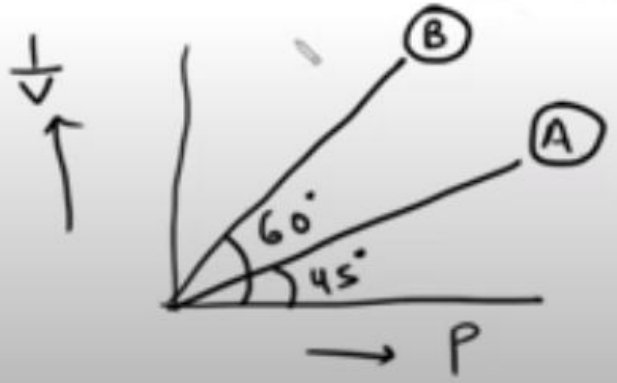
8:34 / 18:53

Sample (3), $\log n_3 = -0.48 = \log \frac{1}{3}$

$\Rightarrow n_3 = \frac{1}{3}$ mol

Imp

Q → At constant temperature of 273K, $\frac{1}{V}$ vs P are plotted for two ideal gases A and B as shown. Find out no. of moles of gas A and B.



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Sol": $T = 273\,K$ (constant)

$$PV = nRT \Rightarrow \underbrace{\left(\frac{1}{V}\right)}_{y} = \left(\frac{1}{nRT}\right) \times \underbrace{P}_{x}$$

$y = mx$ (straight line)

$$\text{Slope} = \frac{1}{nR \times 300} = \frac{1}{22.4 \times n}$$

[RT at 273K = 22.4
RT at ?

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#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Sol": $T = 273\,K$ (constant)

$$PV = nRT \Rightarrow \underbrace{\frac{1}{V}}_{y} = \left(\frac{1}{nRT}\right) \times \underbrace{P}_{x}$$

$y = mx$ (straight line)

$$\text{Slope} = \frac{1}{nRx300}$$

$$= \frac{1}{22.4 \times n}$$

$$\left[\begin{array}{l} RT \text{ at } 273K = 22.4 \\ RT \text{ at } 300K = 24.63 \end{array}\right]$$

5 seconds

12:17 / 18:53

$$= \frac{1}{22.4 \times n}$$

[RT at 273K = 22.4
RT at 300K = 24.63]

For gas A, slope $= \frac{1}{22.4 \times n_A} = \tan 45^\circ = 1$

$$\Rightarrow n_A = \frac{1}{22.4} \text{ mol}$$

Ans

For gas B, slope $= \frac{1}{22.4 \times n_B} = \tan 60^\circ = \sqrt{3}$

$$\Rightarrow n_B = \frac{1}{22.4 \times \sqrt{3}} \text{ mol}$$

Ans

5 seconds

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#4-GASEOUS STATE|NUMERICALS ON CHARLE'S LAW||IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Q → (a) Plot the curve between PT vs T at constant volume and number of moles.

(b) Find the number of moles of gas taken when the volume of the vessel is 82.1 ml and

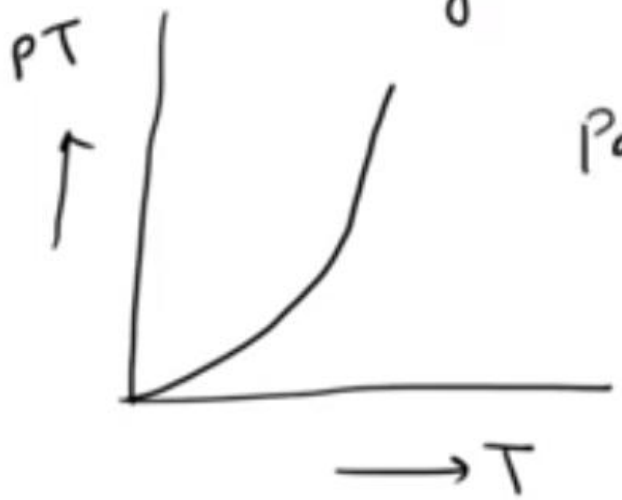
$$\left\{\frac{d[PT)}{dT}\right\}_{at\ T=300K} = 300$$ for above curve.

15:42 / 18:53

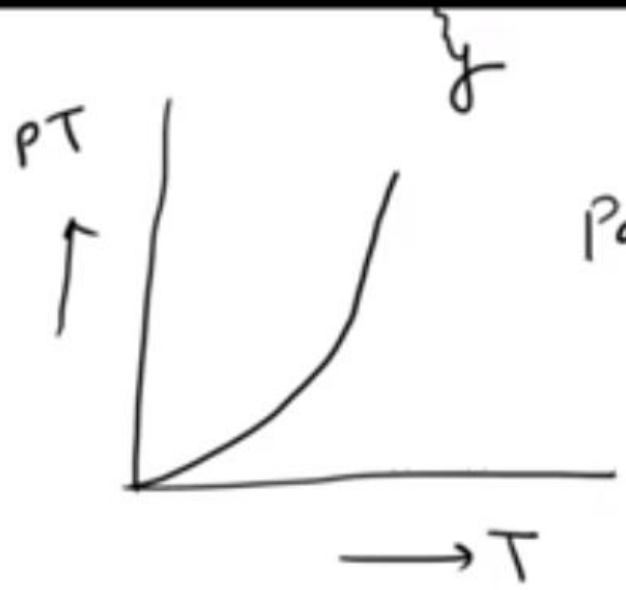
Soln : PT vs T

$\Rightarrow T \times PV = nRT^2$

$\Rightarrow \underbrace{PT}_{y} = \left(\frac{nR}{V}\right) \times \underbrace{T^2}_{x} \quad \Rightarrow y = kx^2$



Parabola



Parabola

(b) $\frac{d(PT)}{dT}$ at 300K

$= 300$

$$\frac{d(PT)}{dT} = \frac{nR}{V} \times 2T$$

$$\left(\frac{d(PT)}{dT}\right)_{300} = \frac{n \times 0.0821}{82.1 \times 10^{-3}} \times 2 \times 300$$

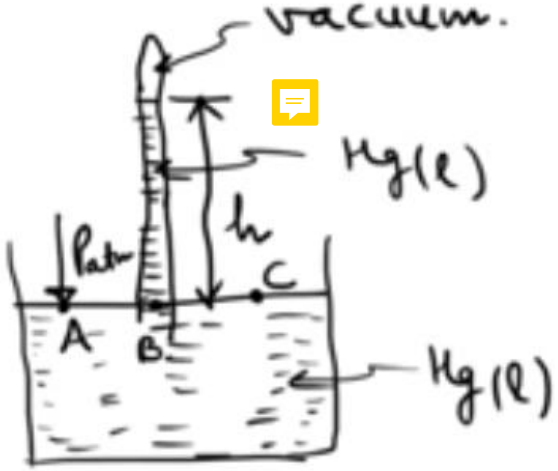
$$300 = 600n$$

$$\Rightarrow n = \frac{1}{2} \text{ mol}$$

#5-GASEOUS STATE|BAROMETER|FAULTY BAROMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

GASEOUS STATE

Barometer : A device used to measure atmospheric (Torricelli) pressure.



5 seconds

Atmospheric

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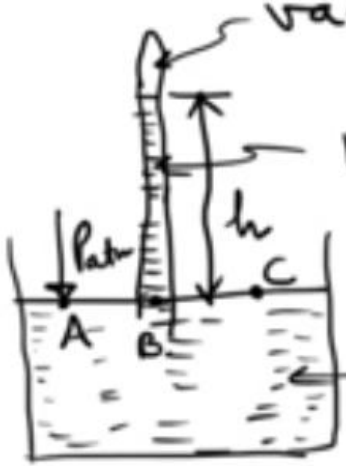
2:38 / 21:15

Atmospheric pressure : Pressure exerted by atmospheric gases under the effect of gravity.

→ In a liquid, pressure in a horizontal level is same every where.

Barometer: A device used to measure atmospheric (Torricelli) pressure.



$\Rightarrow (P_A) = P_B = P_C$

$P_{atm} = P_B$

$= \frac{mg}{A}$

$\Rightarrow P_{atm} = \frac{(d \times V) \times g}{A}$

$= \frac{d_{Hg} \times (A \times h) g}{A}$

$P_{atm} = h\, d_{Hg} \times g$

Atmospheric pressure: Pressure exerted by atmospheric gases under the effect of gravity.

is same every where.

$P_{atm} = h d g$

At ground level, $h = 0.76$ meter

$P_{atm} = 0.76 \times 13.6 \times 10^3 \text{ kg/m}^3 \times 9.81 \text{ m/sec}^2$

$P_{atm} = 1.01325 \times 10^5 \text{ Pa}$

$= 1$ atmosphere or 1 atm at ground level.

#5-GASEOUS STATE|BAROMETER|FAULTY BAROMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

GASEOUS STATE

Imp

→ 1 atm is atmospheric pressure at ground level.

However that that atmospheric pressure is variable and changes with the altitude. At higher altitude atmosphere is less than 1 atm.

⇒ altitude ↑, atm. pr. ↓

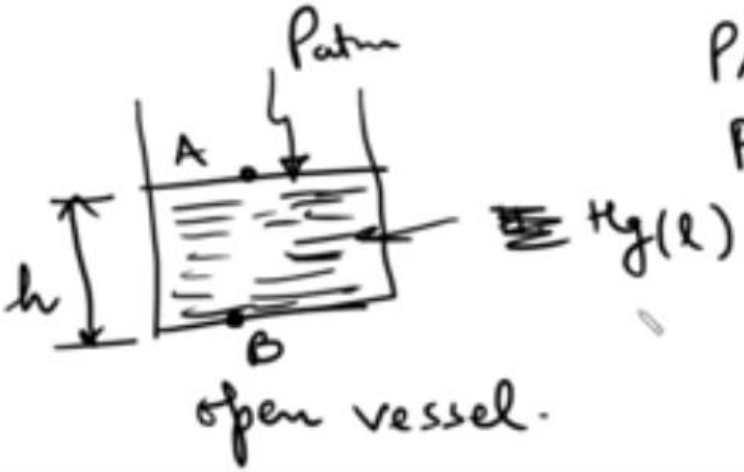
11:25 / 21:15

#5-GASEOUS STATE|BAROMETER|FAULTY BAROMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

GASEOUS STATE

$\Rightarrow$ altitude $\uparrow$, atm. pr. $\downarrow$

$\Rightarrow$



open vessel.

$P_A = P_{atm}$

$P_B = P_{atm} + h d g$

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13:11 / 21:15

#5-GASEOUS STATE|BAROMETER|FAULTY BAROMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

GASEOUS STATE

Note: why $Hg_{(\ell)}$ is used as barometric fluid.

Because

(i) It does not stick with walls of glass tube (capillary tube)

(ii) density of $Hg_{(\ell)}$ is considerably high, so lesser length of tube is

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15:12 / 21:15

tube (capillary tube)

(ii) density of $Hg_{(l)}$ is considerably high, so lesser length of tube is needed to measure atmospheric pressure.

(iii) $Hg_{(l)}$ is almost non- v

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measure atmospheric pressure.

Imp (iii) $Hg_{(l)}$ is almost non-volatile under room temperature. So maintains vacuum above $Hg_{(l)}$ in capillary tube in empty space.

$\Rightarrow$

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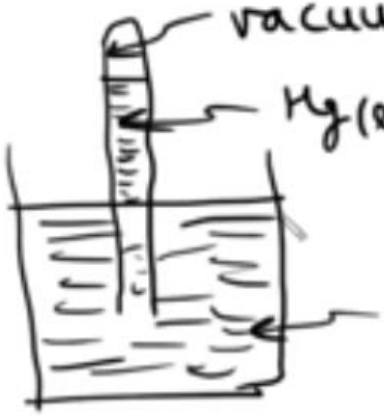
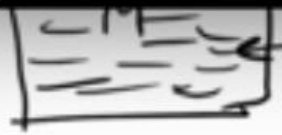
above Hg(ℓ) in capillary tube in empty space.

⇒ Faulty Barometer :



(instead of vacuum if any gas is trapped then the barometer reading will be less than the actual atm pressure.



Hg(ℓ) will be less ... the actual atm pressure.

⇒ If any other liquid (which is volatile) is used then reading of barometer will be wrong because vapours of liquid will be present in vacant space above that liquid in capillary tube. (Faulty Barometer)

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#6-GASEOUS STATE|MANOMETER|U TUBE MANOMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY ...

Manometer: A device used to measure pressure of a gas system.

U-tube manometer:

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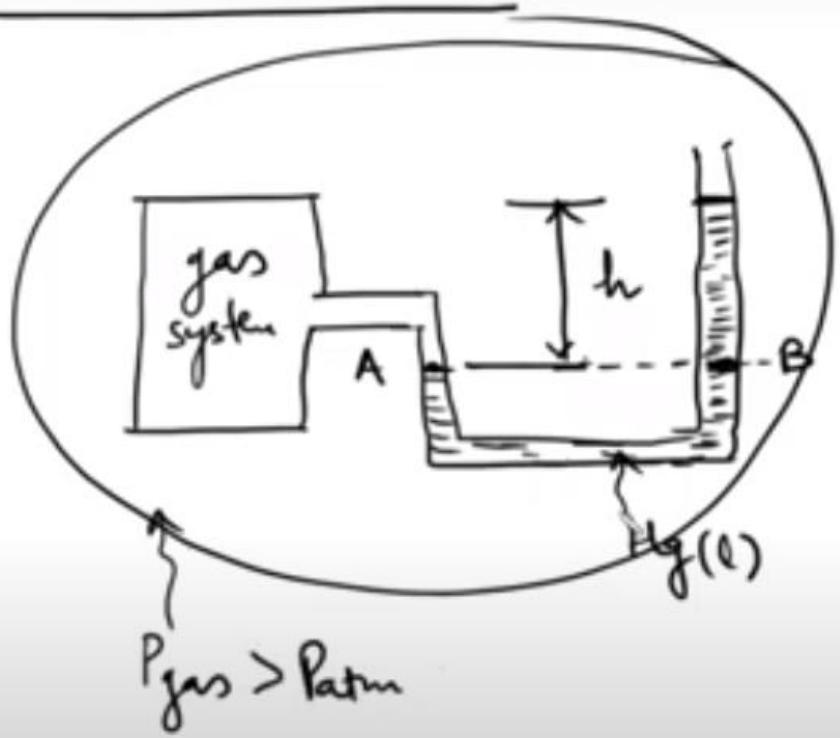
1:08 / 18:38

#6-GASEOUS STATE|MANOMETER|U TUBE MANOMETER|IIT ADVANCED|JEE MAIN|CHEMISTRY ...

U-tube manometer:



$P_{gas} > P_{atm}$

$P_A = P_B$

$P_{gas} = P_{atm} + h d g$

$d_{Hg(l)} = 13.6\ g/ml$

$= 13.6 \times 10^3\ kg/m^3$

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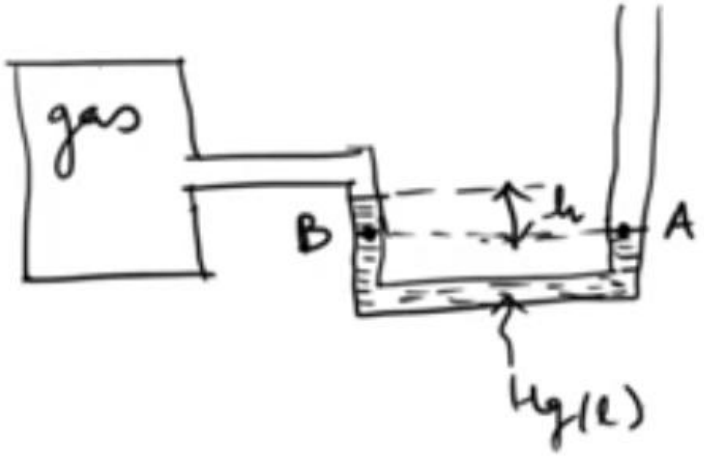
4:19 / 18:38

$P_{gas} > P_{atm}$

$= 13.6 \times 10^3 \text{ kg/m}^3$



$P_A = P_B$

$P_{atm} = h d g + P_{gas}$

$\Rightarrow P_{gas} = P_{atm} - h d g$

$\Rightarrow P_{gas} < P_{atm}$

(Partial vacuum)

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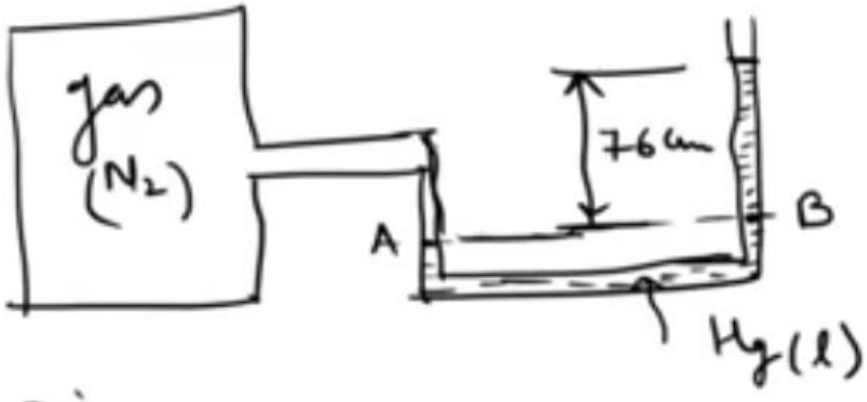
Subscribe

(Partial vacuum)

Q →



Find gas pressure. Given atmospheric pr = 1atm

Soln:

$P_{N_2} = P_B = 76$ cm of Hg + P_{atm}

$= 76 + 76 =$ 152 cm of Hg

$= \frac{152}{76} = 2$ atm

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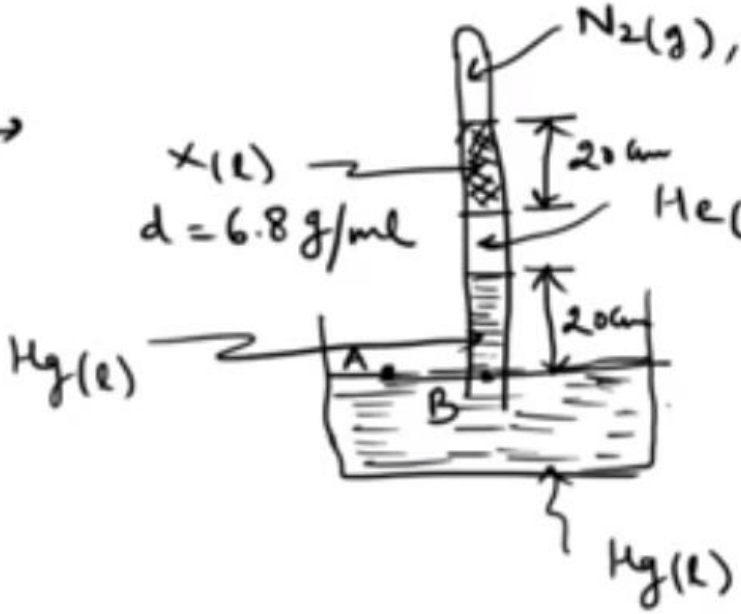
Q →



Find atmospheric pressure.

Soln: $P_A = P_B$

$P_{atm} = P_{Hg\,column} + P_{He} + P_{liq\,x} + P_{N_2}$

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Soln: $P_A = P_B$

$P_{atm} = P_{Hg\,column} + P_{He} + P_{liq\,x} + P_{N_2}$

$= 20\,cm\,of\,Hg + 0.2 \times 76\,cm\,of\,Hg + P_{x(\ell)} + 0.2 \times 76$

$\Rightarrow h_1 d_1 = h_2 d_2$

$20 \times 0.8 = h_{Hg(\ell)} \times 13.6$

$h_{Hg(\ell)} = 10\,cm$

20 cm of liq 'x' exerts same pr. as 10 cm of Hg.

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$20 \times 0.8 = h_{Hg(l)} \times 13.6$

$h_{Hg(l)} = 10\ cm$ | 20 cm of liq 'x' exerts same pr. as 10 cm of Hg.

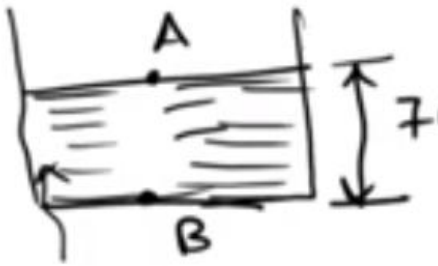
$P_{atm} = 20 + 0.2 \times 76 + 10 + 0.2 \times 76$

$= 20 + 10 + 0.4 \times 76$

$= 30 + 30.4 = 60.4$ cm of Hg

$= \frac{60.4}{76}$ atm Ans

$= \frac{60.4}{76}$ atm Ans

Q→ 

, Given P_{atm} = 1 atm

Hg(l)

To create same pressure at 'B', what height of water$_{(l)}$ ($d = 1$ g/ml) should be used.

Soln: $P_B = P_{atm} + 76 \times d_{Hg} \times g = P_{atm} + h_{H_2O(l)} \times 1 \times g$

$\Rightarrow h_{H_2O(l)} = 76 \times 13.6$ cm

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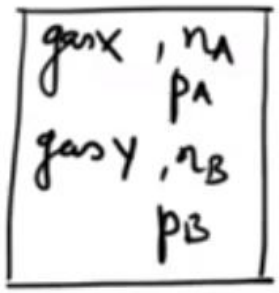
#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

Dalton's law of partial pressure:

→ At a given temperature, total pressure of a gas system consisting of two or more non-reacting gases is the sum of partial pressure of constituent gases.

1:52 / 22:04

T, V



Total Pr,

$P_T = p_A + p_B$

p_A and p_B — partial pr of gas A and B.

$P_T \times V = (n_A + n_B)\ RT$

$p_T = (n_A + n_B) \times \frac{RT}{V}$ — (1)

#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

$P_T =$

⇒ Partial pressure of a component gas of a gas mixture is the pressure exerted by that gas alone if present in same volume vessel at same temp.

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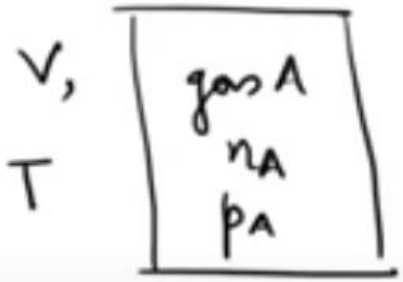
4:46 / 22:04

#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

by that gas alone if present in same volume vessel at same temp.



$$p_A = \frac{n_A}{V} \times RT \quad -(2)$$

$$p_B = \frac{n_B}{V} \times RT \quad -(3)$$

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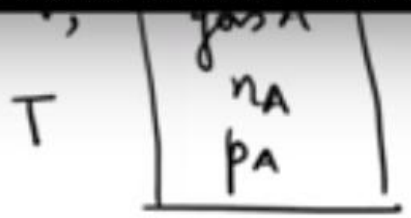
#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...



$$p_B = \frac{n_B}{V} \times RT \quad -(3)$$

$$p_A + p_B = \frac{(n_A + n_B)}{V} \times RT \quad -(4)$$

$$\Rightarrow \boxed{p_T = p_A + p_B}$$

$$\Rightarrow p_T = p_A + p_B + p_C + \dots$$

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6:24 / 22:04

$$\Rightarrow \boxed{P_T = P_A + P_B} \quad (1)$$

$$\Rightarrow P_T = P_A + P_B + P_C + \text{- - - - - -}$$

(2) ÷ (1),

$$\frac{P_A}{P_T} = \frac{\frac{n_A}{V} \times RT}{\frac{n_A + n_B}{V} \times RT}$$

$$\Rightarrow P_A = \left(\frac{n_A}{n_A + n_B}\right) \times P_T$$

$$\Rightarrow P_A = x_A \times P_T$$

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#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

$$\Rightarrow p_A = \left(\frac{n_A}{n_A + n_B}\right) \times p_T$$

$$\Rightarrow \boxed{p_A = x_A \times p_T} \quad \text{---}$$

Similarly, $\boxed{p_B = x_B \times p_T}$ ---

$$\Rightarrow \boxed{x_A + x_B + x_E + \cdots = 1}$$

↳ sum of mole fractions of all con...

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8:27 / 22:04

$\Rightarrow \boxed{P_B = x_B \times P_T}$

$\Rightarrow \boxed{x_A + x_B + x_E + \text{- - - - - } = 1}$

↳ sum of mole fractions of all components in a homogeneous mixture = 1

Note

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in a homogeneous mixture = 1

Note: When an inert gas is added into a a gas mixture isothermally and isochorically then partial pr. of existing gases remain unchanged. However total pr. of gas system will increase.

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Q → A vessel contains He, CH_4 & O_2 gases in 1:4:8 molar ratio. Find partial pr. of each gas if total pressure is 10 atm.

Soln:

1:4:8 molar ratio. Find partial pr. of each gas if total pressure is 10 atm.

Sol^n: He CH_4 O_2

1 : 4 : 8 molar ratio

$\frac{1}{13}$: $\frac{4}{13}$: $\frac{8}{13}$

x_{He} x_{CH_4} x_{O_2}

$p_{He} = \frac{1}{13} \times 10$ atm

$p_{CH_4} = \frac{4}{13} \times 10$ atm

$p_{O_2} = \frac{8}{13} \times 10$ atm

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Problems on open vessel:

$PM = \downarrow dRT \uparrow$

heated, $d \downarrow$, air escapes.

→ In problems of open vessel, P remains constant
V remains constant.

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#7-DALTON'S LAW OF PARTIAL PRESSURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

Q → An open flask contains air at 27°C. Calculate the temperature at which it should be heated so that

(a) $\frac{1}{5}$rd of air measured at 27°C escapes out.

(b) $\frac{1}{5}$rd of air measured at final temp. escapes out.

Sol^{n}: $n_{escaped} = n_i - n_f$

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Subscribe

18:39 / 22:04

Sol^n : $n_{escaped} = n_i - n_f$

$\frac{PxV}{RxT}$ $\quad \frac{1}{5} n_i = n_i - n_f$

$$\frac{1}{5} \times \frac{PxV}{Rx300} = \frac{PxV}{Rx300} - \frac{PxV}{RxT}$$

$$\frac{1}{T} = \frac{1}{300} - \frac{1}{900}$$

$$= \frac{2}{900} = \frac{1}{450}$$

$\Rightarrow T = 450\ K$ Ans

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$$= \frac{2}{400} = \frac{1}{450}$$

$$\Rightarrow T = 450\ K \quad \text{Ans}$$

(b) $n_{es} = n_i - n_f$

$$\frac{1}{3} \times (n_f) = \frac{P \times V}{R \times 300} - \frac{P \times V}{R \times T_f}$$

$$\frac{1}{3} \times \frac{P \times V}{R \times T_f} = \frac{P \times V}{R \times 300} - \frac{P \times V}{R \times T_f}$$

$$\frac{1}{T_f} + \frac{1}{3T_f} = \frac{1}{300} \Rightarrow \frac{4}{3T_f} =$$

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$$\frac{1}{5} \times (n_f) = \frac{P \times V}{R \times 300} - \frac{P \times V}{R \times T_f}$$

$$\frac{1}{3} \times \frac{P \times V}{R \times T_f} = \frac{P \times V}{R \times 300} - \frac{P \times V}{R \times T_f}$$

$$\frac{1}{T_f} + \frac{1}{3T_f} = \frac{1}{300} \Rightarrow \frac{4}{3T_f} = \frac{1}{300}$$

$$\Rightarrow T_f = 400\ K$$

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#8-GASEOUS STATE|GRAHAM'S LAW OF DIFFUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

Graham's law of diffusion:

Effusion: Escape of gas molecules from a vessel through a tiny hole or orifice due to pressure difference.

Diffusion

1:44 / 30:08

#8-GASEOUS STATE|GRAHAM'S LAW OF DIFFUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

vessel through a tiny hole or orifice due to pressure difference.

Diffusion : Intermixing of gases due to pressure difference.

2:20 / 30:08

#8-GASEOUS STATE|GRAHAM'S LAW OF DIFFUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

Graham's Law of diffusion:

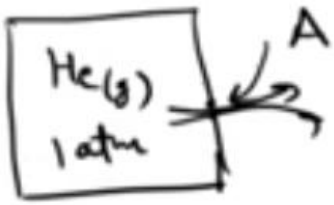
Effusion: Escape of gas molecules from a vessel through a tiny hole or orifice due to pressure difference into vaccuum.

Diffusion: Intermixing of gases due to pressure difference.

2:43 / 30:08

#8-GASEOUS STATE|GRAHAM'S LAW OF DIFFUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

Diffusion : Intermixing of gases due to pressure difference.



A tiny hole is punctured.

effusion.

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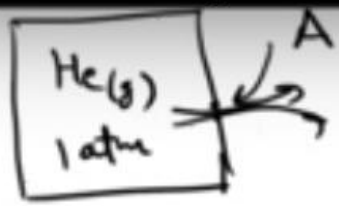

A hole is pierced

effusion.

Graham's law of diffusion or effusion:

At a given pressure and temperature,

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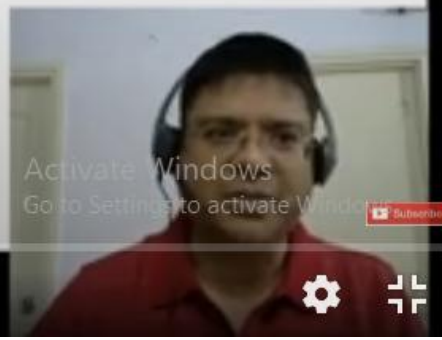
At a given pressure and temperature, rate of diffusion of gas is inversely proportional to the square root of molar mass of gas.



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$\Rightarrow$ rate of diffusion, $r \propto \frac{1}{\sqrt{M}}$

$\Rightarrow PM = dRT$

$M \propto d$

$\Rightarrow r \propto \frac{1}{\sqrt{d}}$ where

d = density of gas

M = Molar mass of gas.

d = density of gas

M = Molar mass of gas.

Rate of effusion 'r' = no of moles of gas effused per unit time = n'/t

or

= ~~no of~~ volume of gas effused per unit time = V'/t

= ~~no of~~ Volume of gas effused per unit time = V'/t

= pressure drop per unit time

$= \left(-\frac{dp}{dt}\right)$ or $\left(\frac{\Delta p}{\Delta t}\right)$

instantaneous rate of effusion

average rate of effusion

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instantaneous rate of effusion

average rate of effusion

* Under same P&T, heavier gases diffuse slower than the lighter gases.

Relative rate of effusion :

$$\frac{r_A}{r_B} = \sqrt{\frac{M_B}{M_A}}$$

$$\Rightarrow \frac{n_A'/t}{n_B'/t} = \sqrt{\frac{M_B}{M_A}}$$

$r_B \propto \sqrt{M_A}$

$$\Rightarrow \frac{n_A'/t}{n_B'/t} = \sqrt{\frac{M_B}{M_A}} \quad \text{or} \quad \frac{V_A'/t}{V_B'/t} = \sqrt{\frac{M_B}{M_A}}$$

$$\text{or} \quad \frac{(\Delta p/\Delta t)_A}{(\Delta p/\Delta t)_B} = \sqrt{\frac{M_B}{M_A}}$$

rate of effusion of effusion.

* Under same P&T, heavier gases diffuse slower than the lighter gases.

Relative rate of effusion :

$$\frac{r_A}{r_B} = \sqrt{\frac{M_B}{M_A}} \quad \left\{ \text{At a given P\&T} \right\}$$

$$\Rightarrow \frac{n_A'/t}{n_B'/t} = \sqrt{\frac{M_B}{M_A}} \quad \text{or} \quad \frac{V_A'/t}{V_B'/t} = \sqrt{\frac{M_B}{M_A}}$$

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$(\Delta P/\Delta t)_B$ ∝

n_A', n_B' → no. of moles of gases A & B effused out.

V_A', V_B' → vol. of gases A and B effused out.

In g

$(\Delta p/\Delta t)_B$ ∝

n_A', n_B' → no. of moles of gases A & B effused out.

V_A', V_B' → vol. of gases A and B effused out.

In general, rate of effusion, $r \propto \frac{p A_o}{\sqrt{MT}}$

where p = partial pr. difference of gas

A_o = Crossection area of hole or orifice

T = Temperature in Kelvin.

$$r \propto \frac{pA_0}{\sqrt{MT}}$$

For a given volume of vessel, $p \propto T$

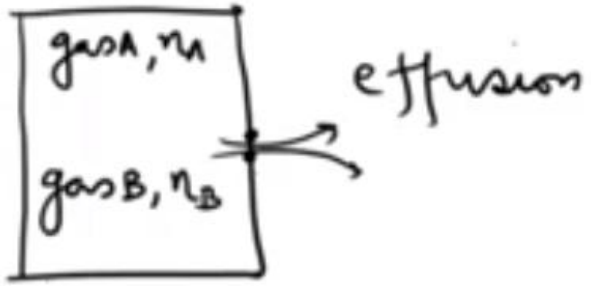
$\Rightarrow T \uparrow, r \uparrow$

#8-GASEOUS STATE|GRAHAM'S LAW OF DIFFUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

for a given volume of vessel, P

$\Rightarrow T \uparrow, n \uparrow$

e.g.



→ For two gases in same vessel effusion through an orifice, A_o → same, T → same } for both

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gas B, M_B

→ For two gases in same vessel effusion through an orifice, A_o → same, T → same } for both gases

$$r \propto \frac{P A_o}{\sqrt{M T}} \Rightarrow r \propto \frac{P}{\sqrt{M}}$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{p_1}{p_2}\sqrt{\frac{M_2}{M_1}} \quad \Big| \quad \frac{r_A}{r_B} = \frac{p_A}{p_B}\sqrt{\frac{M_B}{M_A}}$$

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$$\frac{V_A'/t}{V_B'/t} = \frac{n_A'/t}{n_B'/t} = \frac{n_A}{n_B}\sqrt{\frac{M_B}{M_A}}$$

* From Dalton's law

$$\frac{p_A}{p_B} = \frac{x_A \times p_T}{x_B \times p_T} \Rightarrow \frac{p_A}{p_B} = \frac{n_A}{n_B}$$

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Q → A vessel contains H_2 and He gases in molar ratio 2:3. Find volume of He gas that will be collected in 5 min, if volume of H_2 gas collected in 5 min is 20 ml if both gases are allowed to effuse.

both gases are allowed to effuse through a tiny hole.

Sol^{n}:

$$\frac{r_{He}}{r_{H_2}} = \frac{p_{He}}{p_{H_2}}\sqrt{\frac{M_{H_2}}{M_{He}}} \Rightarrow \frac{V_{He}/5}{20/5} = \frac{3}{2}\times\sqrt{\frac{2}{4}}$$

$$\Rightarrow \frac{V_{He}}{20} = \frac{3}{2\sqrt{2}} \Rightarrow V_{He} = \frac{30}{\sqrt{2}}\ ml$$

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$$\Rightarrow \frac{V_{He}}{\frac{26}{10}} = \frac{3}{1\sqrt{2}} \Rightarrow V_{He} = \frac{30}{\sqrt{2}}\ ml$$

Q → A vessel initially contains He and CH_4 in mass ratio 3 : 1. If gases are allowed to effuse through a hole, find ratio of no. of moles effuses initially.

of no of moles effuses initially

Soln: $He : CH_4$

$3 : 1$ (Mass ratio)

$\frac{3}{4} : \frac{1}{16}$ $\Rightarrow$ 12 : 1 molar ratio

$$\frac{n'_{He}/t}{n'_{CH_4}/t} = \frac{12}{1} \times \sqrt{\frac{16}{4}} = 24$$

3 : 1 (Mass ratio)

$\frac{3}{x} : \frac{1}{16_4}$ ⇒ 12 : 1 molar ratio

$$\frac{n'_{He}/x}{n'_{CH_4}/x} = \frac{12}{1} \times \sqrt{\frac{16_4}{x}} = 24$$

$n'_{He} : n'_{CH_4} = 24:1$ Ans

$r_{He} : r_{CH_4} = 2 : 1$ (by)

Q → A vessel contains a mixture of He, D_2, H_2 in mass ratio 1 : 2 : 3. Find intial ratio of their rate of effusion through a tiny hole.

Sol^n:

He ; D_2 ; H_2

1 ; 2 ; 3 mass ratio

$\frac{1}{4} : \frac{2}{4} : \frac{3}{2} \Rightarrow 1 : 2 : 6$ molar ratio

$r_{He} : r_{D_2} : r_{H_2} = \frac{1}{\sqrt{4}} : \frac{2}{\sqrt{4}} : \frac{6}{\sqrt{2}}$

$= 1 : 2 : 6\sqrt{2}$ Ans

Note: ① Rate of effusion of a gas depends on partial pressure difference and not on total pressure.

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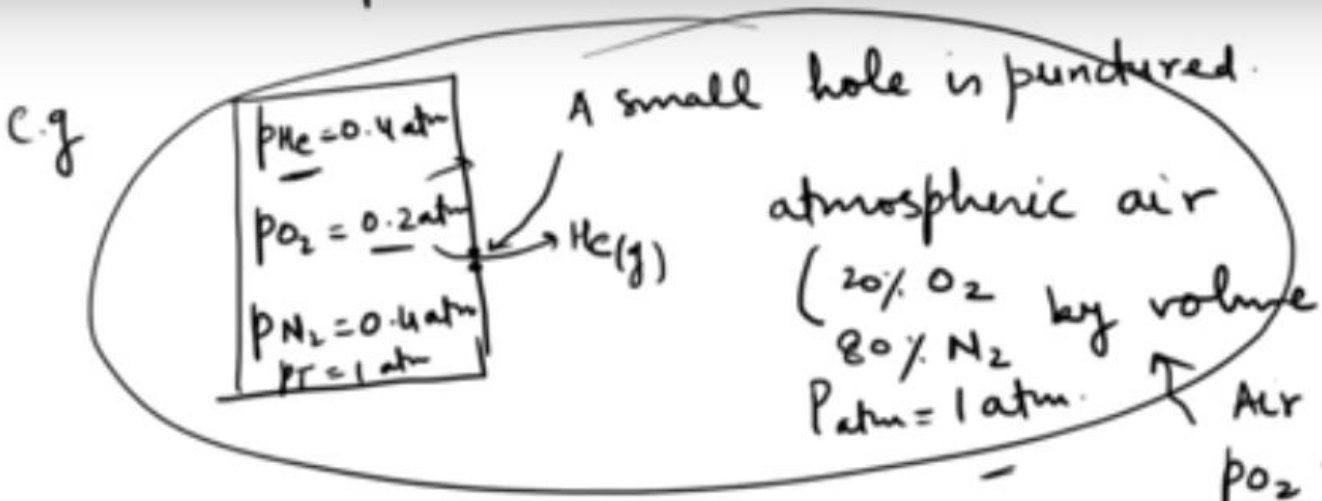
total pressure



Finally, In the vessel,

$p_{He} \cong 0$, $p_{O_2} = 0.2$ atm, $p_{N_2} = 0.8$ atm

6:24 / 34:57

$p_{He} \cong 0$, $p_{O_2} = 0.2$ atm , $p_{N_2} = 0.8$ atm

(2) In previous video lecture, rate of effusion of a gas was assumed constant over a period of time and accordingly questions were solved.

→ But fact is that rate of effusion of gas continuously decrease as pressure of gas decrease continuously with time.

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time and accordingly questions were solved.

→ But fact is that rate of effusion of gas continuously decrease as pressure of gas decrease continuously with time during the process of effusion.

Imp

5 seconds

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decrease continuously with time during the process of effusion.

Considering decrease in pressure with time,

Rate of effusion, $-\frac{dp}{dt}$

pressure drop per unit time

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Rate of effusion, $\left(\frac{-dp}{dt}\right) \propto \frac{p}{\sqrt{M}}$

pressure drop per unit time

$$\Rightarrow \quad -\frac{dp}{dt} = \frac{k\,p}{\sqrt{M}}$$

$$-\int_{p_0}^{p} \frac{dp}{p} = \frac{k}{\sqrt{M}} \times \int_0^t dt \quad \Rightarrow \quad -\left[\ln p\right]_{p_0}^{p}$$

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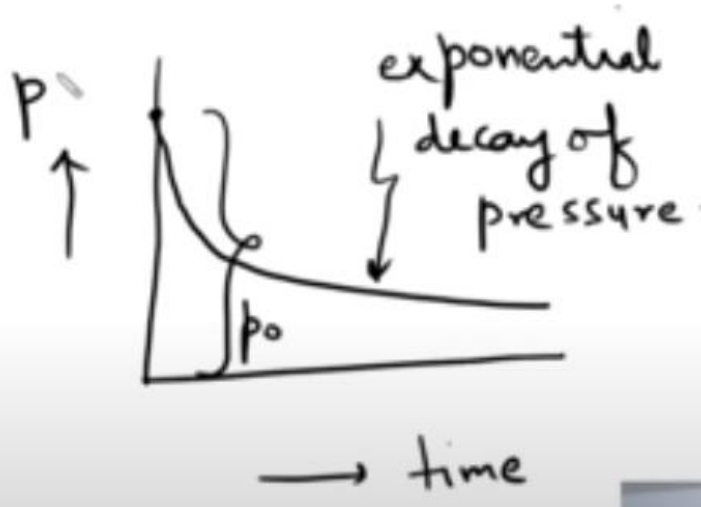
$$\frac{dt}{\sqrt{M}}$$

$$-\int_{p_0}^{p}\frac{dp}{p} = \frac{k}{\sqrt{M}} \times \int_0^t dt \Rightarrow -\left[\ln p\right]_{p_0}^{p} = \frac{k}{\sqrt{M}} \times t$$

$$\Rightarrow -\ln\frac{p}{p_0} = \frac{kt}{\sqrt{M}}$$

$$\boxed{p = p_0 e^{-\frac{kt}{\sqrt{M}}}}$$



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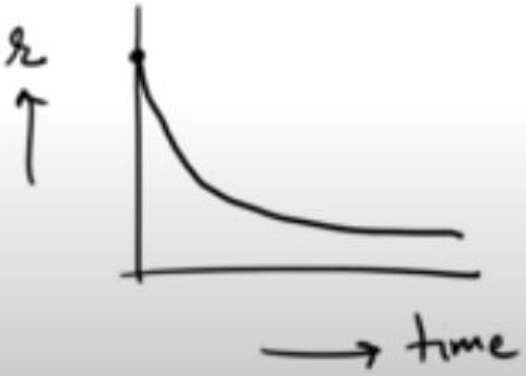
$$\Rightarrow -\ln \frac{p}{p_0} = \frac{kt}{\sqrt{M}}$$

$$\boxed{p = p_0 e^{-\frac{kt}{\sqrt{M}}}}$$

$$\Rightarrow r \propto p$$



13:13 / 34:57

→ time

Q → Find rate of effusion He gas wrt a gas mixture (He & H_2 in 1:2 molar ratio) under same temp and same total pressure.

Sol: 1st approach

$$\frac{r_{He}}{r_{mix}} = \sqrt{\frac{(M_{av})_{mix}}{M_{He}}}$$

$$M_{av} = \frac{1\times4 + 2\times2}{1+2} = \frac{8}{3}$$

$$= \sqrt{\frac{8}{3\times4}} = \sqrt{\frac{2}{3}} = \frac{1.41}{1.7}$$

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$$\frac{r_{mix}}{r_{He}} = \sqrt{\frac{M_{He}}{M_{mix}}} = \frac{5}{3}$$

$$= \sqrt{\frac{8}{5\times4}} = \sqrt{\frac{2}{3}} = \frac{1.41}{1.73} \approx \theta_1$$

2nd approach: $He : H_2 = 1:2$ molar mixture

$p_{He} = p/3$, $p_{H_2} = 2p/3$, Total $pr = p$

$$\frac{r_{He}}{r_{mix}} = \frac{r_{He}}{(r_{He}+r_{H_2})} \Rightarrow \frac{r_{mix}}{r_{He}} = \frac{(r_{He})_{mix}}{r_{He}} + \frac{(r_{H_2})_{mix}}{r_{He}}$$

$$\frac{r_{mix}}{r_{He}} = \frac{p/3}{p}\sqrt{\frac{4}{4}} + \frac{2p/3}{p}\sqrt{\frac{x^2}{2}} = \frac{1}{3} + \frac{2\sqrt{2}}{3}$$

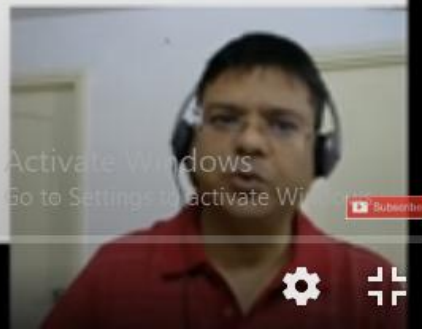
$$\Rightarrow \frac{r_{He}}{r_{mix}} = \frac{5}{1+2\sqrt{2}} = \frac{3}{1+2\times1.4} = \frac{3}{3.8} = 0.8$$

#9-APPLICATIONS OF RELATIVE RATE OF FUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

Application of relative rate of effusion :

↳ U.S. atomic commission → to enrich U^{235} over U^{238}.

$^{235}UF_{6(g)}$ $U^{238}F_{6(g)}$



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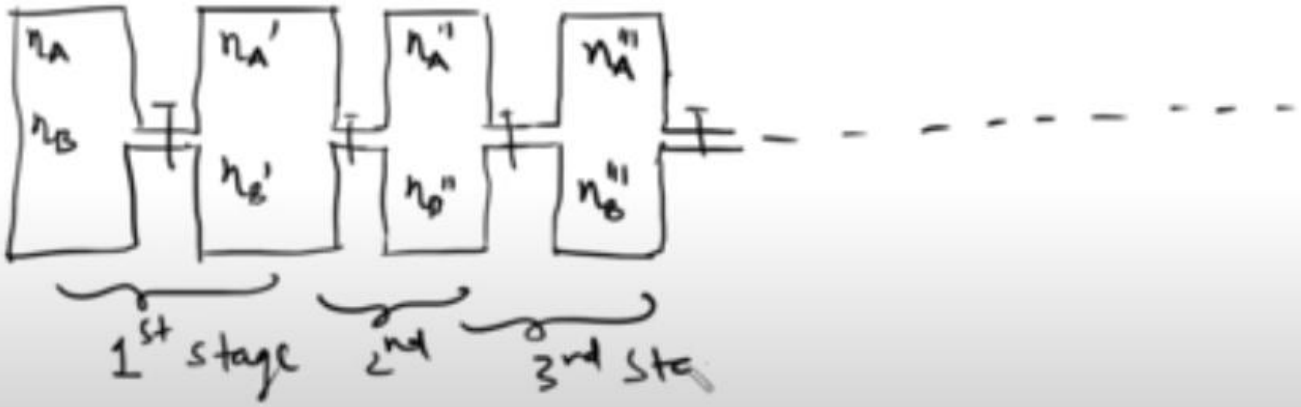
↳ U.S. atomic commission ⇒ [illegible]

$^{235}UF_{6(g)}$ $U^{238}F_{6(g)}$ ⇌

e.g. To enrich a gas mixture.



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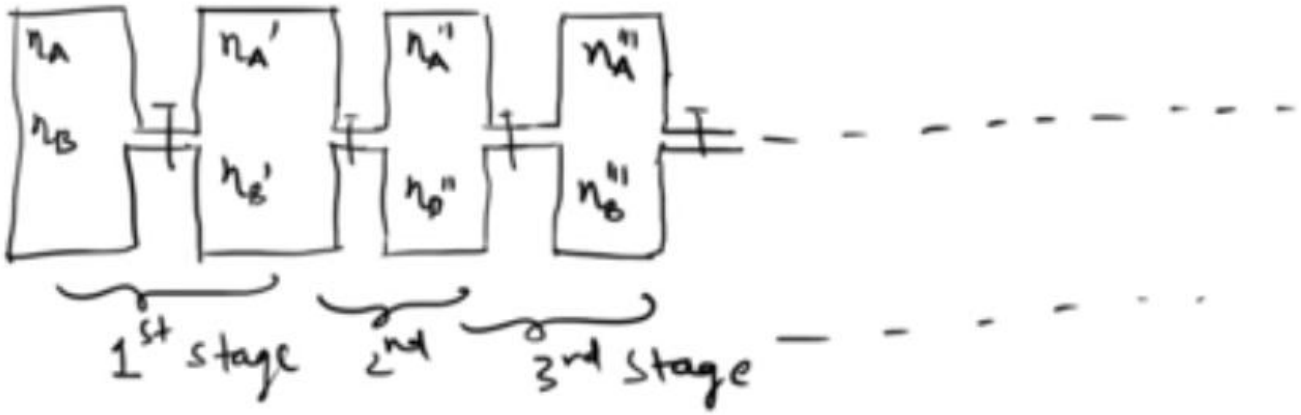
e.g. To enrich a gas mixture.



For 1st stage of effusion, $\frac{n_A'}{n_B'} = \frac{n_A}{n_B}\sqrt{\frac{M_B}{M_A}}$

2nd stage, $\frac{n_A''}{n_B''} = \frac{n_A'}{n_B'}\sqrt{\frac{M_B}{}}$

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Subscribe

1st stage 2nd 3rd stage — - - - - .

For 1st stage of effusion, $\frac{r_A'}{r_B'} = \frac{r_A}{r_B}\sqrt{\frac{M_B}{M_A}}$

2nd stage, $\frac{r_A''}{r_B''} = \frac{r_A'}{r_B'}\sqrt{\frac{M_B}{M_A}}$

3rd stage, $\frac{r_A'''}{r_B'''} = \frac{r_A''}{r_B''}\sqrt{\frac{M_B}{}}$

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1st stage of effusion, $\frac{r_A}{r_B'} = \frac{r_A}{r_B}\sqrt{\frac{M_B}{M_A}}$ — (1)

2nd stage, $\frac{r_A''}{r_B''} = \frac{r_A'}{r_B'}\sqrt{\frac{M_B}{M_A}}$ — (2)

3rd stage, $\frac{r_A'''}{r_B'''} = \frac{r_A''}{r_B''}\sqrt{\frac{M_B}{M_A}}$ — (3)

(1) × (2) $\quad \frac{r_A''}{r_B''} = \frac{r_A}{r_B} \times \left(\frac{M_B}{M_A}\right)^{2/2}$

(1) × (2) × (3) $\quad \frac{r_A'''}{r_B'''} = \frac{r_A}{r_B} \times \left(\frac{M_B}{M_A}\right)^{3/2}$

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3rd stage, $\frac{n_A}{n_B'''} = \frac{n_A''}{n_B''}\sqrt{\frac{M_B}{M_A}}$ — (3)

(1) × (2) $\quad \frac{n_A''}{n_B''} = \frac{n_A}{n_B} \times \left(\frac{M_B}{M_A}\right)^{2/2}$

(1) × (2) × (3) $\quad \frac{n_A'''}{n_B'''} = \frac{n_A}{n_B} \times \left(\frac{M_B}{M_A}\right)^{3/2}$

$$\boxed{\frac{n_A^x}{n_B^x} = \frac{n_A}{n_B} \times \left(\frac{M_B}{M_A}\right)^{x/2}}$$

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#9-APPLICATIONS OF RELATIVE RATE OF FUSION|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

$$\frac{n_A^x}{n_B^x} = \left(\frac{n_A}{n_B}\right)\left(\frac{M_B}{M_A}\right)^{x/2}$$

where x = no. of diffusion stages.

$\Rightarrow$ final ratio of number of moles after x diffusion steps.

initial ratio of no. of moles

29:45 / 34:57

Q → Intial a Uranium sample contains U and U in 1:9 molar ratio. It has to be enriched as 9:1 molar ratio by using the concept of relative rati of effusion of $U^{235}F_6$ and $U^{238}F_6$. How many diffusion step are needed. [$\log 81 = 1.9$, $\log \frac{352}{349} = 3.7 \times 10^{-3}$]

F → 19

Sol^n : Let x diffusion steps needed.

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step are needed. [$\log 81 = 1.9$, $\log \frac{352}{349} = 5.7 \times 10^{-3}$]

$F \to 19$

Sol^n : Let x diffusion steps needed.

$$\frac{9}{1} = \frac{1}{9} \times \left(\sqrt{\frac{352}{349}}\right)^{x} \Rightarrow 9^2 = \left(\frac{352}{349}\right)^{x/2}$$

$$\log 81 = \frac{x}{2} \log \frac{352}{349}$$

$$\Rightarrow x = \frac{2 \times 1.9}{3.7 \times 10^{-3}} = 380$$

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step are needed. [$\log 81 = 1.9$, $\log \frac{352}{349} = 3.7 \times 10^{-3}$]

F → 19

Soln : Let x diffusion steps needed.

$$\frac{9}{1} = \frac{1}{9} \times \left(\sqrt{\frac{352}{349}}\right)^{x} \Rightarrow 9^2 = \left(\frac{352}{349}\right)^{x/2}$$

$$\log 81 = \frac{x}{2} \log \frac{352}{349}$$

$$\Rightarrow x = \frac{2 \times 1.9}{3.7 \times 10^{-3}} = \frac{3800}{3.7} \text{ Ans}$$

$$=$$

#10-GASEOUS STATE|PAYLOAD CAPACITY OF BALLOON|IIT ADVANCED|JEE MAIN|CHEMISTRY...

Payload Capacity of balloon :

⇒ Payload is defined as the maximum mass a balloon can carry in atmospheric air to the desired height.

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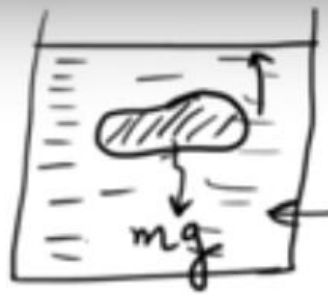
#10-GASEOUS STATE|PAYLOAD CAPACITY OF BALLOON|IIT ADVANCED|JEE MAIN|CHEMISTRY...



F_b (Buoyant force)

$H_2O_{(l)}$

In general,

if $mg > F_b$ (object sinks)

if $mg < F_b$ (object floats)

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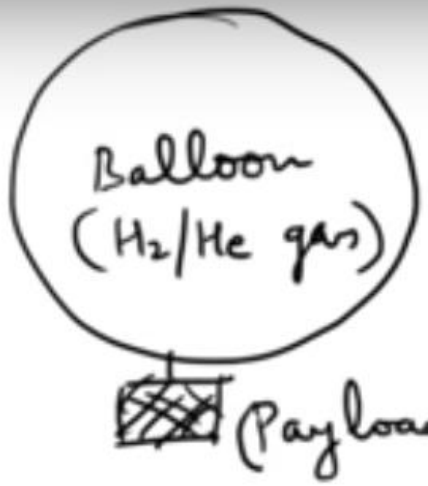
#10-GASEOUS STATE|PAYLOAD CAPACITY OF BALLOON|IIT ADVANCED|JEE MAIN|CHEMISTRY...



→ For balloon, air is the medium.

⇒ If balloon has to rise up in atmospheric air,

Total weight of balloon ≤ F_b (Buoyant

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Total weight of balloon $\leq F_b$ (Buoyant force)

$\Rightarrow$ Buoyant force = weight of medium displaced by balloon

$$(m_{\text{balloon material}} + m_{\text{gas}} + m_{\text{payload}}) \times g \leq m_{\text{air displaced}} \times g$$

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#10-GASEOUS STATE|PAYLOAD CAPACITY OF BALLOON|IIT ADVANCED|JEE MAIN|CHEMISTRY...

displaced by balloon

$$(m_{\text{balloon material}} + m_{\text{gas}} + m_{\text{payload}}) \times g \leq m_{\text{air displaced}} \times g$$

$$\Rightarrow \boxed{(m_{\text{balloon}} + m_{\text{gas}} + m_{\text{payload}}) \leq d_{\text{air}} \times V_{\text{balloon}}}$$

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8:59 / 31:23

$$\leq m_{air\ displaced} \times g$$

$$\Rightarrow \boxed{(m_{balloon} + m_{gas} + m_{payload}) \leq d_{air} \times V_{balloon}}$$

* In numerical problems, mass of balloon material can be ignored if not given in the question.
* In general we assume constant density of air with altitude. So F_b (Buoyant force is taken constat with altitude).

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#10-GASEOUS STATE|PAYLOAD CAPACITY OF BALLOON|IIT ADVANCED|JEE MAIN|CHEMISTRY...

constat with altitude).

* However, the fact is that Buoyant force continuously decrease as density ~~of~~ of air decrease with altitude.

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12:15 / 31:23

(constat with altitude).

* However, the fact is that Buoyant force continuously decrease as density of air decrease with altitude.

Q→ Find payload of 100 lit balloon filled with He gas at 1atm and 27°C. Given density of air = 1.3 g/lit. Take mass of balloon material = 20 g.

Soln: $m_{Payload} + 20 + m_{gas} \leq 1.3 \times 100$

$n_{He} = \frac{1 \times 100}{\frac{1}{12} \times 300}$

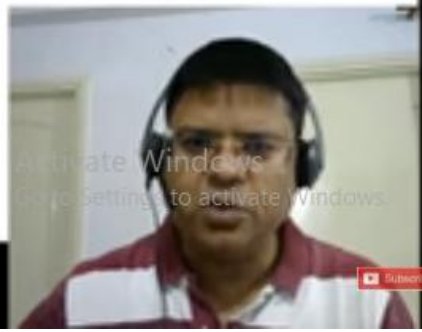
$= 4\,g$

$\Rightarrow m_{Payload} = 130 - 20 - 16 = \underline{94\,g}$ Ans

Q→ Mr. X wants to fly in the sky with the help of balloons each containing 50 mol of H_2 gas at 0.5 atm and 27°C. How many balloons Mr. X (weight = 80 Kg) must attach with himself to fly if density of air is 1.25 g/lit.

Soln: x balloons

$$80\times10^3 + 100\times x + 0 \leq x\times\left\{\frac{\left(50\times\frac{1}{12}\times300\right)}{0.5}\right\}\times 1.25$$



to fly if density of air is 1.25 g/lit.

Soln: x balloons

$$80\times10^3 + 100\times x + 0 \leq x\times\left\{\frac{\left(50\times\frac{1}{12}\times300\right)}{0.8}\right\}\times 1.25$$

$$\Rightarrow \quad 80\times10^3 = (2500\times1.25\,x - 100x)$$

$$\Rightarrow \quad (31.25x - x) = 800$$

$$\Rightarrow \quad x = \frac{800}{30.25} = 26.\ldots$$

Ans: 27 balloons must be attached with Mr. X.

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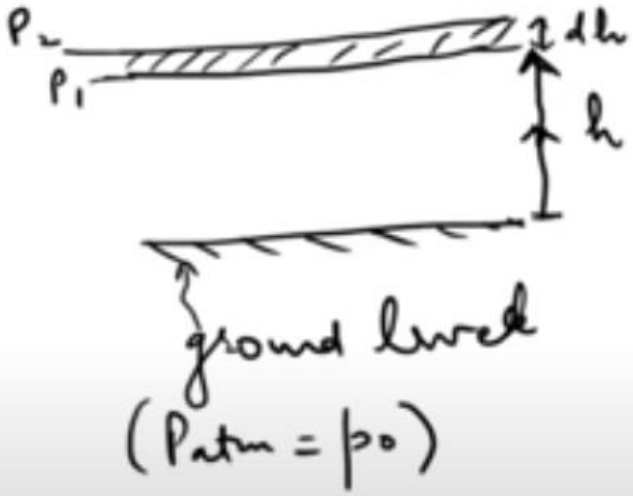
Variation of atmospheric pressure with altitude :

$P_2 - P_1 = (dh) \times \underline{d}g$

$-dp = dh \times dg$

$\Rightarrow PV = nRT$ (Air is treated ideal gas)

$\Rightarrow PM = dRT$



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$\Rightarrow x = \frac{}{30.25}$

Ans : 27 balloons must be attached with Mr. X.

Variation of atmospheric pressure with altitude :

(Barometric distribution)

$P_2 - P_1 = (dh) \times \underline{d}g$

$-dp = dh \times dg$

ground lev

$(P_{atm} = p_0)$

$\Rightarrow PV = nRT$ (Air is treated ideal gas)

$\Rightarrow PM = \rho RT$

$\frac{PM}{}$

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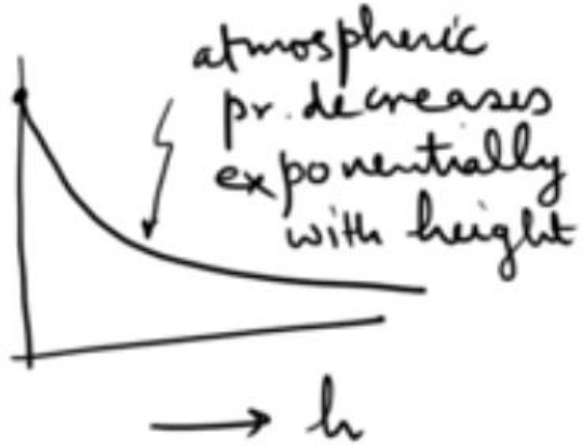
$(P_{atm} = p_0)$

$$-dp = dh \times \frac{PM}{RT} \times g$$

$$\int_{p_0}^{p} \frac{dp}{p} = \frac{Mg}{RT} \int_{0}^{h} dh$$



$$-\ln \frac{p}{p_0} = \frac{mg}{RT} \times h$$

$$\Rightarrow \boxed{p = p_0 e^{-(Mgh/RT)}}$$

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$$-\ln \frac{p}{p_0} = \frac{mg}{RT} \times h$$

$\longrightarrow h$

$$\Rightarrow \boxed{p = p_0 e^{-(mgh/RT)}}$$

* M_{air}, g, T are assumed constant

$$\Rightarrow \boxed{p = p_0 e^{-(mgh/RT)}}$$

* M_{air}, g, T are assumed constant.

$$PM = dRT$$

$$d \propto p$$

$$\Rightarrow \frac{d}{d_0} = \frac{p}{p_0} = e^{-(mgh/RT)}$$

$$\Rightarrow \boxed{d_{air} = d_0 e^{-(mgh/RT)}}$$

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$PM = dRT$

$d \propto p$

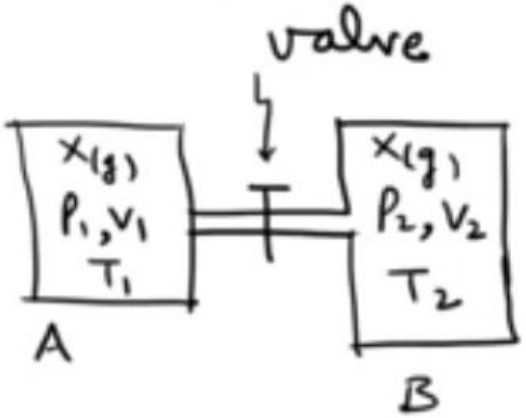
$$\Rightarrow \frac{d}{d_0} = \frac{p}{p_0} = e^{-(mgh/RT)}$$

$$\Rightarrow \boxed{d_{air} = d_0 \, e^{-(mgh/RT)}}$$

d_0 : density of air at ground level.

#11-GASEOUS STATE|PROBLEMS ON CONNECTING VESSELS|IIT ADVANCED|JEE MAIN|CHEMIS...

Problems on Connecting vessels :



∴ After open

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2:11 / 28:37

#11-GASEOUS STATE|PROBLEMS ON CONNECTING VESSELS|IIT ADVANCED|JEE MAIN|CHEMIS...

∴ After opening valve, at eqm
(after mixing of gases)

(i) Pressure in all vessels connected will be same.

(ii) Total number of moles (non-reacting gases) will be conserved.

$n_T = n_1 + n_2$

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4:24 / 28:37

#11-GASEOUS STATE|PROBLEMS ON CONNECTING VESSELS|IIT ADVANCED|JEE MAIN|CHEMIS...

(after mixing of gases)

(i) Pressure in all vessels connected will be same.

(ii) Total number of moles (non-reacting gases) will be conserved.

$n_T = n_1 + n_2$

$n_1' + n_2' = n_1 + n_2$

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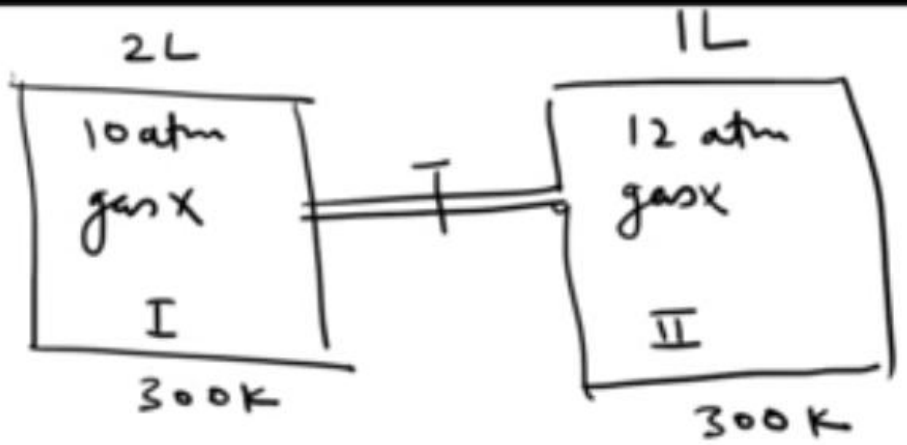
4:37 / 28:37

Q →



If valve is opened & temperature of vessel-II is raised to 500K, find pressure in each vessel finally.

Soln

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is raised to 500 K, find pressure in each vessel finally.

Soln:

$$n_1' + n_2' = n_T = n_1 + n_2$$

$$\frac{P_f \times 2}{R \times 300} + \frac{P_f \times 1}{R \times 500} = \frac{10 \times 2}{R \times 300} + \frac{12 \times 1}{R \times 300}$$

$$\left(\frac{2}{3} + \frac{1}{5}\right) \times P_f = \frac{20}{3} + 4 = \frac{32}{3}$$

$$\frac{10+3}{15} \times P_f = \frac{32}{3} \Rightarrow P_f = \frac{32 \times 5}{13}$$

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$$\frac{P_f \times 2}{R \times 300} + \frac{P_f \times 1}{R \times 500} = \frac{10 \times 2}{R \times 300} + \frac{12 \times 1}{R \times 300}$$

$$\left(\frac{2}{3} + \frac{1}{5}\right) \times P_f = \frac{20}{3} + 4 = \frac{32}{3}$$

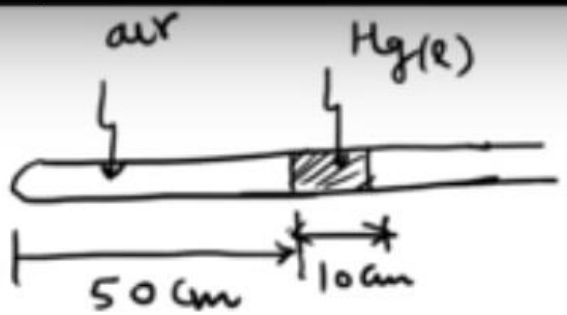
$$\frac{10+3}{15_5} \times P_f = \frac{32}{3} \Rightarrow P_f = \frac{32 \times 5}{13}$$

$$= \frac{160}{13} \text{ atm}$$

$$Q_2$$

#11-GASEOUS STATE|PROBLEMS ON CONNECTING VESSELS|IIT ADVANCED|JEE MAIN|CHEMIS...

Q →



(Horizontal capillary tube with one end open)

Patm (given) = 76 cm of Hg.

Find length of air column if

(i) tube is held vertical with open end up.

(ii) tube is held vertical with open end dow

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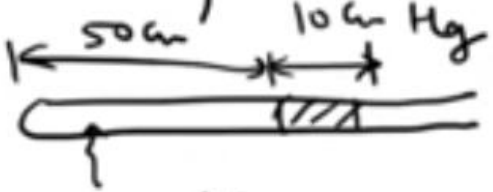
12:29 / 28:37

(ii) tube is held vertical with open end down.

(iii) tube is held at 45° from horizontal with open end up.

Soln:



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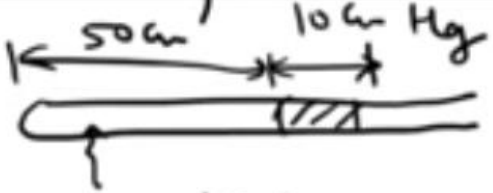
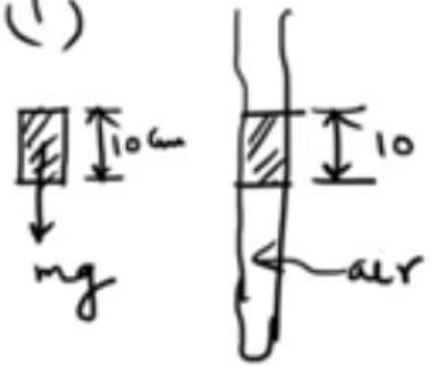
Sol^{n} :



(i)



$P_1V_1 = P_2V_2$

$76 \times (A \times 50) = (76+10) \times (A \times l_1)$

$l_1 = \frac{76 \times 50}{86}$ cm Ans

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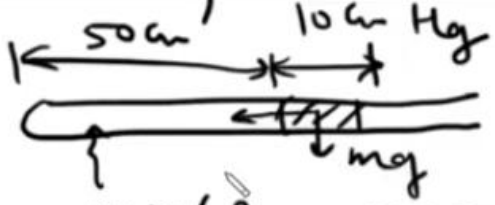
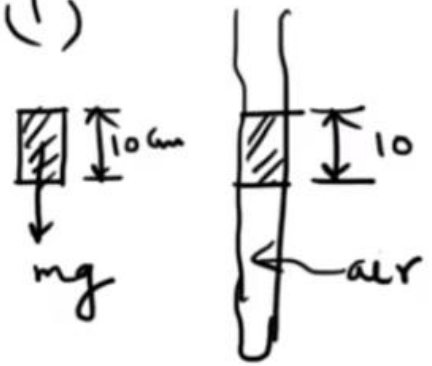
Soln:



(i)



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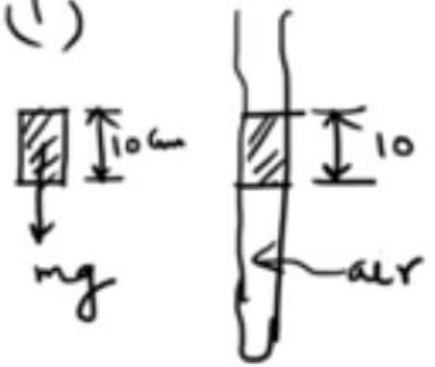
(i)

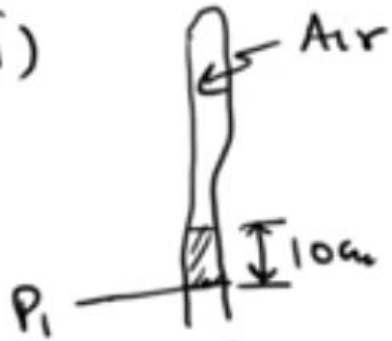


$P_1V_1 = P_2V_2$

$76 \times (A \times 50) = (76+10) \times (A \times l_1)$

$l_1 = \frac{76 \times 50}{86}$ cm Ans

(ii)



$P_1 = 10 + P_{air}$

$\Rightarrow P_{air} = 76-10$

$= 66$ cm of Hg

$76 \times (A \times 50) = (76-10) \times (A \times l_2)$

$l_2 = \frac{76 \times 50}{66}$ cm Ans

mg air

$l_1 = \frac{76 \times 50}{86}$ cm Ans

(ii)

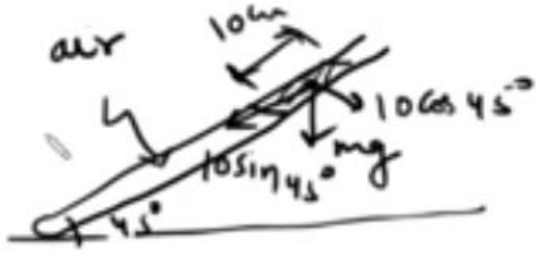
$P_1 = 10 + P_{air}$

$\Rightarrow P_{air} = 76 - 10 = 66$ cm of Hg

$76 \times (A \times 50) = (76-10) \times (A \times l_2)$

$l_2 = \frac{76 \times 50}{66}$ cm Ans

(iii)



$76 \times (A \times 50) = (76 + 10\sin 45) \times A \times l_3$

$\Rightarrow l_3 = \left(\frac{76 \times 50}{76 + \frac{10}{\sqrt{2}}}\right)$ cm Ans

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Jee

Q → A closed vertical container divided into two parts by a frictionless piston, each part contains 1 mol of air. At 300 K, the volume of upper part is 4 times than that of lower part. Calculate the temperature when volume of upper part will be 3 times than that of the lower part.

Soln:

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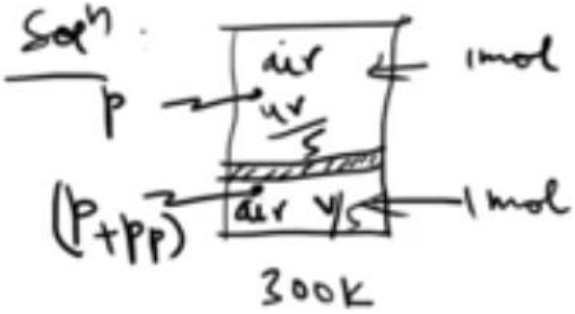
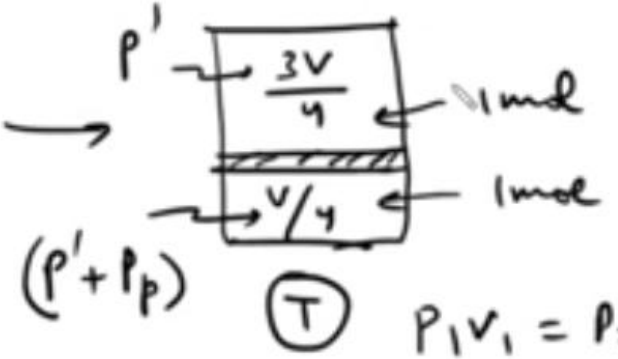
will be 3 times than that of the lower part.

Soln:





$P_1V_1 = P_2V_2$

$(P+P_p) \times \frac{V}{5} = P \times \frac{4V}{5}$

$P_p = 3P$

$P_1V_1 = P_2V_2$

$P' \times \frac{3V}{4} = (P'+3P) \times \frac{V}{4}$

$2P' = 3P$

$P' = \frac{3P}{2}$

$P_f = 3P$

$2P' = 3P$

$P' = \frac{3P}{2}$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ (For upper part)

$$\frac{P \times \frac{4V}{5}}{300} = \frac{\left(\frac{3P}{2}\right) \times \left(\frac{3V}{4}\right)}{T}$$

$$\frac{4}{5 \times 300} = \frac{9}{8T} \Rightarrow T = \frac{9}{8} \times \frac{5 \times 300}{4}$$

$$= \frac{45 \times 75}{8}$$

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#12-GASEOUS STATE|KINETIC THEORY OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

Kinetic theory of Gases : (KTG)

→ KTG provides a theoretical background to various Gas laws like Boyle's law, charle's law, etc.

2:01 / 38:10

#12-GASEOUS STATE|KINETIC THEORY OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

Gas laws like Boyle's law, Charle's law, etc.

Postulates :

→ molecules/atoms of a gas consist of identical particles and have same mass, size and shape (spherical)

5 seconds

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3:08 / 38:10

(spherical).

→ No intermolecular forces between gas molecules.
(P.E = 0)

→ Volume of gas molecule is negligible in comparison to volume of gas.

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→ Since no intermolecular forces, molecules randomly move with high K.E and collide among themselves as well as with the walls of container. Pressure exerted by gas is the consequence of collisions of gas molecules with walls of container.

Consequence of collisions of gas molecules with walls of container.

→ Since no intermolecular forces between gas molecules, molecules move in straight line between collisions.

→ Since no intermolecular forces between gas molecules, molecules move in straight line between collisions.

→ All collisions between molecules and with walls of vessel are perfectly elastic i.e. energy is conserved in collision (i.e KE is conserved as P.E = 0)

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→ Average K.E of gas molecules is directly proportional to the absolute temperature.

→ Effect of gravity on gas molecules is negligible.

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#12-GASEOUS STATE|KINETIC THEORY OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

Kinetic theory of Gases : (KTG) (for Ideal gas)

→ KTG provides a theoretical background to various Gas laws like Boyle's law, charle's law, etc.

Postulates :

→ molecules/atoms of a gas consist of identical particles and have same mass, size and shape (spherical).

→ No intermolecular forces between gas molecules ($\because E = 0$)

11:42 / 38:10

#12-GASEOUS STATE|KINETIC THEORY OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

Expression of Gas pressure (Kinetic gas equation)

→ A cubical container has edge length = l, containing N gas molecules each of mass m, moving with velocities $u_1, u_2, u_3 \ldots\ldots u_N$.

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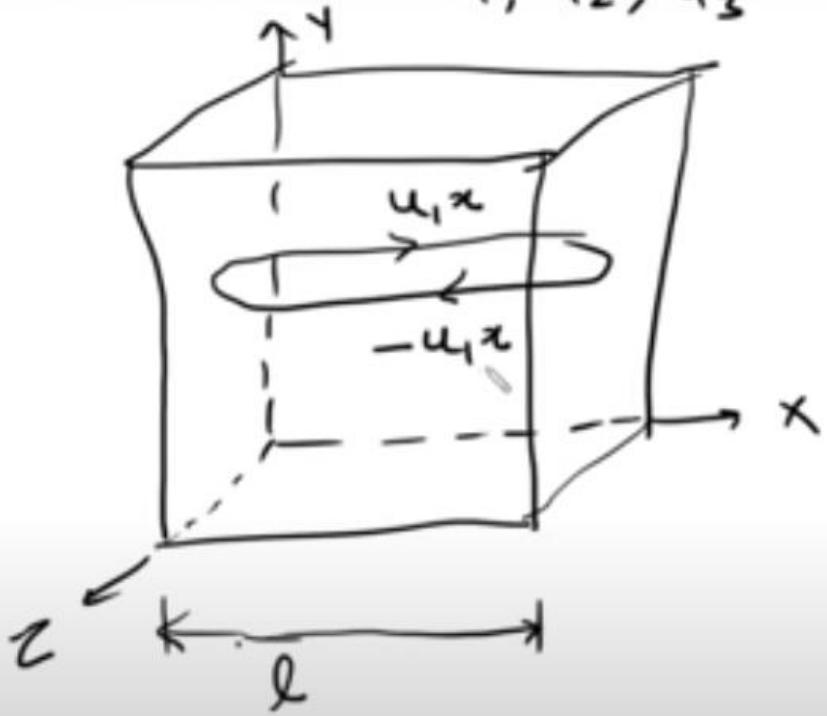
14:16 / 38:10

#12-GASEOUS STATE|KINETIC THEORY OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS...

with velocities $u_1, u_2, u_3 \text{ ----- } u_N$.



$u_1 \ni u_{1x}, u_{1y}, u_{1z}$

$u_1^2 = u_{1x}^2 + u_{1y}^2 + u_{1z}^2$

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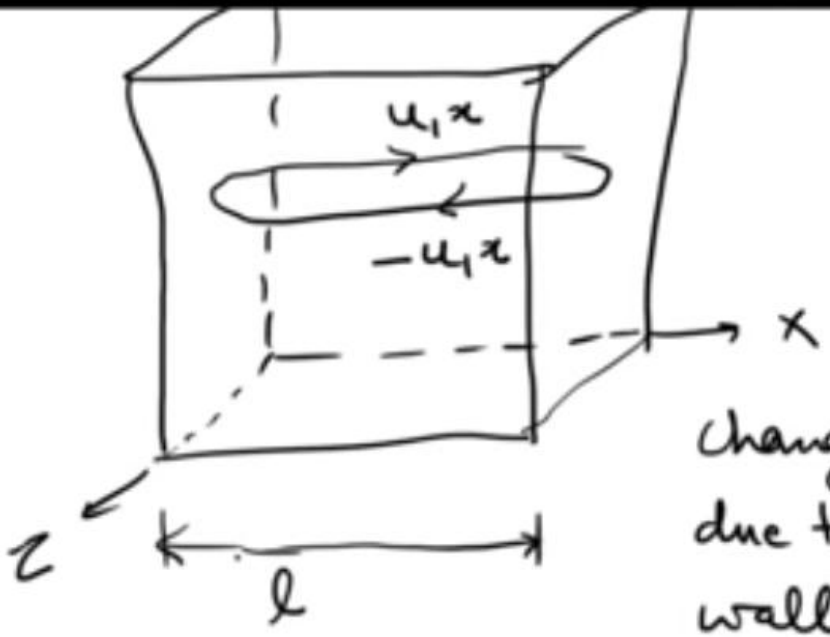
16:57 / 38:10



$u_1 \Rightarrow u_{1x}, u_{1y}, u_{1z}$

$u_1^2 = u_{1x}^2 + u_{1y}^2 + u_{1z}^2$

Change in momentum due to collision with wall $= |-u_{1x} - u_{1x}| \times m$

$= 2u_{1x} \times m$

$= 2mu_{1x}$

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$= 2mu_{1x}$

Time taken for each collision in x-axis,

$$t = \frac{l}{u_{1x}}$$

$\Rightarrow$ Rate of change of momentum $= \frac{2mu_{1x}}{\frac{l}{u_{1x}}}$

Force in x-dir, $F_{u_{1x}} = \frac{2mu_{1x}^2}{l}$

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$= 2mu_{1x}$

Time taken for each collision in x-axis,

$$t = \frac{l}{u_{1x}}$$

$\Rightarrow$ Rate of change of momentum $= \frac{2mu_{1x}}{\frac{l}{u_{1x}}}$

Force in x-dir, $F_{u_{1x}} = \frac{2mu_{1x}^2}{l}$

Total Force in x-direction $= \frac{2m}{l}\left(\frac{u_{1x}^2 + u_{2x}^2 + u_{3x}^2 + \cdots u_{Nx}^2}{N}\right) \times N$

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$\Rightarrow$ Rate of change of momentum $= \frac{2mu_{1x}}{\frac{l}{u_{1x}}}$

Force in x-dirn, $F_{u_{1x}} = \frac{2mu_{1x}^2}{l}$

Total Force in x-direction $= \frac{2m}{l}\left(\frac{u_{1x}^2 + u_{2x}^2 + u_{3x}^2 + \cdots u_{Nx}^2}{N}\right) \times N$

$= \frac{2mN}{l} \times \bar{u}_x^2$ → mean square velocity in x-dirn

Pressure $= \frac{\frac{2mN}{l} \times \bar{u}_x^2}{l^2} = \frac{mN}{V} \times \bar{u}_x^2$

But $\bar{u}_x^2 = \frac{1}{3}\bar{u}^2$

$\bar{u}^2 = \bar{u}_x^2 + \bar{u}_y^2 + \bar{u}_z^2$

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$\Rightarrow$ Rate of change of momentum $= \dfrac{2mu_{1x}}{\frac{l}{u_{1x}}}$

Force in x-dirn, $F_{u_{1x}} = \dfrac{2mu_{1x}^2}{l}$

Total Force in x-direction $= \dfrac{2m}{l}\left(\dfrac{u_{1x}^2 + u_{2x}^2 + u_{3x}^2 + \cdots u_{Nx}^2}{N}\right) \times N$

$= \dfrac{2mN}{l} \times \bar{u}_x^2$ → mean square velocity in x-dirn

Pressure $= \dfrac{\frac{2mN}{l} \times \bar{u}_x^2}{\frac{}{l^2}} = \dfrac{mN}{V} \times \bar{u}_x^2$

$\Rightarrow PV = mN \times \frac{1}{3}\bar{u}^2 = \frac{1}{3}mN\bar{u}^2$

But $\bar{u}_x^2 = \frac{1}{3}\bar{u}^2$

$\bar{u}^2 = \bar{u}_x^2 + \bar{u}_y^2 + \bar{u}_z^2$

$\bar{u}_x^2 = \bar{u}_y^2 = \bar{u}_z^2$

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$$PV = \frac{1}{3} mN \bar{u}^2$$ or $$PV = \frac{1}{3} mN C^2$$ → Kinetic

Where C = Roots mean square velocity

$$= u_{rms} = \bar{u}$$

$$= \sqrt{\frac{u_1^2 + u_2^2 + u_3^2 + \cdots u_N^2}{N}}$$

$$= \sqrt{\frac{u_1^2 + u_2^2 + u_3^2 + \cdots u_N^2}{N}}$$

$$PV = \frac{1}{3} mN c^2 \times \frac{1}{2} \times 2 = \frac{2}{3} \times N \times \frac{1}{2} mc^2$$

Average translational K.E. per molecule

$$\Rightarrow (\text{av. KE})_{\text{per molecule}} = \frac{1}{2} mc^2 = \frac{3}{2} \times \frac{PV}{N}$$

$\Rightarrow$ Since for 1 mol, $PV = RT$

$N =$

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$\Rightarrow (\text{av. KE})_{\text{per molecule}} = \frac{1}{2}mc^2 = \frac{3}{2} \times \frac{PV}{N}$

$\Rightarrow$ Since for 1 mol, $PV = RT$

$N = N_A$

$(\text{av. KE})_{\text{per molecule}} = \frac{3}{2} \times \frac{RT}{N_A}$

$= \frac{3}{2} kT$ where

K = Boltzman Constant

$= \frac{R}{\quad}$

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$$(\text{av. KE})_{\text{per molecule}} = \frac{3}{2} \times \frac{RT}{N_A}$$

$$= \frac{3}{2} kT \quad \text{where}$$

k = Boltzman Constat

$$= \frac{R}{N_A}$$

$$= \frac{8.314}{6.023 \times 10^{23}}$$

$$= 1.38 \times 10^{-23} \text{ J/molecule/K}$$

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$$(\text{av. KE})_{\text{per molecule}} = \frac{3}{2} \times \frac{RT}{N_A}$$

$$= \frac{3}{2} kT \quad \text{where}$$

k = Boltzman Constat

$$= \frac{R}{N_A}$$

$$= \frac{8.314}{6.023 \times 10^{23}}$$

$$R = 1.38 \times 10^{-23} \text{ J/molecule/K}$$

$$\Rightarrow \boxed{(\text{av. KE})_{\text{per mol}} = \frac{3}{2} RT}$$

Translational

$$\Rightarrow \frac{1}{2} M \times C^2 = \frac{3}{2} RT$$

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Translational

6.023×10^{23}

$R = 1.38 \times 10^{-23}$ J/molecule/K

$\Rightarrow \frac{1}{2} M \times C^2 = \frac{3}{2} RT$

$\Rightarrow C = U_{rms} = \sqrt{\frac{3RT}{M}}$

m/sec

8.314 J mol^{-1} K^{-1}

in Kelvin

in kg/mol e.g. O_2 gas $M = 32 \times 10^{-3}$ kg/mol

where M = Molar mass of gas

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$\Rightarrow$ $C = u_{rms} = \sqrt{\frac{3RT}{M}}$

m/sec

in Kelvin

in kg/mol e.g O_2 gas $M = 32 \times 10^{-3}$ kg/mol

where M = Molar mass of gas

e.g Find u_{rms} of O_2 gas at 300K.

$$u_{rms} = \sqrt{\frac{3 \times \frac{25}{8} \times 300^{100}}{32 \times 10^{-3}}} = \frac{5 \times 10}{4} \times \sqrt{\frac{3}{2} \times 1000}$$

$$= 12.5 \times 10 \times \sqrt{15} \text{ m/sec}$$

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$$= \sqrt{\frac{8}{32\times10^{-3}}} = \frac{}{4} \times \sqrt{2}$$

$$= 12.5 \times 10 \times \sqrt{15} \text{ m/sec}$$

SUMMARY

$$PV = \frac{1}{3} m N u_{rms}^2$$ Kinetic gas eq^n

$$u_{rms} = \sqrt{\frac{3RT}{M}}$$

M

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$$u_{rms} = \sqrt{\frac{3RT}{M}}$$

M ← molar mass of gas

$$(\text{av. trans. KE})_{\text{per mol}} = \frac{3}{2}RT$$

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$$\left(\text{av. trans. KE}\right)_{\text{per mol}} = \frac{3}{2}RT$$

- Average KE depends only on temperature
- does not depend on identity of gas.

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→ Average KE depends only on temperature

→ does not depend on identity of gas.

Q → Find total translational KE of CO_2 gas at 300K. (11th)

Soln: Total K.E $= \left(\frac{3}{2}RT\right) \times n$

$$= \frac{11}{44} \times \frac{3}{2} \times \frac{25}{2} \times 300$$

$$= \frac{7500}{8} \text{ Joule}$$

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#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

Maxwell distribution of molecular speed :

→ It is impossible ~~for~~ to track the speed variation of an individual gas molecule as it collides with molecules a billion of times or even more per second.

2:45 / 33:02

#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

→ It is impossible ~~for~~ to track the speed [illegible]
of an idividual gas molecule as it collides with molecules a billion of times or even more per second.

→ Since pr, temp, KE etc of a gas system remains constant with time if undisturbed, then the fraction of gas molecules having speed in a range should be constant at

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4:11 / 33:02

per second.

→ Since pr, temp, KE etc of a gas system remains constant with time if undisturbed, then the fraction of gas molecules having speed in a range should be constant at a given temperature.

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→ Mathematically, he derived an equal from ~~stat~~ statistical approach for a large collection of gas molecules as

$$dN_u = 4\pi N \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2 du$$

#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

$$dN_u = 4\pi N \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2 du$$

↳ This equation describes distribution of molecular speeds.

N = Total no. of gas molecules in vessel

T = Temp. in Kelvin

dN_u = No of gas molecules having speed in a range from u to (u+du)

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8:00 / 33:02

#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

dN_u = No of gas molecules having speed in a range from u to $(u+du)$

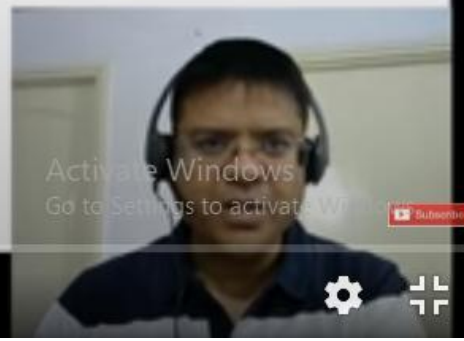
$\frac{dN_u}{N}$ = fraction of gas molecules having speed in a range u to $(u+du)$

eg 300 m/sec to 301 m/sec

↳ $u = 300$ m/sec

$du = 1$ m/sec

↑ speed interval.



9:11 / 33:02

statistical approach for a large collection of gas molecules as

$$dN_u = 4\pi N \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2 du$$

Parabolic factor

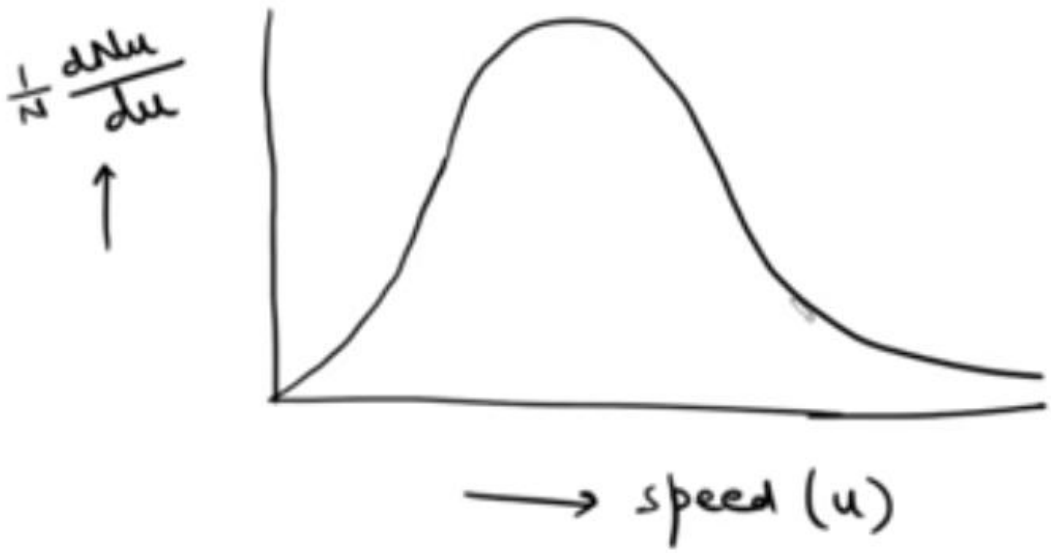
→ This equation describes distribution of molecular speeds.

N = Total no. of gas molecules in vessel

T = Temp. in Kelvin

dN_u = No of gas molecules having speed in a range from u to (u + du)

speed interval.



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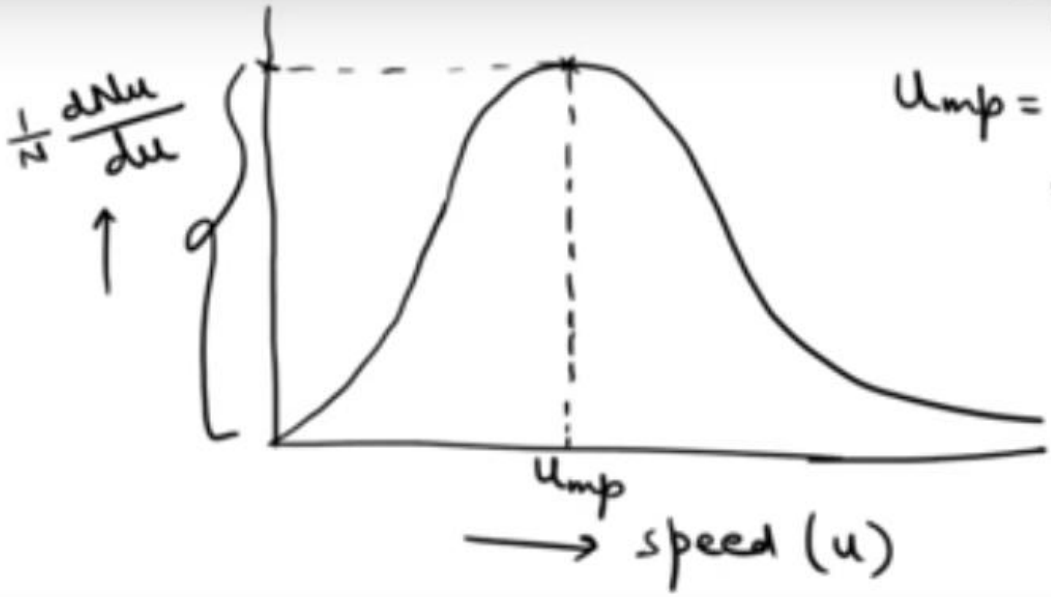
#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

speed interval.

u_{mp} = Most Probable speed

= The speed possessed by max. number of gas molecules.



11:58 / 33:02

Determination of U_{mp} :

$$dN_u = 4\pi N \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2 du$$

$$\frac{d\left(\frac{1}{N}\frac{dN_u}{du}\right)}{du} = \frac{d}{du}\left(4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2\right) = 0$$

$$K\left(e^{-\frac{Mu^2}{2RT}} \times 2u + u^2 \times e^{-\frac{Mu^2}{2RT}} \times \left(-2M\right.\right.$$

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#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

$$\frac{d\left(\frac{1}{N}\frac{dN_u}{du}\right)}{du} = \frac{d}{du}\left(4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mu^2}{2RT}} \times u^2\right) = 0$$

$$K\left(e^{-\frac{Mu^2}{2RT}} \times 2u + u^2 \times e^{-\frac{Mu^2}{2RT}} \times \left(-\frac{2Mu}{2RT}\right)\right) = 0$$

$$2 - \frac{Mu_{mp}^2}{RT} = 0$$

$$\Rightarrow \boxed{u_{mp} = \sqrt{\frac{2RT}{M}}}$$

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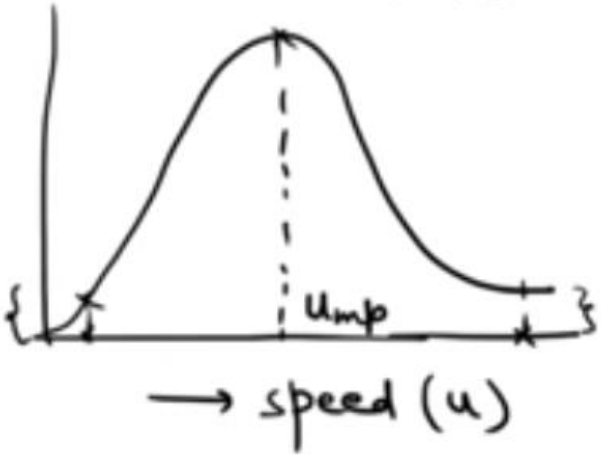
14:57 / 33:02

$$\Rightarrow \boxed{u_{mp} = \sqrt{\frac{2RT}{M}}}$$

Features of Maxwell distribution:

① No. of gas molecules having either very high speed or very low speed are low. Most of the molecules have speed near u_{mp}

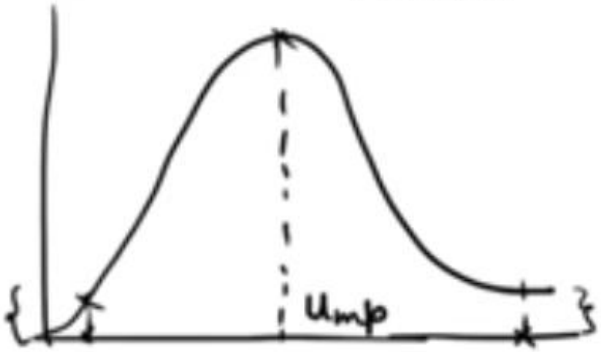
$\frac{1}{N}\frac{dN_u}{du}$ ↑



At a given temp

(1) No. of gas molecules having either very high speed or very low speed are low. Most of the molecules have speed near u_{mp}.

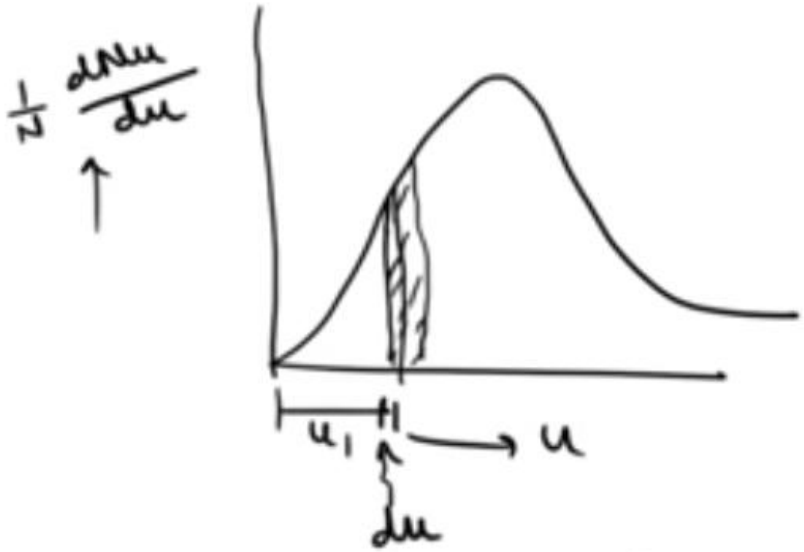
$\frac{1}{N}\frac{dN_u}{du}$ ↑



At a given temp

(2) → Area of $\frac{1}{N}\frac{dN_u}{du}$ vs u graph is constant (equal to 1)

→ Area of $\frac{dN_u}{du}$ vs u graph is equal to N (nu
of gas molecules



Total Area of curve $= \left(\frac{1}{N}\frac{dN_{u_1}}{du} \times du\right) + \frac{1}{N} \times \frac{dN_{u_2}}{du} \times du + \cdots$

$= \frac{1}{N}\left(dN_{u_1} + dN_{u_2} + \cdots\right)$

$= 1$

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#13-MAXWELL DISTRIBUTION OF MOLECULAR SPEED|IIT ADVANCED|JEE MAIN|CHEMISTRY C...

$$= \frac{1}{N}\left(dN_{u_1} + dN_{u_2} + \dots\right) + \dots$$

$$= 1$$

(3)

$\frac{1}{2}\,\frac{dN_u}{du}$ ↑

$T_1 < T_2 < T_3$

T_1, T_2, T_3

$(u_{mp})_{T_1}$ $(u_{mp})_{T_2}$ $(u_{mp})_{T_3}$ → u

For same gas

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By increase in temp,

→ u_{mp} increases but fraction of gas molecules having u_{mp} is decreased.

→ Also no of molecules of very low speed are reduced. Similarly no. of molecules of very high speed increases.

→ Curve gets broad near u_{mp}, so more no. of gas molecules have speed near u_{mp}.

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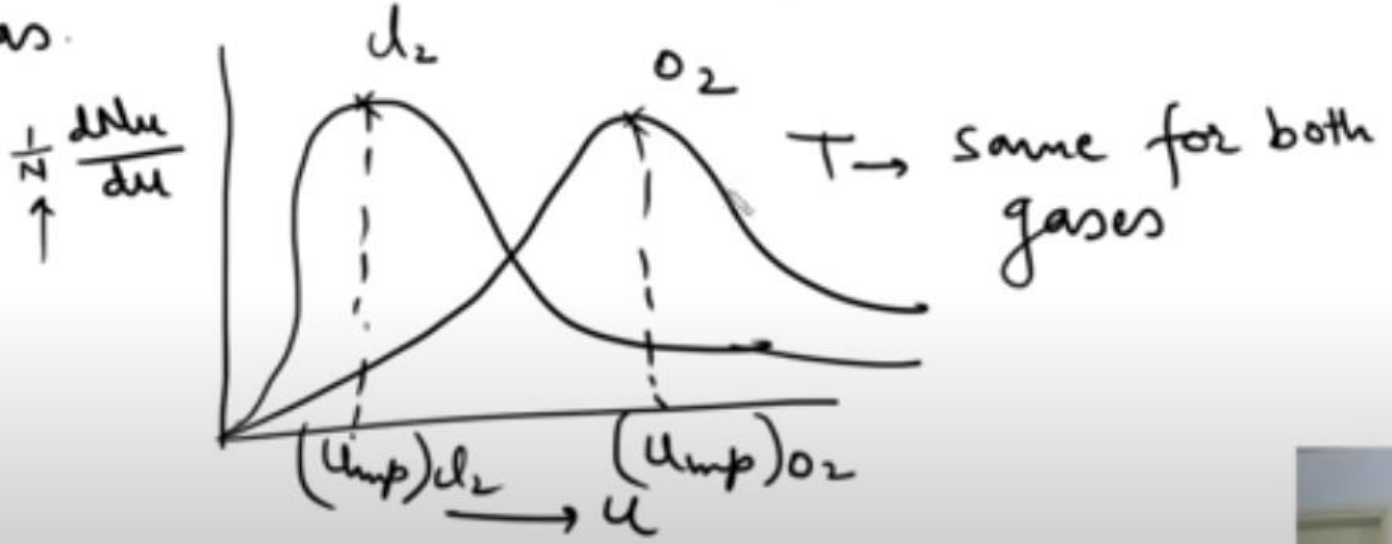
more no. of gas molecules have speed near u_{mp}.

(4) Speed distribution also depends on molar mass of gas.



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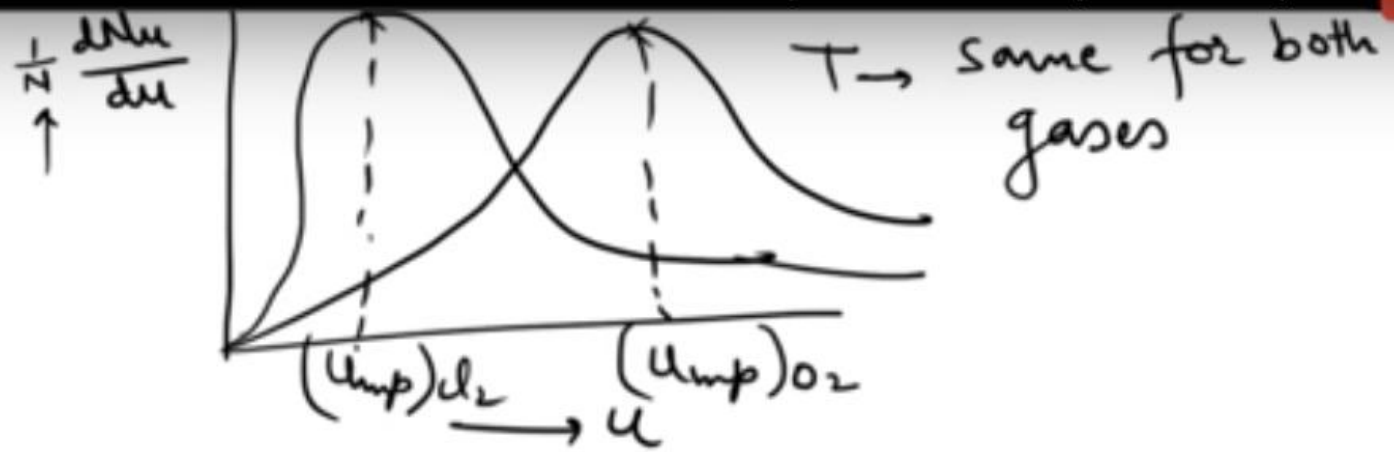

→ Maxwell graph become identical for two different gases when $(u_{mp})_1 = (u_{mp})_2$

$$\frac{T_1}{M_1} = \frac{T_2}{M_2}$$

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$$\frac{r_1}{M_1} = \frac{r_2}{M_2}$$

Various type of speeds of Gas molecules :

① Most probable speed : $u_{mp} = \sqrt{\frac{2RT}{M}}$

② Root mean square speed (u_{rms})

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① Most probable speed : $u_{mp} = \sqrt{\frac{2RT}{M}}$

② Root mean square speed (u_{rms})

$$u_{rms} = \sqrt{\frac{u_1^2 + u_2^2 + u_3^2 + \text{----} u_N^2}{N}}$$

$u_{rms} = \sqrt{\frac{3RT}{M}}$ → Applicable when you have a large collection of gas molecules.

$u_{rms} = \sqrt{\frac{3RT}{M}}$ → Applicable when you have a large collection of gas molecules.

(3) Mean speed or Average speed (u_{av})

$$u_{av} = \frac{u_1 + u_2 + u_3 + - - - u_N}{N}$$

$$\Rightarrow \boxed{u_{av} = \sqrt{\frac{8RT}{\pi M}}}$$

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≈ a large collection of gas molecules

(3) Mean speed or Average speed (u_{av})

$$u_{av} = \frac{u_1 + u_2 + u_3 + - - - u_N}{N}$$

$$\Rightarrow \boxed{u_{av} = \sqrt{\frac{8RT}{\pi M}}}$$

$$u_{mp} : u_{av} : u_{rms} = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}}$$

$$= \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$$

$$\Rightarrow u_{mp} < u_{av} < u_{rms}$$

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#14-COLLISION FREQUENCY AND MEAN FREE PATH|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

Collision frequency (Z_{11}) and mean free path (λ) :

Assumptions :

→ All gas molecules have same size and shape (spherical)

→ All molecules move with same speed i.e. $u_{av} \left(= \sqrt{\frac{8RT}{\pi M}}\right)$

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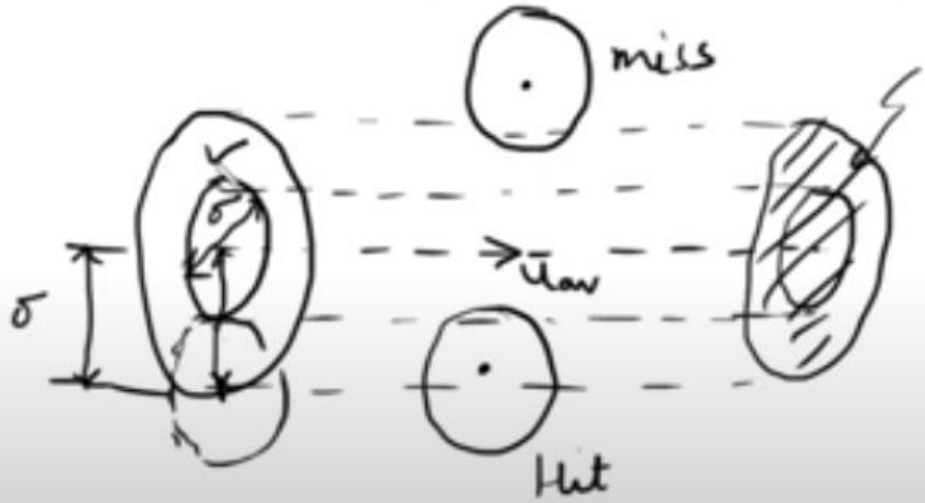
$\left(= \sqrt{\frac{8RT}{\pi M}}\right)$

Collision frequency (Z_{11}) → No of collisions between similar gas molecules per unit volume per unit time.

#14-COLLISION FREQUENCY AND MEAN FREE PATH|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

$\left(-\sqrt{\frac{8RT}{\pi M}}\right)$

Collision frequency (Z_{11}) → no of collisions between similar gas molecules per unit volume per unit time.



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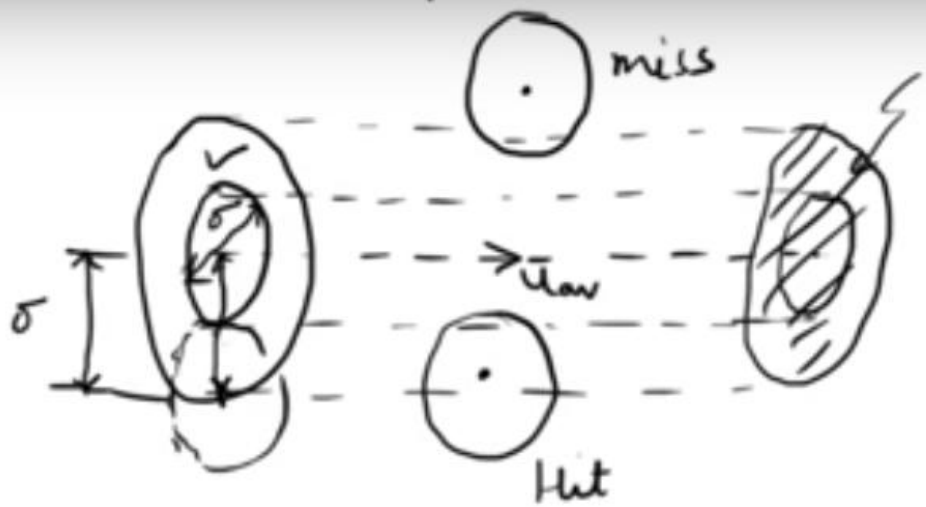

Volume swept by a molecule in 1 sec

$= \pi\sigma^2 v_{av}$

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→ volume swept by a molecule in 1sec

$= \pi\sigma^2 \bar{u}_{av}$

→ $N^* = \frac{N}{V}$

N^* : no. of gas molecules per unit volume.

→ Considering other molecules to be stationary,

no. of collisions done by this molecules in 1sec

$= \pi\sigma^2 \bar{u}_{av} \times N^*$

no. of gas molecules per unit volume.

→ Considering other molecules to be stationary,
no. of collisions done by this molecules in 1sec

$$= \pi \sigma^2 u_{av} \times N^*$$

→ Considering other molecules to be moving as well,
we have take relative velocity, $u_{rel} = \sqrt{2} \times$

$$u_{rel} = \sqrt{2} \times u_{av}$$ [An average of 90° collision angle is taken]

$= \pi \sigma^2 u_{av} \times N^*$

→ Considering other molecules to be moving as well, we have take relative velocity, $u_{rel} = \sqrt{2} \times u_{av}$ [An average of 90° collision angle is taken]

→ Collision number, $\boxed{Z_1 = \sqrt{2}\, \pi \sigma^2 u_{av} \times N^*}$

$Z_1 = \sqrt{2}\,\pi\,\sigma^2\,\bar{u}_{av} \times N^*$

→ Considering bimolecular collisions, total no. of collisions per unit volume per unit time

$$Z_{11} = Z_1 \times \frac{N^*}{2}$$

=

→ Collision Number, $\boxed{Z_1 = \sqrt{2}\,\pi\sigma^2\, u_{av} \times N^*}$

→ Considering bimolecular collisions, total no. of collisions per unit volume per unit time

$$Z_{11} = Z_1 \times \frac{N^*}{2}$$

$$\boxed{Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2\, u_{av} \times N^{*2}}$$

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→ Collision number, $\boxed{Z_1 = \sqrt{2}\,\pi\sigma^2\, u_{av} \times N^*}$

→ Considering bimolecular collisions, total no. of collisions per unit volume per unit time

$$Z_{11} = Z_1 \times \frac{N^*}{2}$$

$$\boxed{Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2\, u_{av} \times N^{*2}}$$

No. of collisions per unit volume per unit time.

Mean free path (λ) : → Average distance travelled by a gas molecule between two successive collisions.

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successive collisions.

$$\text{Mean free path } (\lambda) = \frac{\text{Total distance travelled in 1sec}}{\text{Total collisions by one molecule per sec}}$$

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successive collisions.

$$\text{Mean free path } (\lambda) = \frac{\text{Total distance travelled in 1sec}}{\text{Total collisions by one molecule per sec}}$$

$$= \frac{u_{av}}{Z_1} = \frac{u_{av}}{\sqrt{2}\,\pi\sigma^2 u_{av} \times N^*}$$

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per sec

$$= \frac{u_{av}}{z_1} = \frac{u_{av}}{\sqrt{2}\pi\sigma^2 u_{av} \times N^*}$$

$$\Rightarrow \boxed{\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 \times N^*}}$$

per sec

$$= \frac{u_{av}}{z_1} = \frac{u_{av}}{\sqrt{2}\pi\sigma^2 u_{av} \times N^*}$$

$$\Rightarrow \boxed{\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 \times N^*}} \quad \because \quad N^* = \frac{N}{V}$$

$$PV = nRT = \frac{N}{N_A} \times RT$$

$$\Rightarrow N^* = \frac{N}{V} = \frac{P \times N_A}{RT}$$

$$\Rightarrow N^* \propto \frac{P}{T}$$

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#14-COLLISION FREQUENCY AND MEAN FREE PATH|IIT ADVANCED|JEE MAIN|CHEMISTRY CLA...

$\sigma \rightarrow$ collision diameter

total collisions by one molecule per sec

$$= \frac{u_{av}}{z_1} = \frac{u_{av}}{\sqrt{2}\pi\sigma^2 u_{av} \times N^*}$$

$$\Rightarrow \boxed{\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 \times N^*}} \quad \because \quad N^* = \frac{N}{V}$$

$$PV = nRT = \frac{N}{N_A} \times RT$$

$$\Rightarrow N^* = \frac{N}{V} = \frac{P \times N_A}{RT}$$

$$\Rightarrow N^* \propto \frac{P}{T}$$

$\lambda \propto \frac{1}{N^*}$

$\lambda \propto \frac{T}{P}$

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$PV = nRT = \frac{}{N_A} \times RT$

$\lambda \propto \frac{1}{N^*}$

$\Rightarrow N^* = \frac{N}{V} = \frac{P \times N_A}{RT}$

$\lambda \propto \frac{T}{P}$

$\Rightarrow N^* \propto \frac{P}{T}$

$\Rightarrow$ At Const pr, $T \uparrow$, $\lambda \uparrow$

At Cont Temp, $P \uparrow$, $\lambda \downarrow$

$\Rightarrow$ At cont volume, $P \propto T \Rightarrow \frac{T}{P} =$ Constant

$\Rightarrow \lambda =$ Constant

$\Rightarrow$ No effect of T or P variation on λ at constant volume.

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#15-REAL GAS| COMPRESSIBILITY FACTOR |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12| K...

(REAL GAS)

→ Gas that does not follow experimental gas laws (ideal gas laws) like Boyle's law, charle's law etc is called Real gas.

→ Every gas that exists is a real gas.

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→ A real gas may show large deviation from ideal behaviour ($PV = nRT$) under certain set of P&T conditions.

IIT

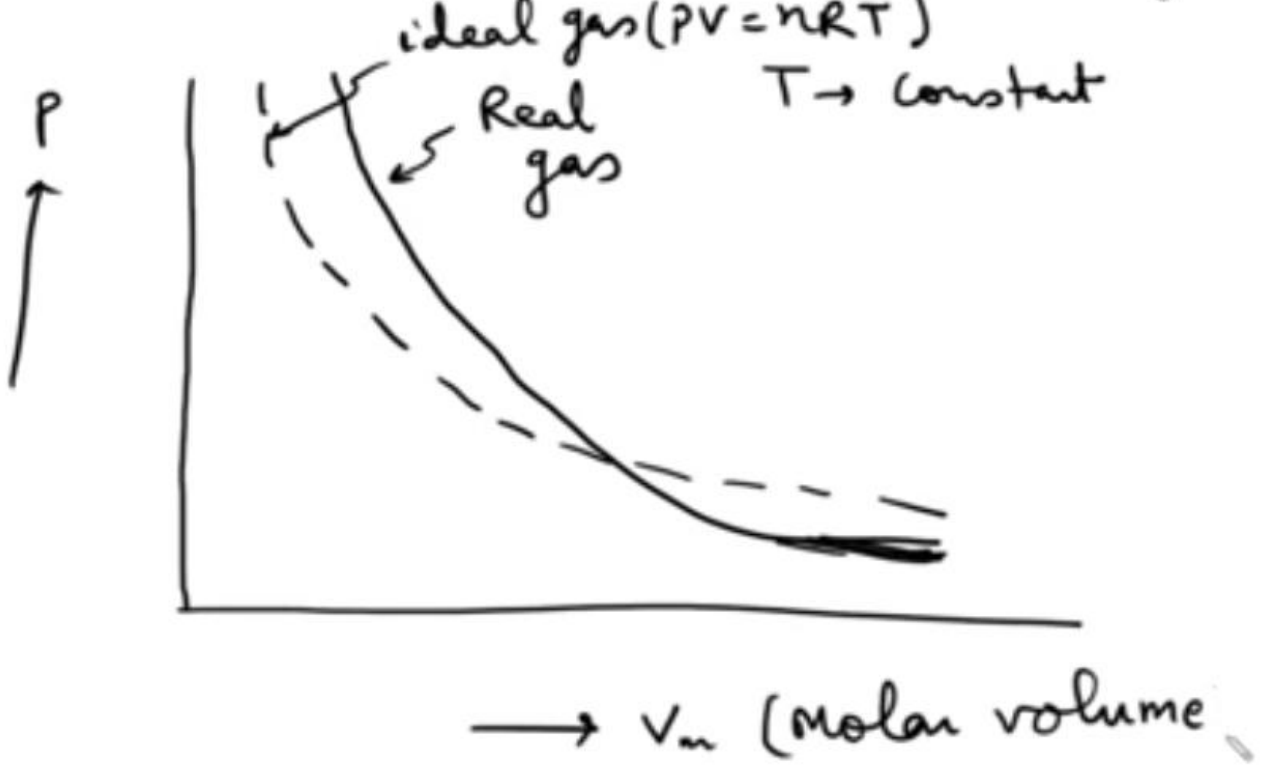
Imp

* However, a real gas behaves very closely to ideal behaviour under low pressure and high temp. conditions.

low pressure and high temp. conditions.

→ Deviation of a real gas from ideal behaviour can be depicted by the following isotherm (or P vs V_m graph)

Ideal behaviour can be depicted by the following isotherm (or P vs V_m graph)



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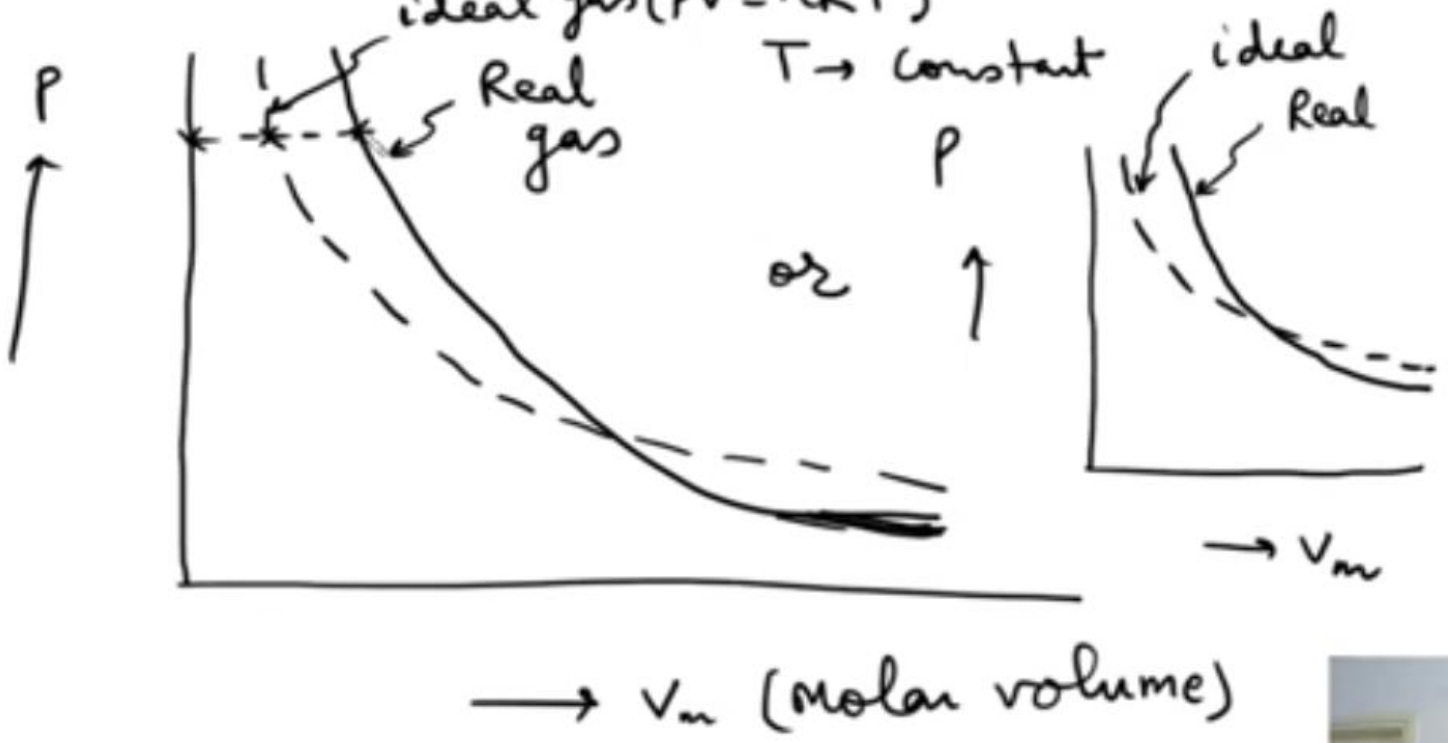
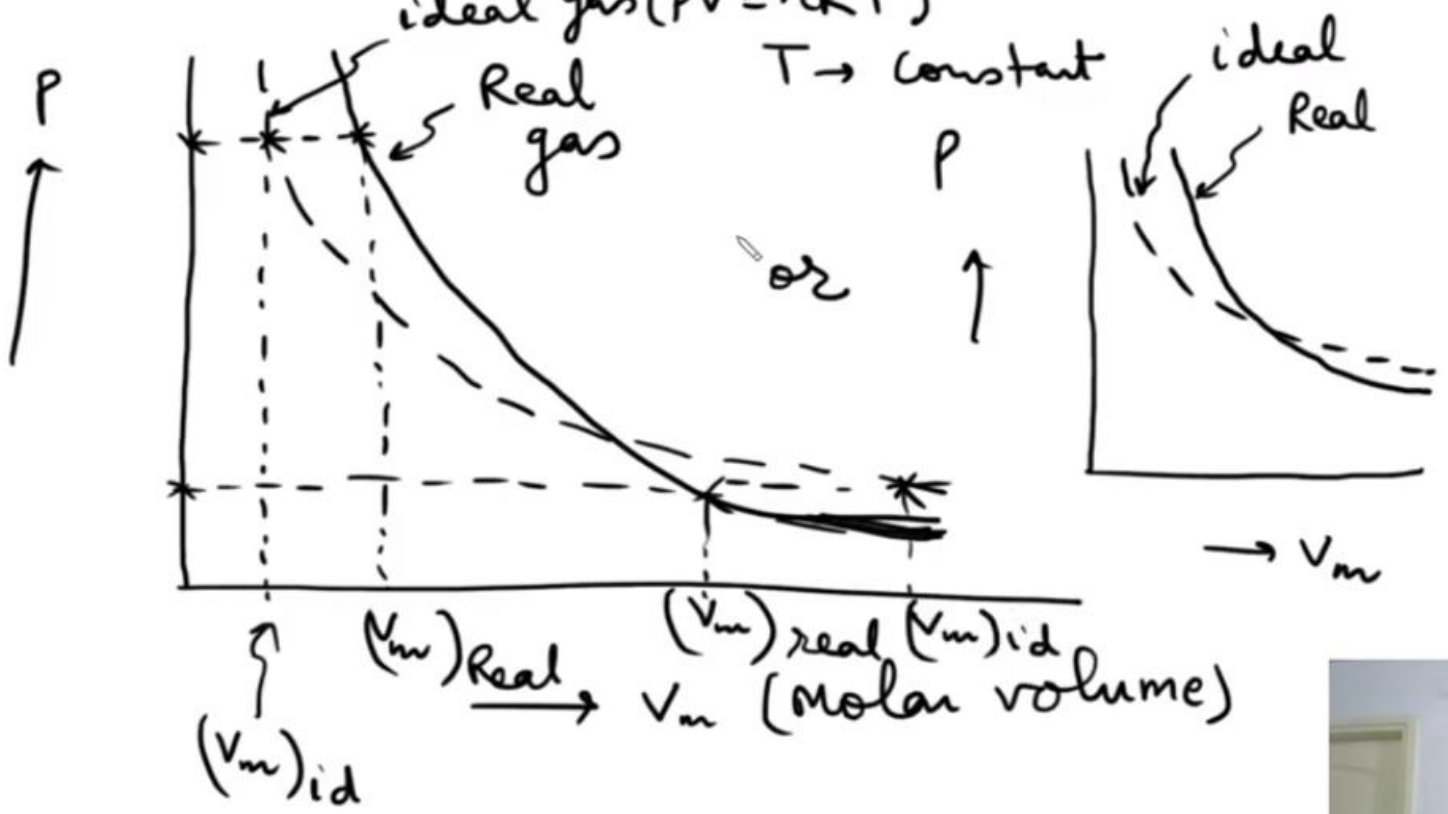
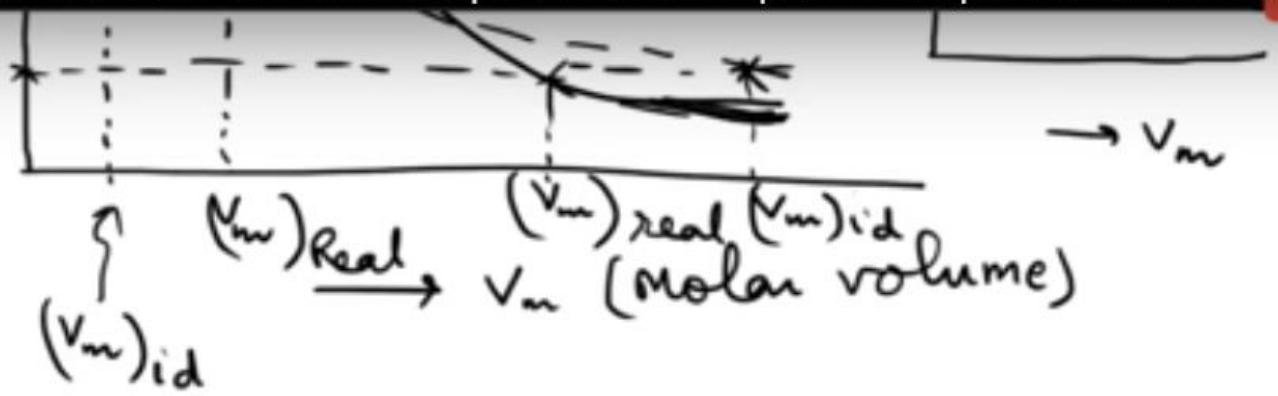
following isotherm (or P vs V_m graph)



following isotherm (or P vs V_m graph)



#15-REAL GAS| COMPRESSIBILITY FACTOR |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12| K...



V_m (Molar volume)

Observation: At high pressure,

$$(V_m)_{real} > (V_m)_{ideal}$$

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#15-REAL GAS| COMPRESSIBILITY FACTOR |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12| K...

$(V_m)_{id}$

Observation: At high pressure,

$$(V_m)_{real} > (V_m)_{ideal}$$

At low pr,

$$(V_m)_{real} < (V_m)_{ideal}$$

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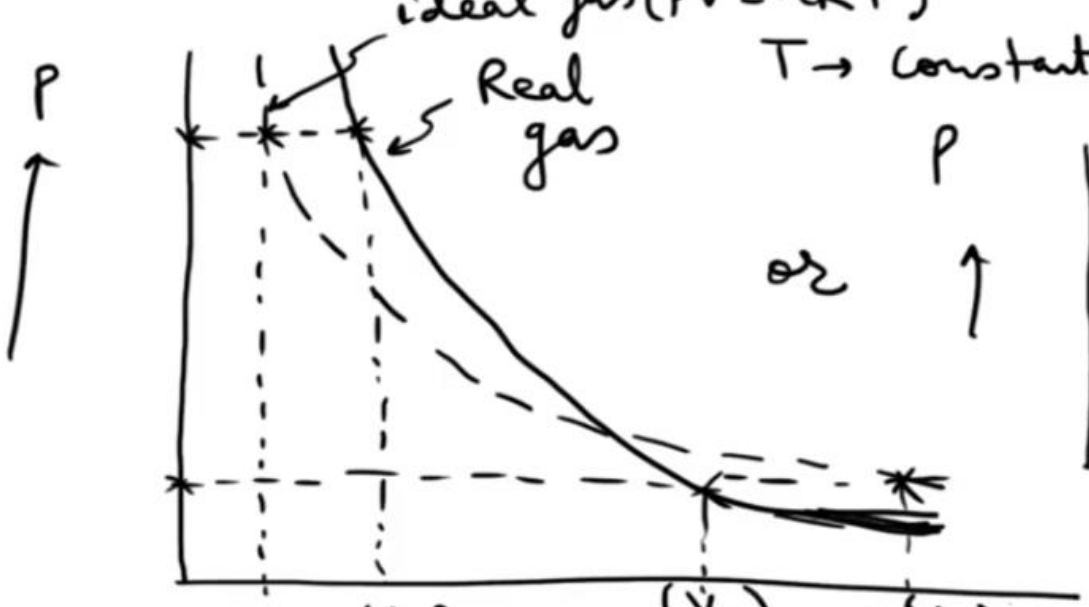
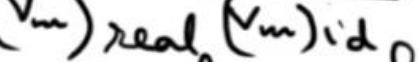
ideal gas ($PV = nRT$)



$T \rightarrow$ constant



$\longrightarrow V_m$ (Molar volume) $= \left(\frac{V}{n}\right)$

Observation: At high pressure,

$(V_m)_{real} < (V_m)_{ideal}$

→ Deviations of a real gas from ideal behaviour can be more accurately or clearly expressed by Z vs P graphs

Z ↑ Compressibility factor.

5 seconds

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Compressibility factor (Z) (or Compression factor): It is the measure of extent of deviation of a real gas from ideal behaviour.

It is the ratio of molar volume of real gas and ideal gas under same P & T conditions.

behaviour.

It is the ratio of molar volume of real gas and ideal gas under same P & T conditions.

Compressibility factor $Z = \dfrac{(V_m)_{real}}{(V_m)_{ideal}}$

Compressibility factor $Z = \dfrac{(V_m)_{real}}{(V_m)_{ideal}}$ { at same P & T

$\Rightarrow Z = \dfrac{(V/n)}{\left(\frac{RT}{P}\right)}$

$PV_{id} = n \times R \times T$

$\dfrac{V_{id}}{n} = (V_m)_{id} = \dfrac{RT}{P}$

$Z = \dfrac{PV}{nRT}$

$\Rightarrow \boxed{PV = Z \times nRT}$

$$\Rightarrow Z = \frac{\left(\frac{V}{n}\right)}{\left(\frac{RT}{P}\right)}$$

$PV_{id} = n \times R \times T$

$\frac{V_{id}}{n} = (V_m)_{id} = \frac{RT}{P}$

$$Z = \frac{PV}{nRT}$$

$$\Rightarrow \boxed{PV = Z \times nRT}$$

If $Z = 1$ in all range of P&T, gas → ideal gas

If $Z \neq 1$, then real gas.

If $Z > 1$, +ve deviation from ideal

$Z < 1$, −ve deviation from idea

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→ Greater is the departure of z from unity, greater will the deviation from ideal behaviour.

e.g. * $\begin{cases} z = 1.5 \text{ (more deviation)} \\ z = 1.1 \end{cases}$

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behaviour.

e.g. * $\begin{cases} Z = 1.5 \text{ (More deviation)} \\ Z = 1.1 \end{cases}$

e.g. $Z = 0.95$

$Z = 0.73$ (More deviation)

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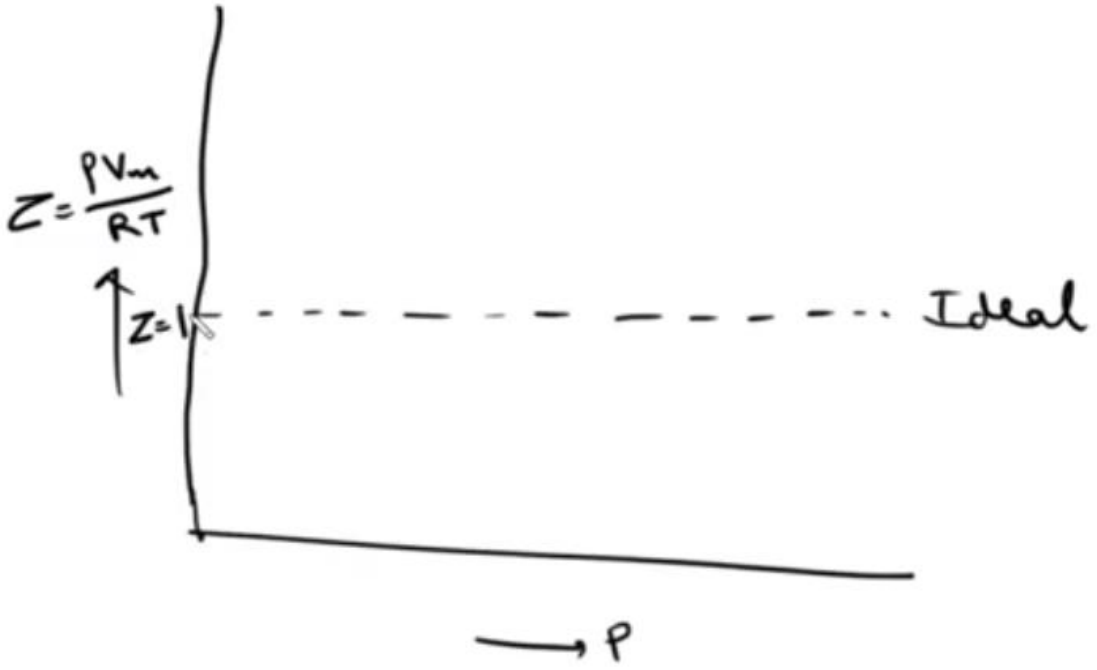
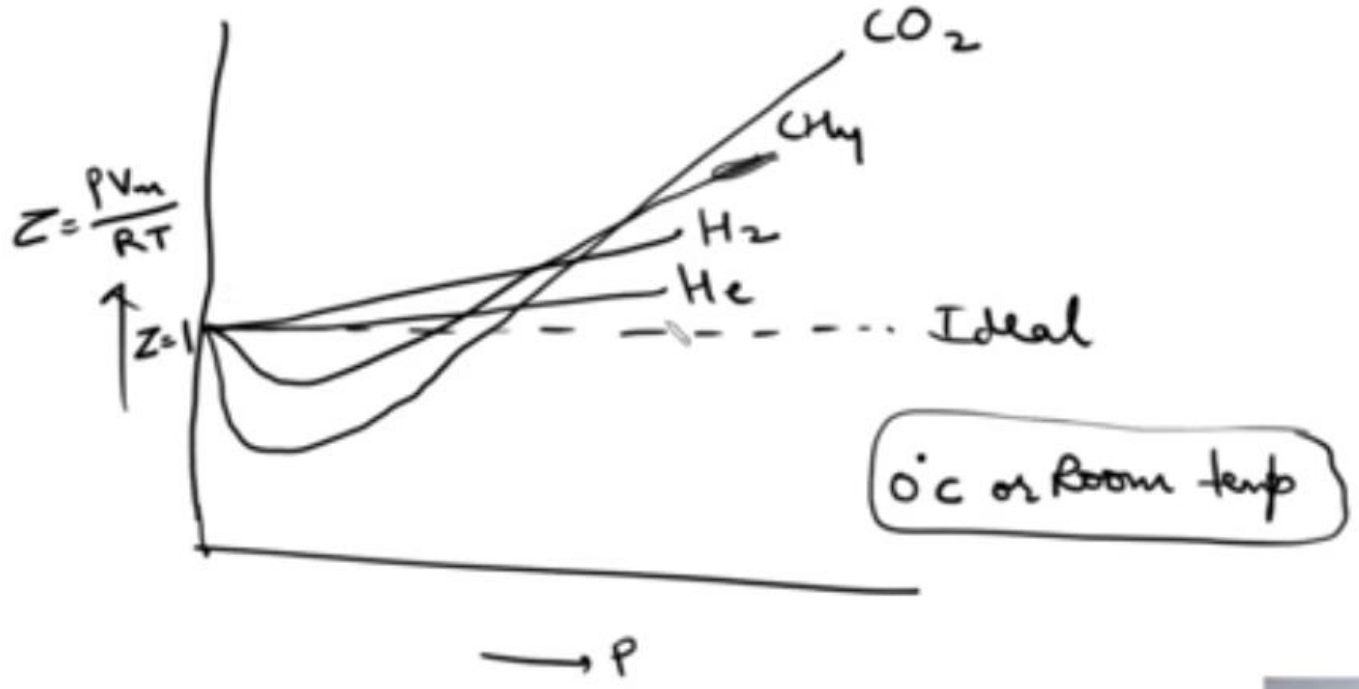
$Z = 0.73$ (More deviation)

Imp

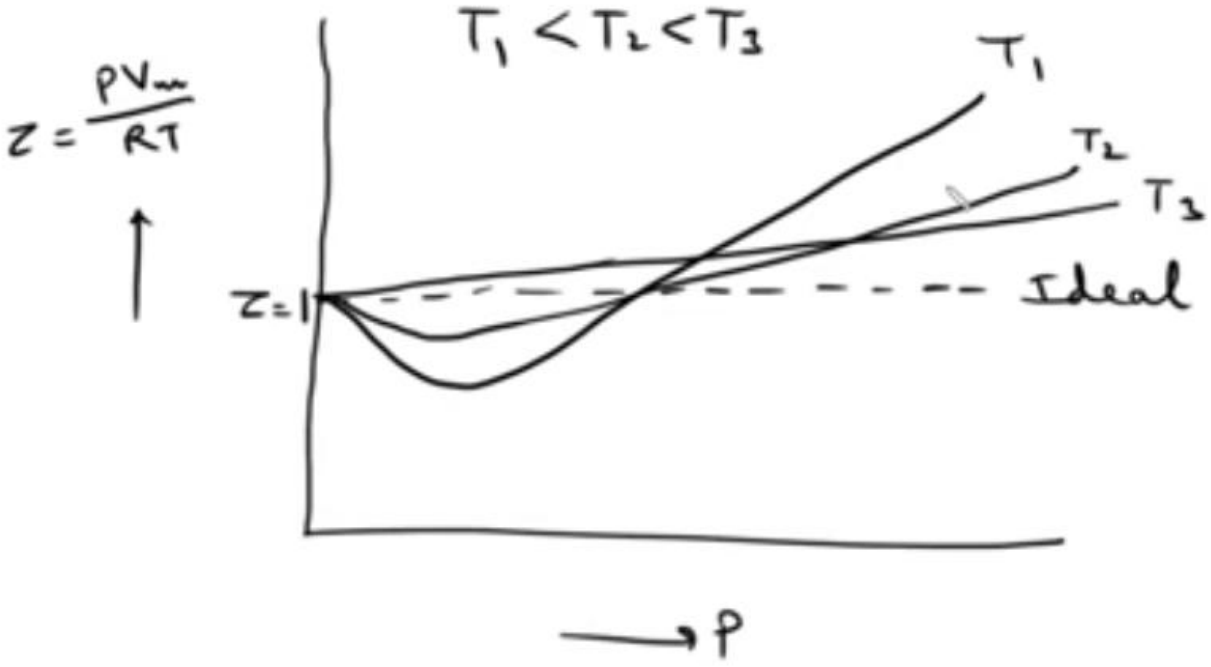
Experimental Z vs P plots:





For different gas (at cont. temp.)

For different gas (at cont. temp.)



For same gas (at diff. temp.)

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VanderWaal's equation of state:

→ Real gases show deviation from ideal behaviour or do not follow ideal gas equation. ($PV_{ideal} = nRT$)

→ VanderWaal's was the first person (physicist) who suggested correction term to

5 seconds

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do not follow ideal gas equation. ($PV_{id} = nRT$)

→ VanderWaal's was the first person (physicist) who suggested correction terms to be applied in ideal gas equation so that the resulting equation can be satisfactorily applied on real gases.

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#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

real gases.

→ Ideal gas equation $PV = nRT$ is based on the following two main assumptions

→ Volume occupied by molecules themselves is negligible as molecules in KTG are taken point masses.

→ No intermolecular forces of attraction.

5:19 / 34:32

point masses.

→ No intermolecular forces of attraction.

→ Vander Waal's suggested that the above assumption of KTG do not hold good when we apply on real gases. He applied a volume correction term and a pressure correction term in ideal gas equation ($PV_{id} = nRT$).

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and a pressure correction term in ideal gas equation ($PV_{id} = nRT$).

@ Correction in volume:

→ Since size of molecule is finite (though small) and molecules are impenetrable (considered hard sphere), some volume will be occupied by molecules themselves

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→ Since size of molecule is finite (though small) and molecules are impenetrable (considered hard sphere), some volume will be occupied by molecules themselves

→ Ideal volume or free volume is lesser than the volume of vessel.

Ideal volume or free volume = $V - nb$

of molecules themselves

→ Ideal volume or free volume is lesser than the volume of vessel.

Ideal volume or free volume = $V - nb$

where b = excluded volume occupied by gas molecules per mol

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@ Correction in volume: (a negative correction term)

→ Since size of molecule is finite (though small) and molecules are impenetrable (considered hard sphere), some volume will be occupied by molecules themselves

→ Ideal volume or free volume is lesser than the volume of vessel.

Ideal volume or free volume = $V - nb$

where b = excluded volume occupied by

gas molecules per mol

excluded volume for the pair of molecule (considering bimolecular collisions) $= \frac{4}{3}\pi (2r)^3$

$= 8 \times \frac{4}{3}\pi r^3$

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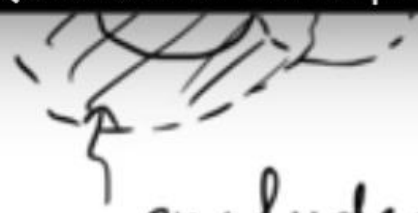

excluded volume for the pair of molecule (considering bimolecular collisions) $= \frac{4}{3}\pi(2r)^3$

$= 8 \times \frac{4}{3}\pi r^3$

→ Excluded volume per molecule

$= \frac{8 \times \frac{4}{3}\pi r^3}{2}$

$= 4 \times \left(\frac{4}{3}\pi r^3\right)$

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→ Excluded volume per molecule

$$= \frac{8 \times \frac{4}{3}\pi r^3}{2}$$

$$= 4 \times \left(\frac{4}{3}\pi r^3\right)$$

→ Excluded volume per mol,

$$b = 4 \times N_A \times \left(\frac{4}{3}\pi r^3\right)$$

where r = radius of gas molecule.

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#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

$$b = 4 \times N_A \times \left(\frac{4}{3}\pi r^3\right)$$

where r = radius of gas molecule.

$$V_{id} = V_{free} = (V - nb)$$

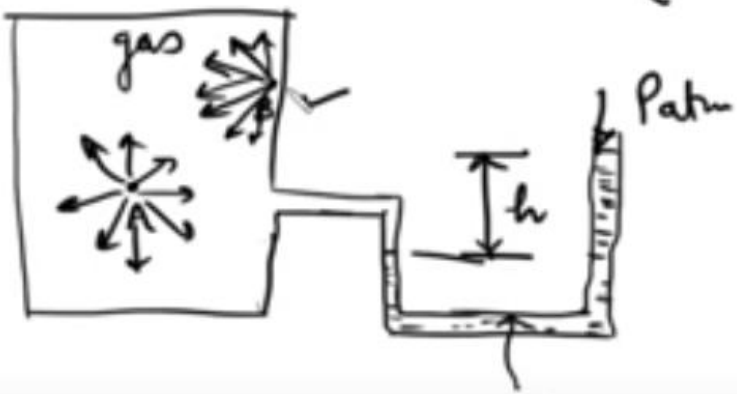
→ b is called vanderwaal's constant which is the measure of finite size of gas molecules.

16:40 / 34:32

#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

(b) Correction in pressure : (+ve correction term)

$P_{ob} = (P_{atm} + h d g)$



Real gas

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#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

Real gas

→ Vander Waal considered attractive forces active in a real gas and observed pressure is less than the ideal gas pressure.

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21:31 / 34:32

#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

→ Vander Waal considered attractive forces active in a real gas and observed pressure is less than the ideal gas pressure due to intermolecular attractive forces.

$P_{id} = P_{ob} +$ pressure correction term.

Pressure correction term depends on following two facto

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two factors.

1) Pr. correction term $\propto$ no. of gas molecules per unit volume

$$\propto \frac{N}{V}$$

$$\propto \frac{n}{V}$$

More the no. of gas molecules per unit volume, more will be the backward drag force on the molecule about to collide with wall of vessel.

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#16-VANDERWAAL'S EQUATION OF STATE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVP...

More the no. of gas molecules per unit volume, more will be the backward drag force on the molecule about to collide with wall of vessel.

(ii) $\propto$ Pr. correction term is also directly proportional to the no. of collisions with wall.

No. of collisions with wall $\propto$ no. of molecules per unit volume

$\propto$

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26:52 / 34:32

(ii) ∴ Pr. correction term is also directly proportional to the no. of collisions with wall.

No. of collisions with wall ∝ no. of molecules per unit volume

$\propto \frac{n}{V}$

$\Rightarrow$ Pr. correction term $\propto \left(\frac{n}{V}\right)^2$

Consi

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$\Rightarrow$ Pr. correction term $\propto \left(\frac{n}{V}\right)$

Considering both the factors,

Pressure correction term $\propto \frac{n}{V} \times \frac{n}{V}$

$$= a \times \frac{n^2}{V^2}$$

a $\rightarrow$ vanderWaal's constant which is the measure of attractive forces between gas molecules.

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$$P_{id} = P_{ob} + \frac{a n^2}{V^2}$$

$$\boxed{\left(P_{ob} + \frac{an^2}{V^2}\right)(V - nb) = nRT}$$ → VanderWaal's eq^{n} of state

$$P_{id} = P_{ob} + \frac{an^2}{V^2} \qquad P_{real} = P_{ob}$$

$$V_{id} = V - nb \qquad V_{real} = V$$

$$\Rightarrow V_m \text{ (molar volume)} = \frac{V}{n}$$

$V_{id} = V - nb$ || $V_{real} = V$

$\Rightarrow V_m$ (molar volume) $= \frac{V}{n}$

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$V_{id} = V - nb$ || $V_{real} = V$

$\Rightarrow V_m$ (molar volume) $= \frac{V}{n}$

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

vander waal's eqn.

Unit of 'a' : atm $lit^2 mol^{-2}$

$Pa\ m^6 mol^{-2}$ in SI

Unit of b : lit/mol

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$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

vanderWaal's eqⁿ.

Unit of 'a' : atm $lit^2 mol^{-2}$
$Pa\ m^6 mol^{-2}$ in SI

Unit of 'b' : lit/mol
: $m^3 mol^{-1}$ in SI

→ 'a', 'b' are vanderWaal's constants which ~~depend~~ are gas specific but do not depend on Pr, temp etc.

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#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

Vanduwaal's equation of state

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

V.Imp Expression of Z in terms of 'a' and

1:48 / 32:48

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

V.Imp Expression of Z in terms of 'a' and 'b':

(Appropriate approximations)

Case-I At low pressure: At low pressure, V_m is high so 'b' can be insignificant relative to V_m. $\Rightarrow (V_m - b) \cong V_m$

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relative to V_m $\Rightarrow (V_m - b) \cong V_m$

$$\Rightarrow \left(P + \frac{a}{V_m^2}\right)(V_m - 0) = RT$$

$$\frac{PV_m}{RT} + \frac{\frac{a}{V_m}}{RT} = \frac{RT}{RT}$$

Imp

$$\boxed{Z = 1 - \frac{a}{RTV_m}} \quad \text{—— (1)}$$

↳ Applicable for most gases under normal temp conditions except H_2 and He.

relative to V_m $\Rightarrow (V_m - b) \cong V_m$

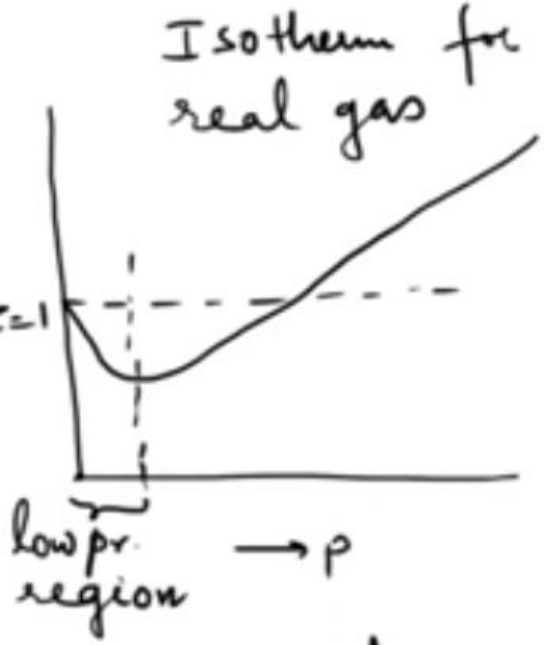
$$\Rightarrow \left(P + \frac{a}{V_m^2}\right)(V_m - 0) = RT$$

$$\frac{PV_m}{RT} + \frac{a}{V_m RT} = \frac{RT}{RT} \qquad Z = \frac{PV_m}{RT}$$

Imp

$$\boxed{Z = 1 - \frac{a}{RTV_m}} \quad \text{—— (1)}$$

↳ Applicable for most gases under normal temp conditions except H_2 and He.

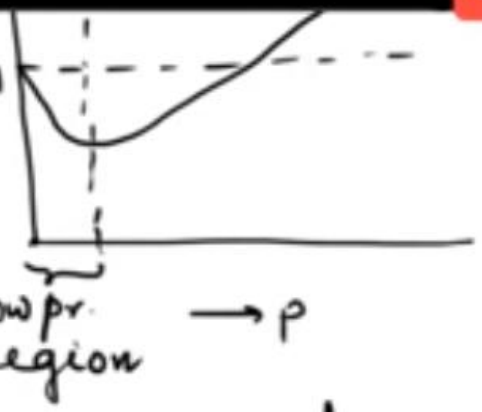


$\frac{PV_m}{RT} = \frac{V_m}{RT} + \frac{a}{RTV_m}$

$Z = 1 + \frac{a}{RTV_m}$

$Z = 1 - \frac{a}{RTV_m}$ — ①

Imp



→ Applicable for most gases under normal temp. conditions except H_2 and He.

Above eq. ① correctly explains variation of Z with pressure in low pr region.

Above eq. ① correctly explains variation of Z with pressure in low pr region.

Case-II **At high pressure**: At high pr, V_m is small and hence 'b' can't be ignored Rather $\frac{a}{V_m^2}$ can be ignored wrt P.

$\Rightarrow$

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$$\Rightarrow \left(P + \frac{a}{V_m^2}\right) \simeq P \quad (\text{at high pr.})$$

$$(P)(V_m - b) = RT$$

$$\frac{PV_m}{RT} - \frac{Pb}{RT} = \frac{RT}{RT} \Rightarrow$$

$$\boxed{Z = 1 + \frac{Pb}{RT}}$$

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$$\Rightarrow \left(P + \frac{a}{V_m^2}\right) \simeq P \quad (\text{at high pr.})$$

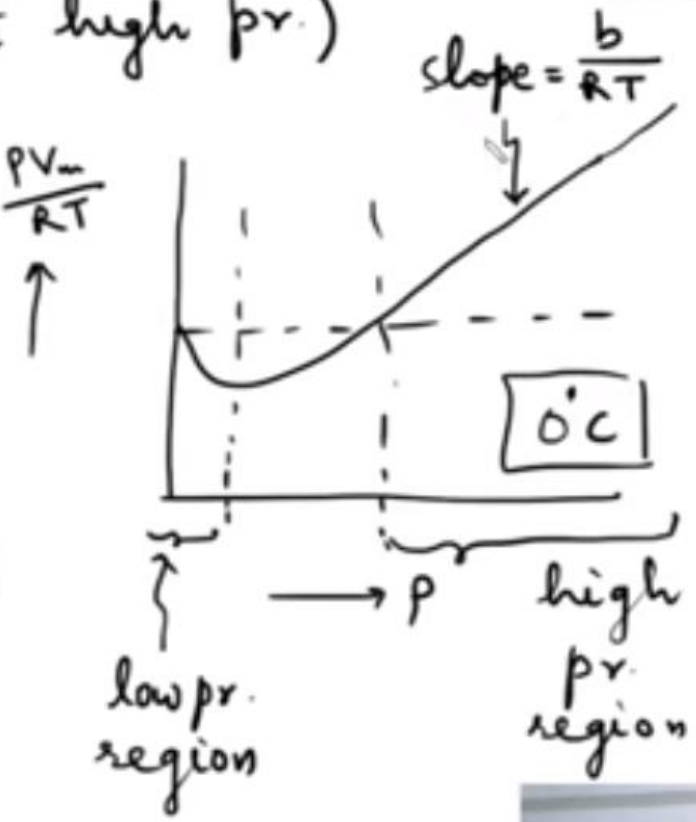
$$(P)(V_m - b) = RT$$

$$\frac{PV_m}{RT} - \frac{Pb}{RT} = \frac{RT}{RT} \Rightarrow$$

Imp $\quad \boxed{Z = 1 + \frac{Pb}{RT}} \quad -(2)$

$$y = mx + c$$

$$\text{slope} = \frac{b}{RT}$$



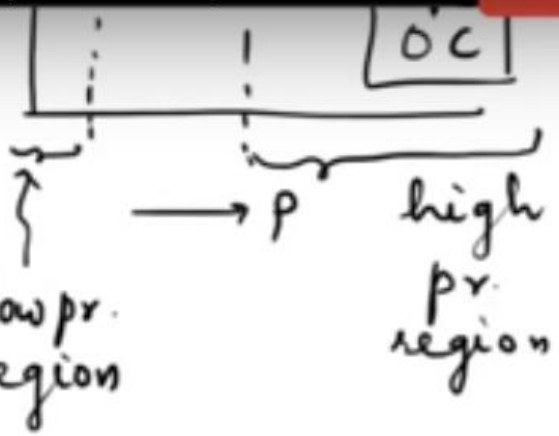
#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

RT RT ~~RT~~

Imp $Z = 1 + \frac{Pb}{RT}$ — (2)

$y = mx + c$

slope $= \frac{b}{RT}$



Above equation (2) ~~can be applied~~ explains behaviour of real gases in high pr. region.

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$y = mx + c$

$\text{Slope} = \frac{b}{RT}$

low pr. region

region

→ Above equation (2) ~~can be applied~~ explains behaviour of real gases in high pr. region.

Case-III At low pr. and high temp. ;

→ At low pr, V_m is large and 'b' can be ignored. ⇒ $(V_m - b) \cong V_m$

→ At high temperature, thermal energy is d

→ At low pr, V_m is large and [illegible]
ignored. ⇒ $(V_m - b) \cong V_m$

→ At high temperature, thermal energy is dominant over intermolecular attractions.

→ At low pr, V_m is large and [illegible] so

ignored. ⇒ $(V_m - b) \cong V_m$

→ At high temperature, thermal energy is dominant over intermolecular attractions. So $\frac{a}{V_m^2}$ can be ignored.

$$\left(P + \frac{a}{V_m^2}\right) \cong P$$

⇒ $PV_m = RT$ or $PV = nRT$

ideal behaviour under low pr. and high temp. conditio

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$\text{Slope} = \frac{b}{RT}$

→ Above equation (2) explains behaviour of real gases in high pr. region.

Case-III At low pr. and high temp. ; (Ideal gas condition)

→ At low pr, V_m is large and 'b' can be ignored. ⇒ $(V_m - b) \cong V_m$

→ At high temperature, thermal energy is d... over intermolecular attractions. So $\frac{a}{V_m^2}$ co...

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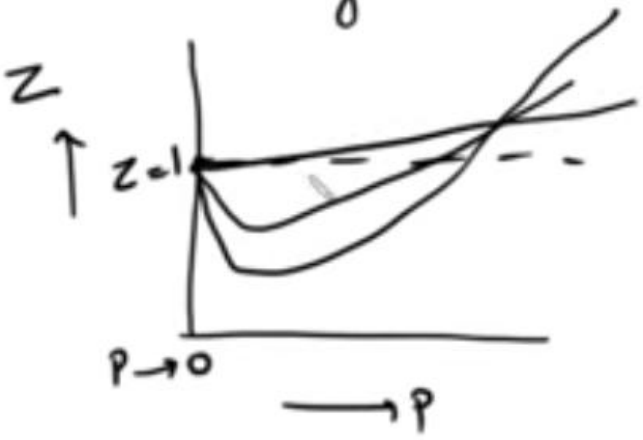
$\Rightarrow$ $PV_m = RT$ or $PV = nRT$

ideal behaviour under low pr. and high temp. conditions.

(V. Imp)

Note: At very low pressure or at high very high molar volume, $Z \to 1$ (Ideal)

(IIT)



when $P \to 0$, $V_m \to \infty$,

$Z \to 1$

#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

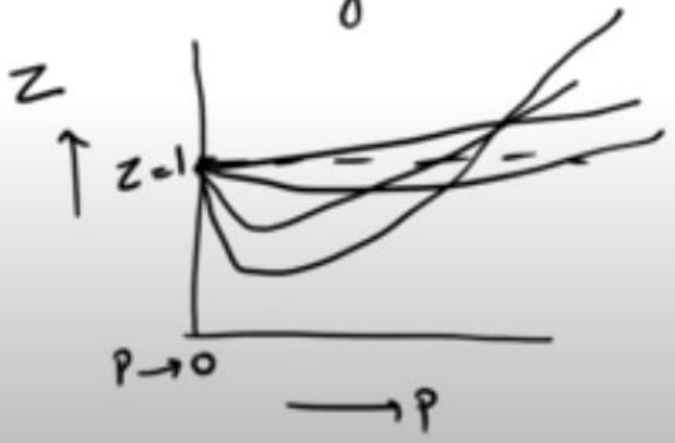
$\Rightarrow$ $PV_m = RT$ or $PV = nRT$

ideal behaviour under low pr. and high temp. conditions.

(V. Imp)

Note: At very low pressure or at high very high molar volume, $Z \to 1$ (Ideal)

(IIT)



when $P \to 0$, $V_m \to \infty$,

$Z \to 1$

$\Rightarrow$ $PV = nRT$ is applic

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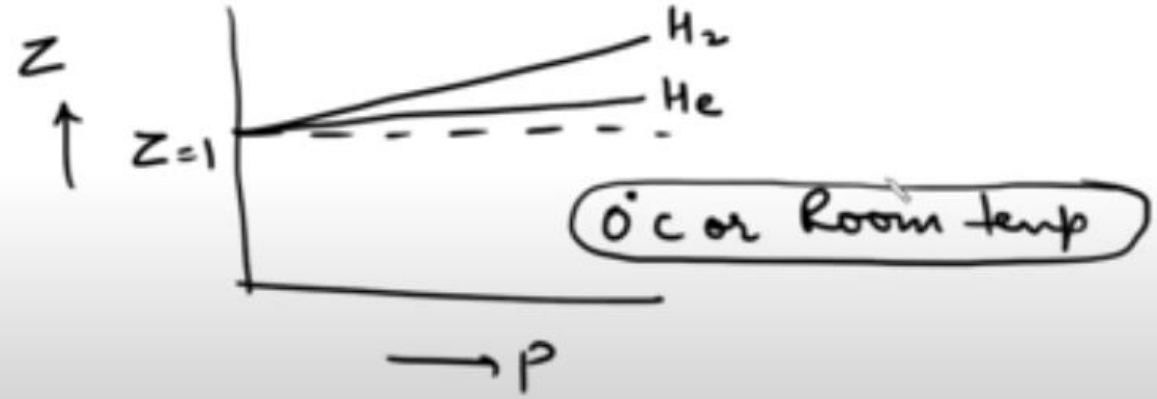
#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

Case-IV : Special Case of H_2 and He :

→ Under normal temp. conditions, Z is more than 1 even in low, moderate or high pr.



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→ P

→ Attractive forces are very weak (insignificant) between molecules/atoms of H_2 or He under normal temp. conditions. This is due to very low no. of polarisable electrons and hence dispersion forces or London forces are very feeble for He and H_2.

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normal temp. conditions. This is due to very low no. of polarisable electrons and hence dispersion forces or London forces are very feeble for He and H_2. So 'a' is very small for these two gases. 'a' is smallest for He.

$$a_{He} < a_{H_2}$$

low no. of polarisable electrons and hence dispersion forces or London forces are very feeble for He and H_2. So 'a' is very small for these two gases. 'a' is smallest for He.

$$a_{He} < a_{H_2}$$

So we can ignore $\frac{a}{V_m^2}$ wrt 'P'.

$$\Rightarrow (P+0)(V_m - b) = RT$$

i

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$a_{He} < a_{H_2}$

So we can ignore $\frac{a}{V_m^2}$ wrt 'P'.

$$\Rightarrow (P+0)(V_m - b) = RT$$

$$\frac{PV_m}{RT} - \frac{Pb}{RT} = \frac{RT}{RT}$$

$$\boxed{Z = 1 + \frac{Pb}{RT}}$$

→ For H_2 & He under all pr conditions at 0°C or room temp.

#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

$a_{He} < a_{H_2}$

So we can ignore $\frac{a}{V_m^2}$ wrt 'P'.

$\Rightarrow (P+0)(V_m - b) = RT$

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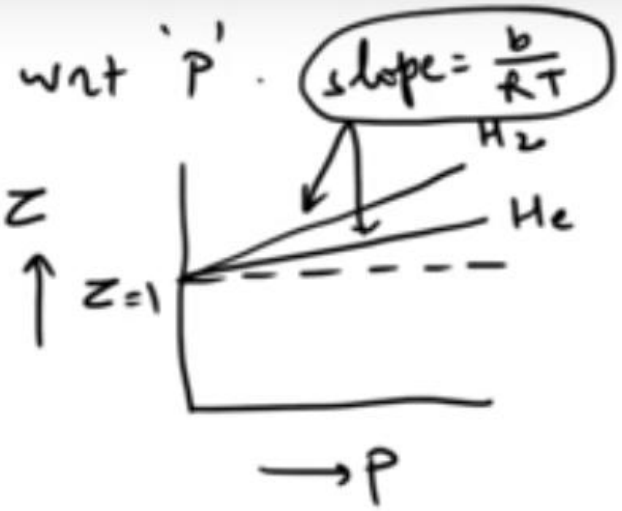

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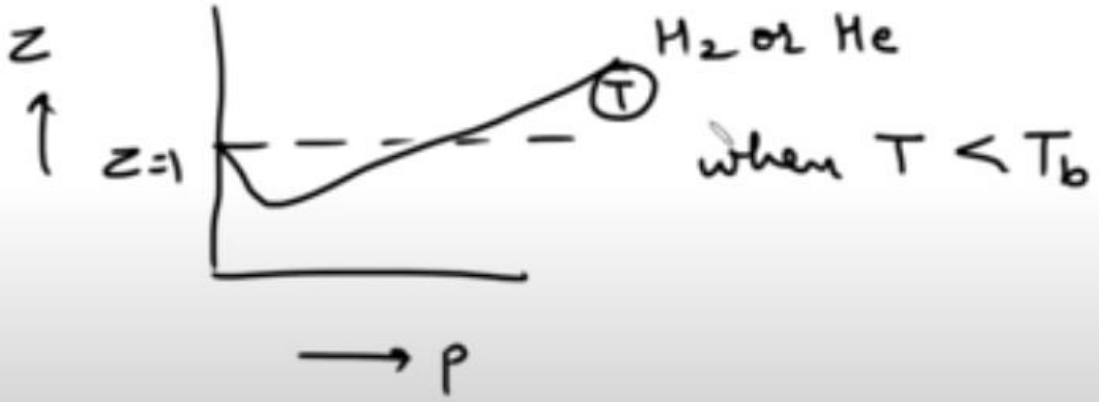
#17-EXPRESSION OF Z IN TERMS OF a AND b |IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|...

→ Variation Value of z becomes less than one (-ve deviation) for H_2 and He also when temp is less than Boyle's temp (T_b).



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#18-PHYSICAL SIGNIFICANCE OF VANDERWAAL'S CONSTANT a,b|IIT ADVANCED|JEE MAIN|CH...

(Imp) Physical Significance of Vander Waal's constants 'a' & 'b'

@ At low pr : $Z = 1 - \frac{a}{RTV_m}$ i.e. $Z < 1$,

$(V_m)_{real} < (V_m)_{ideal}$

→ Attractive forces are dominant over repulsive forces and such deviation ($Z < 1$) is called −ve deviation.

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—ve deviation.

In such conditions, gas is more easily liquified (or said to have greater liquifiability). Also gas is more compressible (or greater compressibility) than the ideal gas.

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than the ideal gas.

→ 'a' is the measure of attractive forces between gas molecule

(b)

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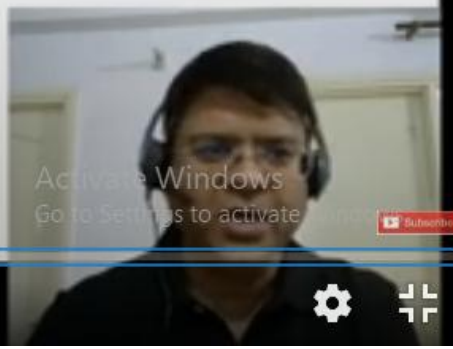
#18-PHYSICAL SIGNIFICANCE OF VANDERWAAL'S CONSTANT a,b|IIT ADVANCED|JEE MAIN|CH...

(b) At high pressure, $Z = 1 + \frac{Pb}{RT}$, $(V_m)_{real} > (V_m)_{ideal}$, repulsive forces are dominant over attractive forces. (+ve deviation)

→ Gas is said to have lesser compressibility than ideal gas under such conditions

→ Lesser liquifiability, difficult to liquify.



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than ideal gas under such conditions

→ Lesser liquifiability, difficult to liquify.

→ b is the measure of finite size of gas molecule and indirectly is the measure of repulsive forces.

#18-PHYSICAL SIGNIFICANCE OF VANDERWAAL'S CONSTANT a,b|IIT ADVANCED|JEE MAIN|CH...

Q→ Gas X, Gas Y

a_1, b_1 a_2, b_2

If $a_1 = a_2$, but $b_1 > b_2$

which gas is more compressible.

A: Lesser 'b' means greater compressibility

Ans: 'Y' gas is more compressible

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#18-PHYSICAL SIGNIFICANCE OF VANDERWAAL'S CONSTANT a,b|IIT ADVANCED|JEE MAIN|CH...

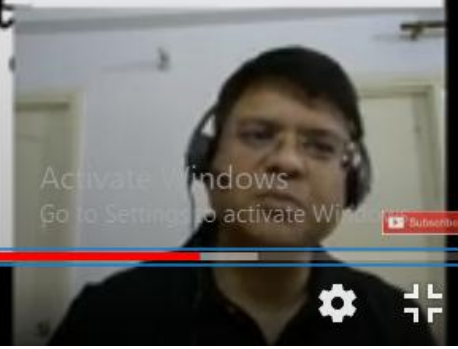
Q → x, a_1, b_1 | y gases a_2, b_2

If $b_1 = b_2$, $a_1 > a_2$

which is more easily liquified

Ans: 'a' → higher, means greater liquifiability

Ans: 'x'



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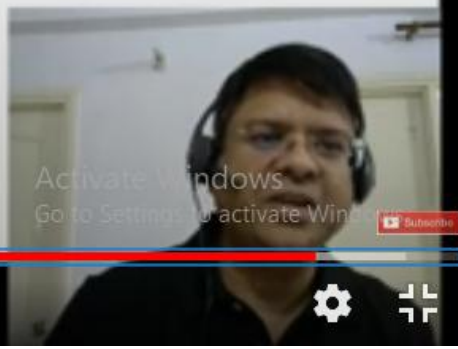
#18-PHYSICAL SIGNIFICANCE OF VANDERWAAL'S CONSTANT a,b|IIT ADVANCED|JEE MAIN|CH...

P_{ex} : 'x'

Note : Ideal gas never liquify.



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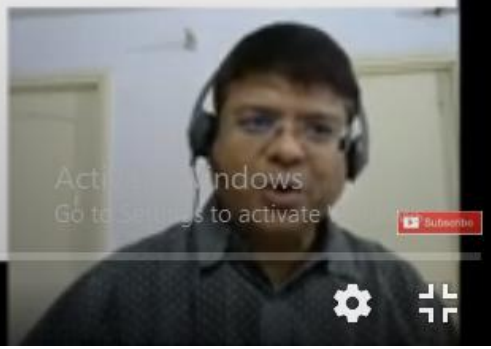
#19-BOYLE'S TEMPERATURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

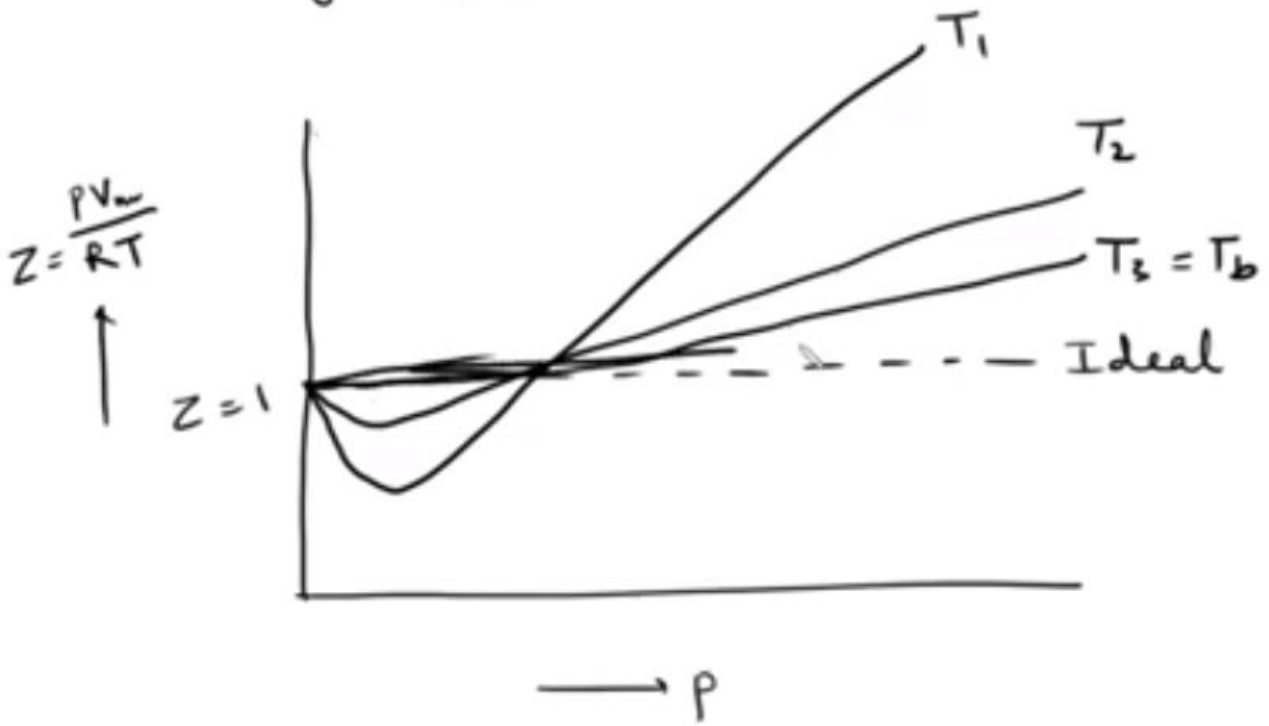
Boyle's temperature (T_b)

→ The temperature of real gas at which real gas follows Boyle's law i.e. $PV = \text{cont}$ over a range of pressure from zero pressure.



1:37 / 14:39

a range of pressure from zero pressure.



For same gas at different temp

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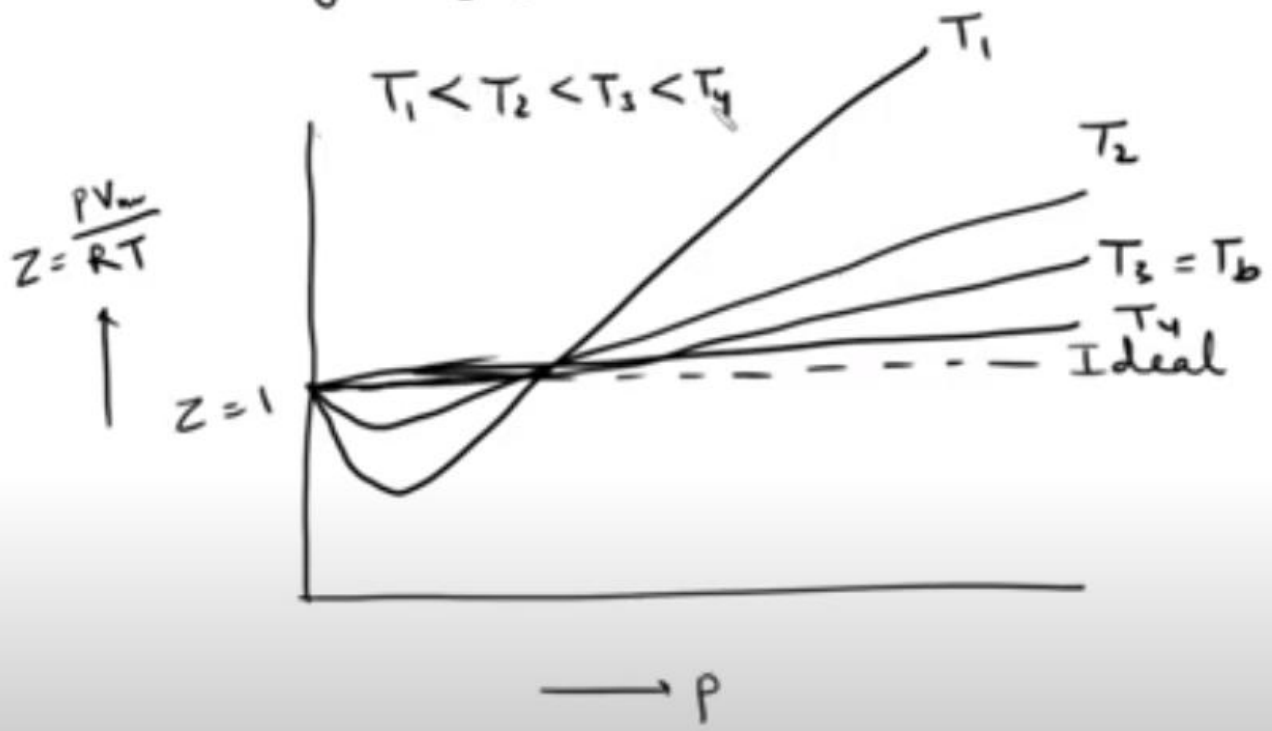
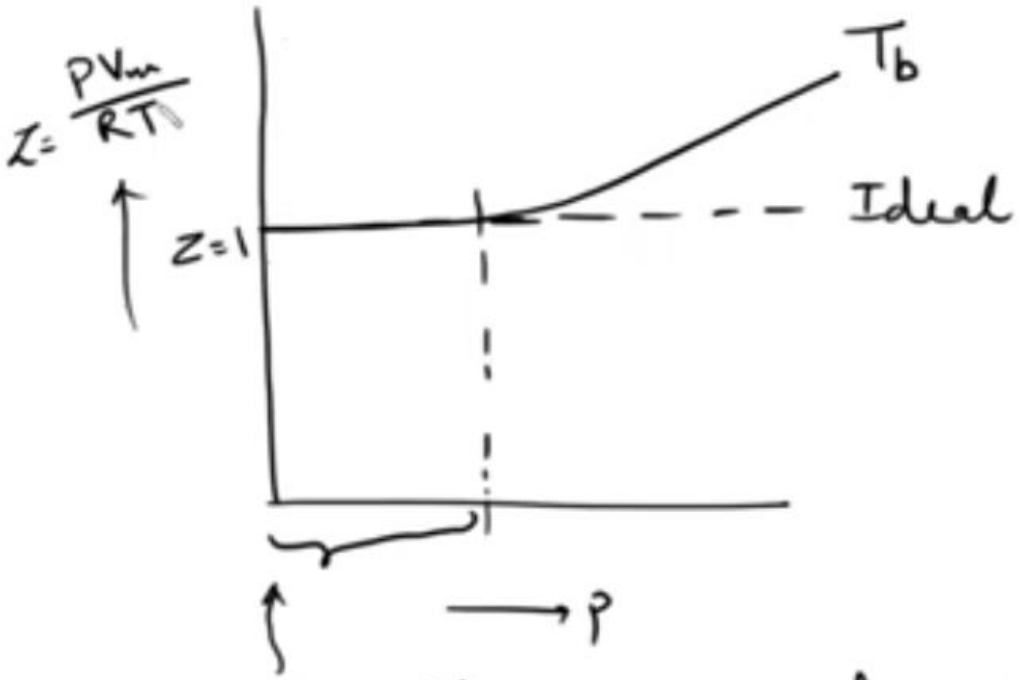
#19-BOYLE'S TEMPERATURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

a range of pressure from zero pressure.



For same gas at different temp

3:37 / 14:39



gas follows Boyle's law i.e. PV = Constant

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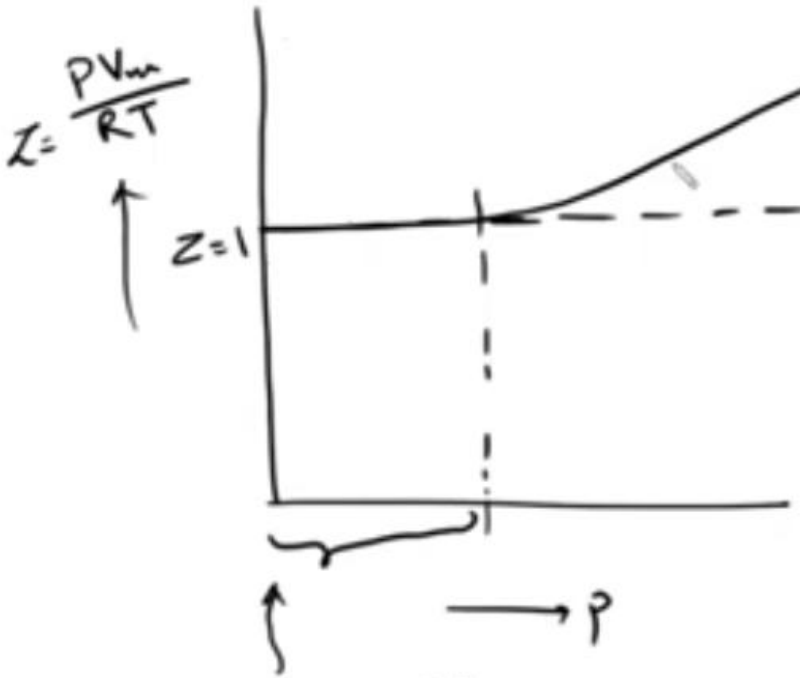

(Isotherm at Boyle's temp)

gas follows Boyle's law i.e. PV = Constant

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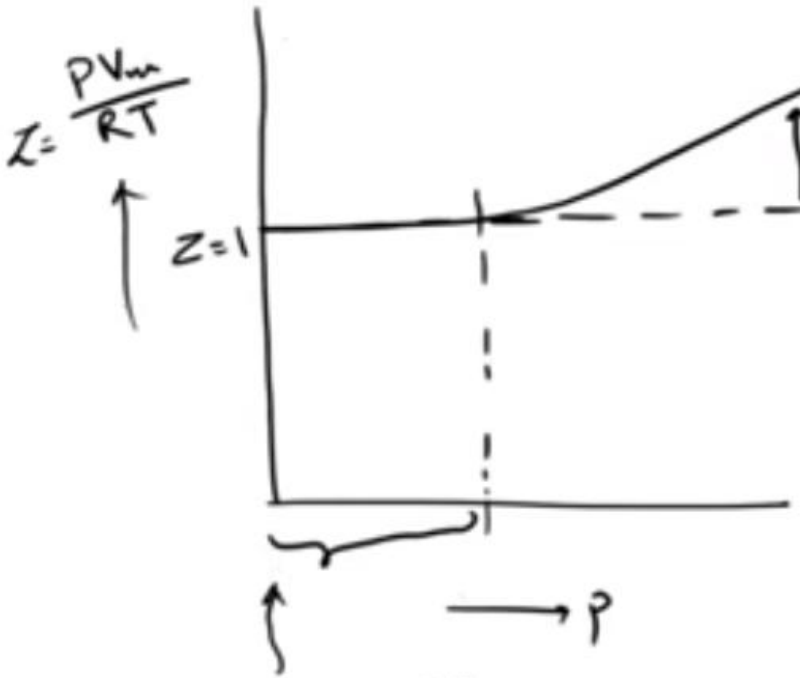

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↑ ⟶ P

gas follows Boyle's law i.e. $PV = \text{Constant}$

$$\lim_{P \to 0} \left\{ \frac{d(PV_m)}{dV_m} \right\}_{T=T_b} = 0$$

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$PV_m = \frac{RTV_m}{V_m - b} - \frac{a}{V_m}$$

$$\frac{d(PV_m)}{dV_m} = (V_m - b) \times R \times 0 = RT \times 1$$

$$\lim_{P\to 0}\left\{\frac{d(PV_m)}{dV_m}\right\}_{T=T_b} = 0$$

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$PV_m = \frac{RTV_m}{V_m - b} - \frac{a}{V_m}$$

$$\frac{d(PV_m)}{dV_m} = (V_m - b)\times RT\times -1 = -RT$$

$$\left\{\frac{d(PV_m)}{dV_m}\right\}_{T=T_b} = RT_b\left\{\frac{(V_m - b)\times 1 - V_m\times 1}{(V_m - b)^2}\right\} + \frac{a}{V_m^2} = 0$$

$\Rightarrow$

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$$\left\{\frac{d(pV_m)}{dV_m}\right\}_{T=T_b} = RT_b\left\{\frac{(V_m-b)\times 1 - V_m\times 1}{(V_m-b)^2}\right\} + \frac{a}{V_m^2} = 0$$

$$\Rightarrow \frac{RT_b \times b}{(V_m-b)^2} = \frac{a}{V_m^2}$$

$$\Rightarrow T_b = \frac{a}{Rb}\left[\frac{V_m-b}{V_m}\right]^2$$

$$T_b = \frac{a}{Rb}\left[1-\frac{b}{V_m}\right]^2$$

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#19-BOYLE'S TEMPERATURE|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

$$T_b = \frac{a}{Rb}\left[1 - \frac{b}{V_m}\right]^2$$

when $P \to 0$, $V_m \to$ large

$\frac{b}{V_m} \to$ very small

$$\boxed{T_b = \frac{a}{Rb}}$$

9:43 / 14:39

when $P \to 0$, $\frac{b}{V_m} \to$ very small

$V_m \to$ large

$$T_b = \frac{a}{Rb}$$

(i) At Boyle's temp (T_b), real gas behaves ideally (or PV = constant) in a certain range of pressure because the effect caused by vanderWaal's constats 'a' & 'b' compensate (nearly) each other.

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(i) At Boyle's temp (T_b), real gas behaves ideally (or PV = constant) in a certain range of pressure because the effect caused by vanderWaal's constants 'a' & 'b' compensate (nearly) each other.

(ii) At T_b, gas shows +ve deviation at high pressure.

by vanderWaal's constants 'a' & 'b' compensate (nearly) each other.

(ii) At T_b, gas shows +ve deviation at high pressure.

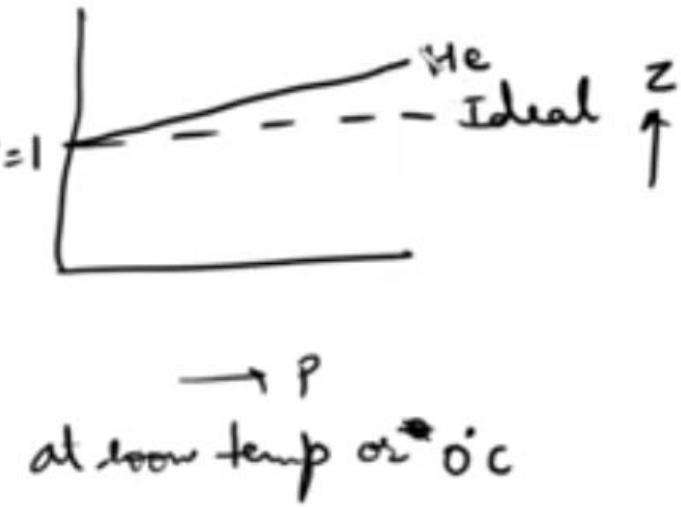
(iii) Even H_2 and He show -ve deviation if $T < T_b$.

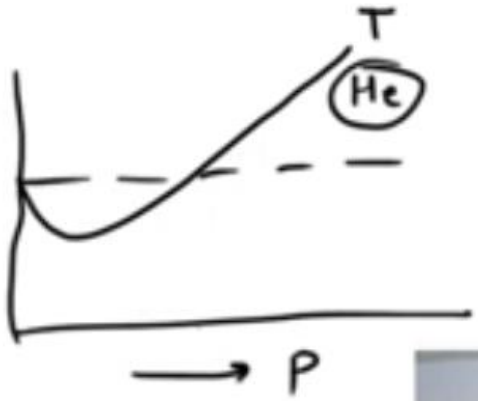
each other.

(ii) At T_b, gas shows +ve deviation at high pressure.

(iii) Even H_2 and He show -ve deviation if $T < T_b$.

* T_b is very low for He and H_2.





GASEOUS STATE

Other equations of state :

Vander Waal's eq^n of state

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

① Dieterici equation of state :

$$P e^{\frac{a}{RTV_m}} (V_m - b) = RT$$

where a, b are Dieterici constant

(2) Berthelot eqⁿ:

$$\left(P + \frac{a}{TV_m^2}\right)(V_m - b) = RT$$

a, b ⟶ Berthelot constant

a, b → Berthelot constant

(3) Virial equation of state :

A general eqⁿ for a real gas,

$$Z = \frac{PV_m}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \cdots$$

GASEOUS STATE

A general eq^{n} for a real gas,

$$Z = \frac{PV_m}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \text{- - - - -}$$

where B = Second Virial Coefficient

C = Third " "

D = Fourth " "

D = Fourth " "

Determination of Virial Coeff. in terms of 'a' & 'b':

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$$P = \frac{RT}{(V_m - b)} - \frac{a}{V_m^2} \Rightarrow P \times \frac{V_m}{RT} = \frac{V_m}{V_m - b} - \frac{a}{RT\,V_m}$$

$P \times \frac{V_m}{RT}$ → Z

$\Rightarrow Z =$

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$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$$P = \frac{RT}{(V_m - b)} - \frac{a}{V_m^2} \Rightarrow \left(P \times \frac{V_m}{RT}\right) = \frac{V_m}{V_m - b} - \frac{a}{RT V_m}$$

Z

$$\Rightarrow Z = \frac{1}{\left(1 - \frac{b}{V_m}\right)} - \frac{a}{RT V_m}$$

$$= \left(1 - \frac{b}{V_m}\right)^{-1} - \frac{a}{RT V_m}$$

$$=$$

$$\Rightarrow Z = \frac{1}{\left(1 - \frac{b}{V_m}\right)} - \frac{a}{RTV_m}$$

$$= \left(1 - \frac{b}{V_m}\right)^{-1} - \frac{a}{RTV_m}$$

$$= \left(1 + \frac{b}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \cdots\right) - \frac{a}{RTV_m}$$

$$Z = 1 + \frac{\left(b - \frac{a}{RT}\right)}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \cdots$$

GASEOUS STATE

$$Z = 1 + \frac{\left(b - \frac{a}{RT}\right)}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \cdots$$

General Virial eqn, $Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \cdots$

$\Rightarrow$ On Comparing, $B = b - \frac{a}{RT}$

$C = b^2$

$D = b^3$

General Virial eqⁿ, $Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \cdots$

$\Rightarrow$ On Comparing, $B = \left(b - \frac{a}{RT}\right)$

Remember

$C = b^2$

$D = b^3$

* Second virial Coeff. becomes zero at Boyle's temp (T_b).

$$(B)_{T=T_b} = b - \frac{a}{R \times T_b} = b - \frac{\cancel{a}}{\cancel{R} \times \frac{\cancel{a}}{\cancel{R}b}}$$

$$= 0$$

$= 0$

Alternate derivation of T_b :

$$Z = 1 + \frac{\left(b - \frac{a}{RT}\right)}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \cdots$$

At Boyle temp, $Z = 1$ in low pr. range.

$\Rightarrow$ V_m^2, V_m^3 etc are very large at low pr, so $\frac{b^2}{V_m^2}$, $\frac{b^3}{V_m^3}$ etc can be ignored.

$$Z = 1 + \frac{\left(b - \frac{a}{RT}\right)}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \cdots$$

At Boyle temp, $Z = 1$ in low pr. range.

$\Rightarrow$ V_m^2, V_m^3 etc are very large at low pr, so $\frac{b^2}{V_m^2}, \frac{b^3}{V_m^3}$ etc can be ignored.

For $Z = 1$, $b - \frac{a}{RT_b} = 0$

$\Rightarrow$ $T_b = \frac{a}{Rb}$

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Liquifaction of Gases: It is done by

(i) Applying pressure and/or

(ii) Reducing temperature.

Condensing influence of intermolecular forces

vs dispersing influence of thermal energy

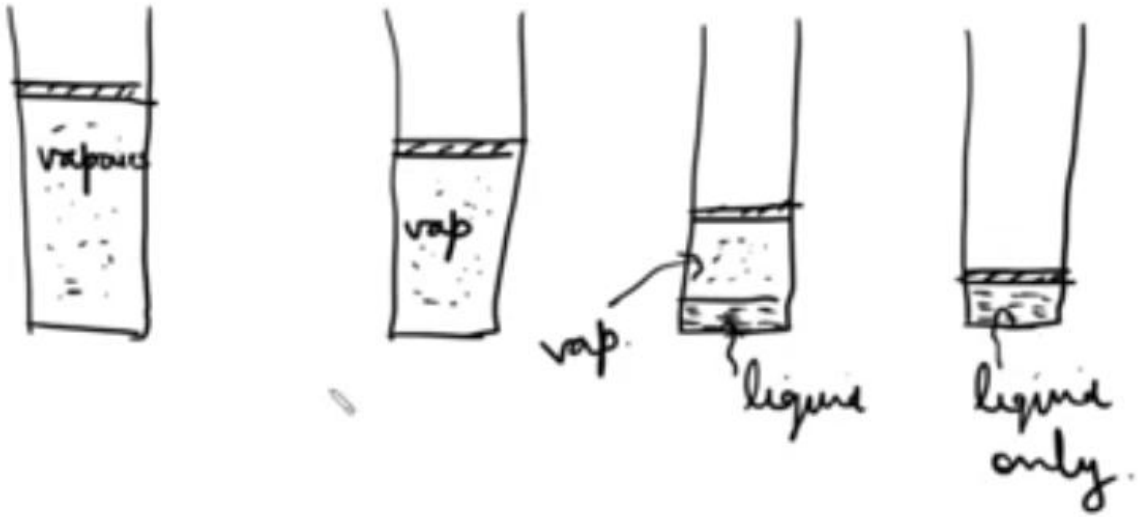
#21-LIQUIFACTION OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

vs dispersing influence of thermal energy

→ Thomas Andrews liquified CO_2 gas.
(Andrews' Isotherms)

5:03 / 24:47

(Andrews' Isotherms)



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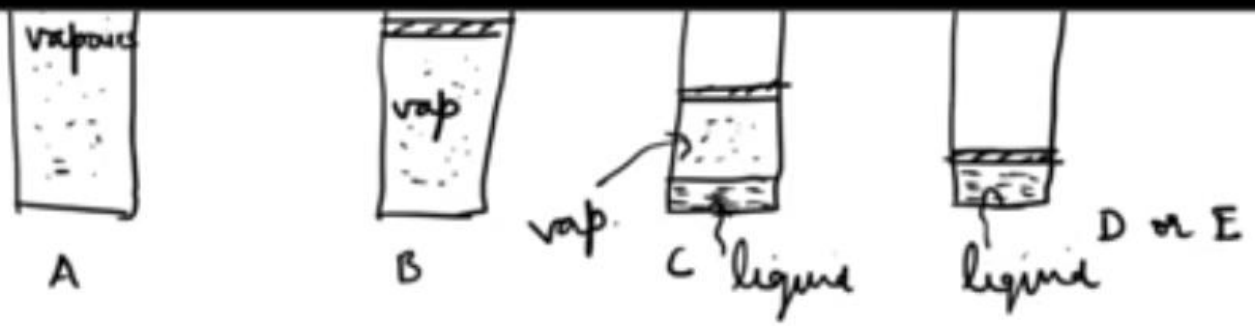
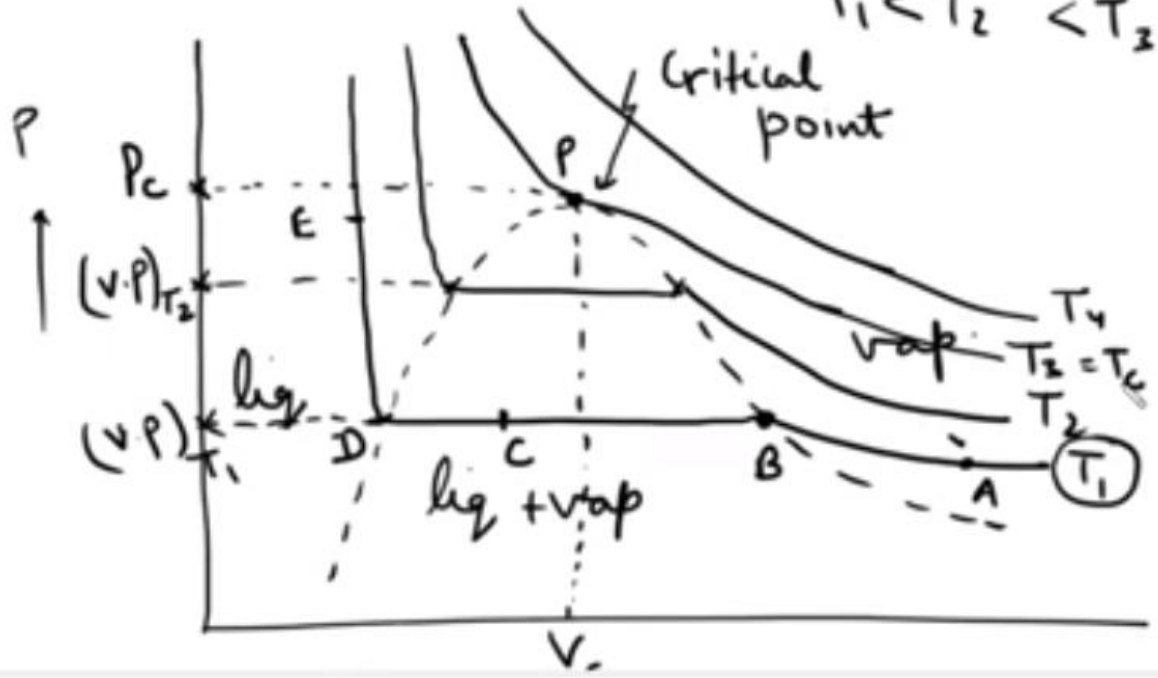

vapours
vap
vap
A
B
C
liquid
liquid
D or E

$T_1 < T_2 < T_3 < T_4$


P
Critical point
P_c
E
$(v.p)_{T_2}$
liq
$(v.p)_{T_1}$
D
C
B
A
liq + vap
vap
T_4
$T_2 = T_c$
T_2
T_1
V

$\longrightarrow V_m$

Critical temperature (T_c): The max. temperature at which vapour can be converted into liquid by application of pressure is called Critical temp.

At temp $> T_c$, vapour or gas can't be converted into liquid what ever be the pressure applied.

#21-LIQUIFACTION OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

called Critical temp.

At temp > T_c, vapour or gas can't be converted into liquid what ever be the pressure applied.

Critical pressure (P_c) : The minimum pressure required to be applied to convert vapour into liquid at critical temp.

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16:13 / 24:47

#21-LIQUIFACTION OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD

applied.

Critical pressure (P_c) : The minimum pressure required to be applied to convert vapour into liquid at critical temp.

Critical volume (V_c) : It is the molar volume at critical point.

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16:52 / 24:47

to convert vapour into liquid at critical temp.

Critical volume (V_c): is the molar volume at critical point.

→ P_c, V_c, T_c are called Critical constants

→ Critical point is the characteristic point of every substance at which P_c, V_c, T_c are ~~constants~~ characteristic constants.

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→ Critical point is the characteristic point of every substance at which P_c, V_c, T_c are ~~constants~~ characteristic constants.

→ At critical point, there is no distinction between vapour and liquid. Density of vapour and liquid are equal at critical point.

→

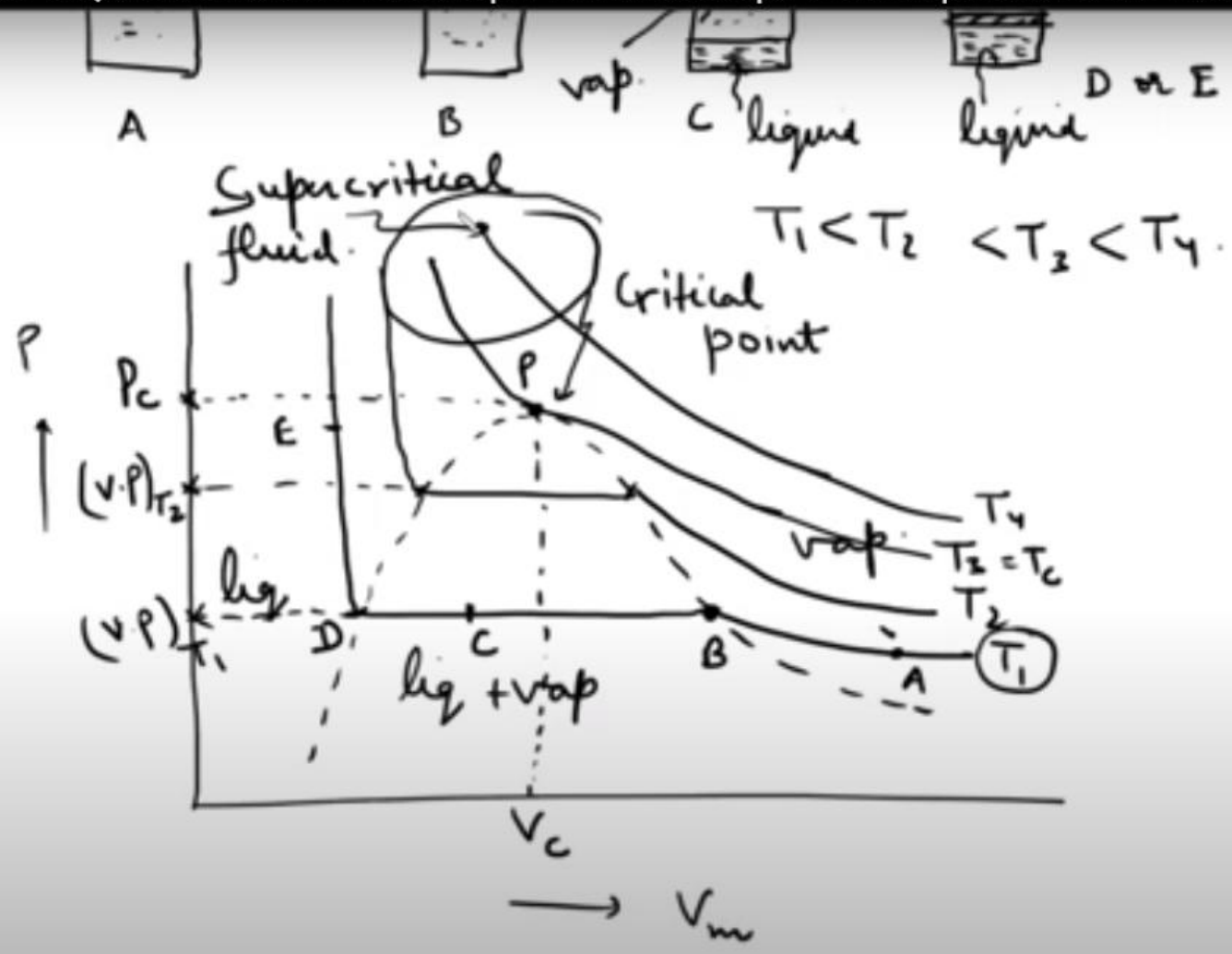
of ~~every substance~~ _ _ _

~~Constants~~ characteristic constants.

→ At critical point, there is no distinction between vapour and liquid. Density of vapour and liquid are equal at critical point.

→ If $T > T_c$, $P > P_c$, then substance is called supercritical fluid ~~substance~~ at which density is greater than that of liquid.

#21-LIQUIFACTION OF GASES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|OLYMPIAD



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21:04 / 24:47

and liquid are equal at critical point.

→ If $T > T_c$, $P > P_c$, then substance is called supercritical fluid at which density is greater than that of liquid.

→ The term vapour is used when $T < T_c$ and the term gas is used when $T > T_c$ under low pr. conditions.

supercritical fluid ~~substance~~ at which density is greater than that of liquid.

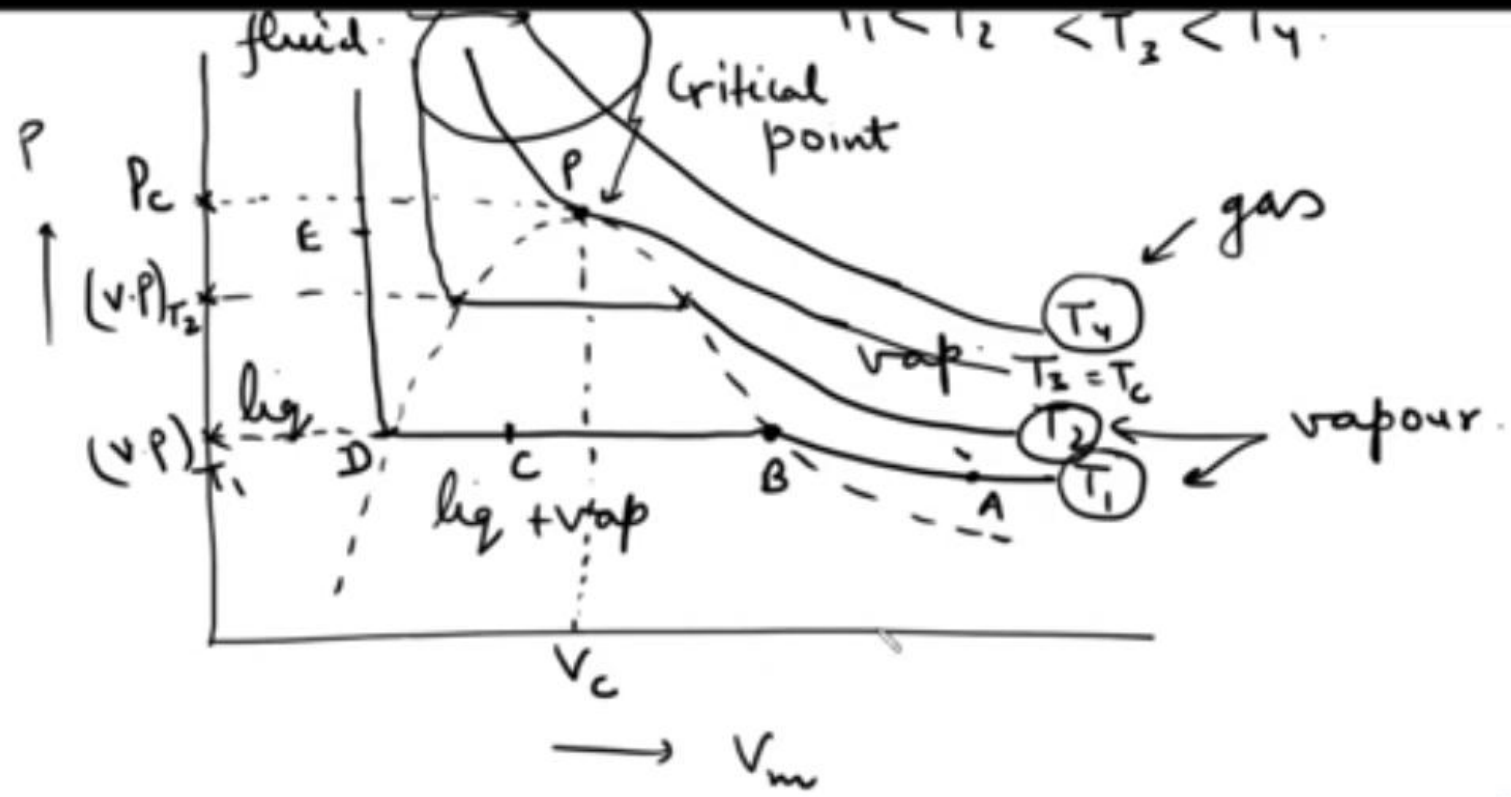
→ The term vapour is used when $T < T_c$ and the term gas is used when $T > T_c$ under low pr. conditions.

* So vapour can be converted into liquid by applying pressure but not gas.

supercritical fluid at which density is greater than that of liquid.

→ The term vapour is used when $T < T_c$ and the term gas is used when $T > T_c$ under low pr. conditions.

* So vapour can be converted into liquid by applying pressure but not gas.
* To convert gas into liquid, first decrea temp below T_c and apply pressure.



Critical temperature (T_c) : The max. temperature which vapour can be

#22-DETERMINATION OF CRITICAL CONSTANTS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS ...

Determination of Critical Constants (P_c, V_c, T_c):

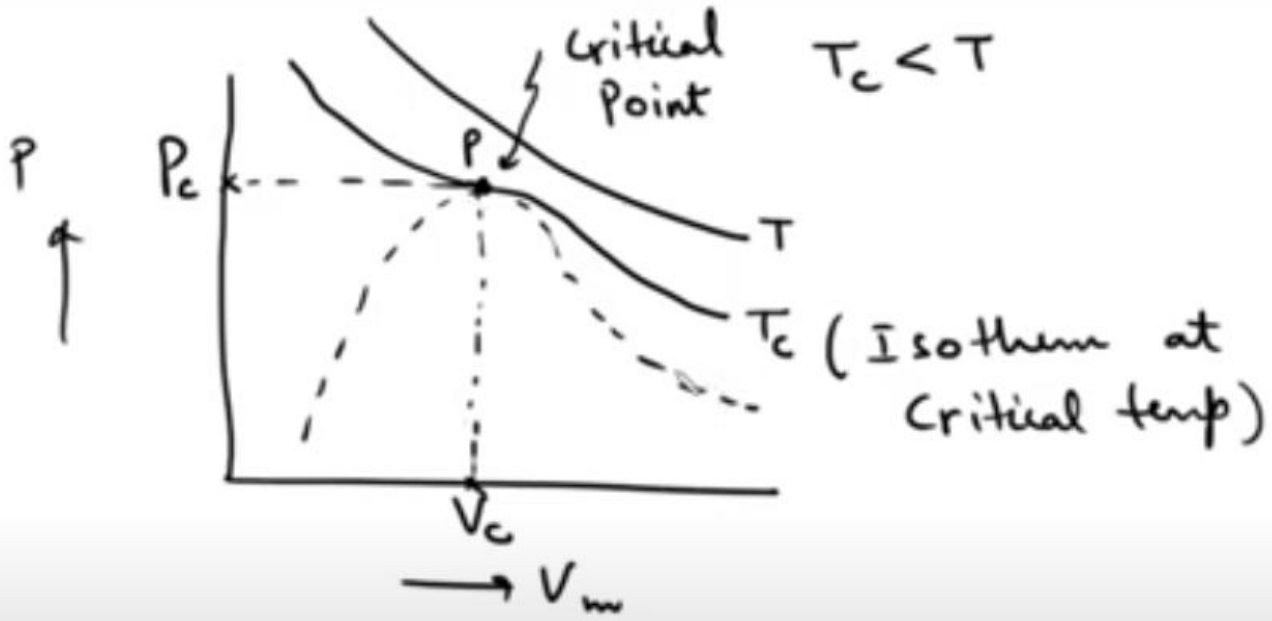
(in terms of vanderWaal's constants)

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

1:01 / 14:53

#22-DETERMINATION OF CRITICAL CONSTANTS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS ...

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$



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3:21 / 14:53

$\longrightarrow V_m$

→ At critical point 'P', slope of P vs V_m is not only equal to zero but maximum also.

$$\text{slope} = \left(\frac{dP}{dV_m}\right)_{T=T_c} = 0$$

$$\left(\frac{d(\text{slope})}{dV_m}\right)$$

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$$\left(\frac{d(\text{slope})}{dV_m}\right)_{T=T_c} = 0$$

$$\left\{\frac{d}{dV_m}\left(\frac{dP}{dV_m}\right)\right\}_{T=T_c} = 0$$

$$\left(\frac{d^2P}{dV_m^2}\right)_{T=T_c} = 0$$

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#22-DETERMINATION OF CRITICAL CONSTANTS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS ...

$$\left\{\frac{d}{dV_m}\left(\frac{dp}{dV_m}\right)\right\}_{T=T_c} = 0$$

$$\left(\frac{d^2p}{dV_m^2}\right)_{T=T_c} = 0 \quad -(2)$$

$$\left(\frac{dp}{dV_m}\right)_{T=T_c} = 0 \quad -(1)$$

Appl

5:32 / 14:53

$$\left(\frac{d(\text{slope})}{dV_m}\right)_{T=T_c} = 0$$

$$\left\{\frac{d}{dV_m}\left(\frac{dp}{dV_m}\right)\right\}_{T=T_c} = 0$$

$$\left(\frac{d^2p}{dV_m^2}\right)_{T=T_c} = 0 \quad -(2)$$

$$\left(\frac{dp}{dV_m}\right)_{T=T_c} = 0 \quad -(1)$$

Applicable at critical point

#22-DETERMINATION OF CRITICAL CONSTANTS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS ...

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT \Rightarrow P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\left(\frac{dP}{dV_m}\right)_T = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3}$$

$$\left(\frac{d^2P}{dV_m^2}\right)_T = \frac{2RT}{V_m - 1}$$

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7:23 / 14:53

$$\left(\frac{dp}{dV_m}\right)_T = -\frac{RT}{(V_m-b)^2} + \frac{2a}{V_m^3}$$

$$\left(\frac{d^2p}{dV_m^2}\right)_T = \frac{2RT}{(V_m-b)^2} - \frac{6a}{V_m^4}$$

At Critical Point, $\quad \frac{-RT_c}{(V_c-b)^2} + \frac{2a}{V_c^3} = 0$

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$$\left(\frac{dp}{dV_m}\right)_T = -\frac{RT}{(V_m-b)^2} + \frac{2a}{V_m^3}$$

$$\left(\frac{d^2p}{dV_m^2}\right)_T = \frac{2RT}{(V_m-b)^2} - \frac{6a}{V_m^4}$$

At Critical Point,

$$-\frac{RT_c}{(V_c-b)^2} + \frac{2a}{V_c^3} = 0$$

$$\frac{2RT_c}{(V_c-b)^2} - \frac{6a}{V_c^4} = 0$$

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$$\frac{2RT_c}{(V_c-b)^2} - \frac{6a}{V_c^4} = 0 \quad \text{— (4)}$$

$\Rightarrow$ From (3) and (4), $\boxed{V_c = 3b}$

$$\boxed{T_c = \frac{8a}{27Rb}}$$

To determine P_c, applying vanderwaal's eq^n at 'P'.

$$\left(P_c + \frac{a}{V_c^2}\right)(V_c - b) = RT_c$$

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$$T_c = \frac{8a}{27Rb}$$

To determine P_c, applying vanderwaal's eq^n at 'P'.

$$\left(P_c + \frac{a}{V_c^2}\right)(V_c - b) = RT_c$$

$$\left(P_c + \frac{a}{9b^2}\right)(3b - b) = R \times \frac{8a}{27Rb}$$

$$\left(P_c + \frac{a}{9b^2}\right) \times 2b = \frac{4 \cdot 8a}{27b} \Rightarrow \left(P_c + \frac{a}{9b^2}\right) = \frac{4a}{27b^2}$$

$$\Rightarrow \boxed{P_c = \frac{a}{27b^2}}$$

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$$\Rightarrow \boxed{P_c = \frac{a}{27b^2}}$$

Critical Constants:

$$P_c = \frac{a}{27b^2}$$

$$T_c = \frac{8a}{27Rb}$$

$$V_c = 3b$$

$$\Rightarrow \boxed{P_c = \frac{a}{27b^2}}$$

Critical Constants:

$$P_c = \frac{a}{27b^2}$$

$$T_c = \frac{8a}{27Rb}$$

$$V_c = 3b$$

$\Leftarrow$

Note: To calculate size of gas molecule, use the data P_c and T_c instead of V_c.

How: $\frac{T_c}{P_c} = \frac{8a}{27Rb} \times \frac{27b^2}{a} = \frac{8b}{R} = \frac{8}{R} \times 4 \times N_A \times \frac{4}{3}$

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#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

Alternate method to determine P_c, V_c and T_c :

Vander Waal's equation of state

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

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1:22 / 15:20

#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

$$\left(P+\frac{a}{V_m^2}\right)(V_m-b)=RT$$

$$(PV_m^2+a)(V_m-b)=RTV_m^2$$

$$PV_m^3+aV_m-PbV_m^2-ab-RTV_m^2=0$$

P

1:57 / 15:20

#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

$$\left(P+\frac{a}{V_m^2}\right)(V_m-b) = RT$$

$$(PV_m^2 + a)(V_m - b) = RTV_m^2$$

$$PV_m^3 + aV_m - PbV_m^2 - ab - RTV_m^2 = 0$$

$$PV_m^3 - (Pb+RT)V_m^2 + aV_m - ab = 0$$

$$V_m^3 - \left(b+\frac{RT}{P}\right)V_m^2 + \frac{a}{P}V_m - \frac{ab}{P} = 0 \quad \text{—— (1)}$$

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2:40 / 15:20

#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

$$pV_m^3 + aV_m - pbV_m^2 - ab - RTV_m^2 = 0$$

$$pV_m^3 - (pb+RT)V_m^2 + aV_m - ab = 0$$

$$V_m^3 - \left(b + \frac{RT}{p}\right)V_m^2 + \frac{a}{p}V_m - \frac{ab}{p} = 0 \quad \text{—— (1)}$$

This equation (1) is cubic in molar volume,

Let three roots are

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3:11 / 15:20

$$V_m^3 - \left(b + \frac{RT}{P}\right)V_m^2 + \frac{a}{P}V_m - \frac{ab}{P} = 0 \quad \text{—— (1)}$$

This equation (1) is cubic in molar volume,

Let three roots are α, β, γ.

$$(V_m - \alpha)(V_m - \beta)(V_m - \gamma) = 0$$

At critical point, all three roots are equal $= V_c$

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#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

Let three roots are α, β, γ.

$$(V_m-\alpha)(V_m-\beta)(V_m-\gamma)=0$$

At critical point, all three roots are equal $= V_c$

$$(V_m-V_c)^3=0$$

$$V_m^3 - V_c^3 - 3V_m V_c (V_m-V_c)=0$$

$$V_m^3 - 3V_c \times V_m^2 + 3V_c^2 \times V_m - V_c^3 = 0$$

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4:33 / 15:20

$$V_m^3 - 3V_c \times V_m^2 + 3V_c^2 \times V_m - V_c^3 = 0 \quad \text{— (2)}$$

Comparing eq. (1) & eq-(2) at critical point.

$$3V_c = b + \frac{RT_c}{P_c}$$

$$\frac{3V_c^2 = \frac{a}{P_c}}{}$$

$$\frac{V_c^3 = \frac{ab}{P_c}}{}$$

$$\frac{3V_c^{\cancel{2}}}{V_c^{\cancel{3}}} = \frac{\cancel{a}}{\cancel{P_c}} \times \frac{\cancel{P_c}}{\cancel{b}}$$

$$\boxed{V_c = 3b}$$

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$\rightarrow \quad 3V_c = b + \frac{RT_c}{P_c}$

$3V_c^2 = \frac{a}{P_c}$

$V_c^3 = \frac{ab}{P_c}$

$\frac{3V_c^{\cancel{2}}}{V_c^{\cancel{3}}} = \frac{\cancel{a}}{\cancel{P_c}} \times \frac{\cancel{P_c}}{\cancel{R}b}$

$\boxed{V_c = 3b}$

$\boxed{P_c = \frac{a}{27b^2}}$

$\boxed{T_c = \frac{8a}{27Rb}}$

$$T_c = \frac{8a}{27Rb}$$

→ Calculations suggest (For vanderWaal's eq^n)

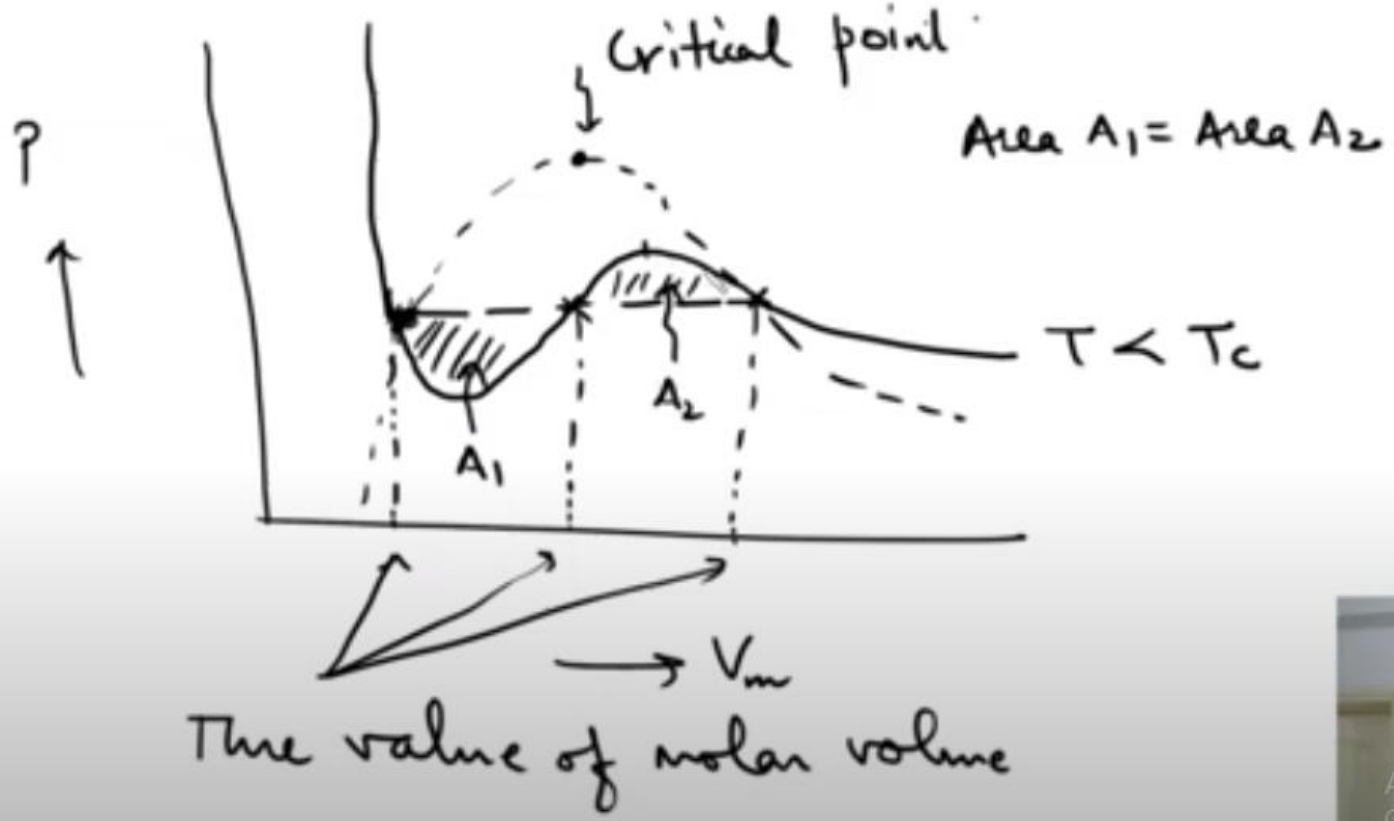
* Above T_c, out of three roots, only one is real & two are imaginary roots.

* At $T = T_c$, all three roots are real & equal.

* Above T_c, out of three roots, only one is real & two are imaginary roots.
* At $T = T_c$, all three roots are real & equal.
* At $T < T_c$, all three roots are real & distinct.

#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

distinct.



The value of molar volume

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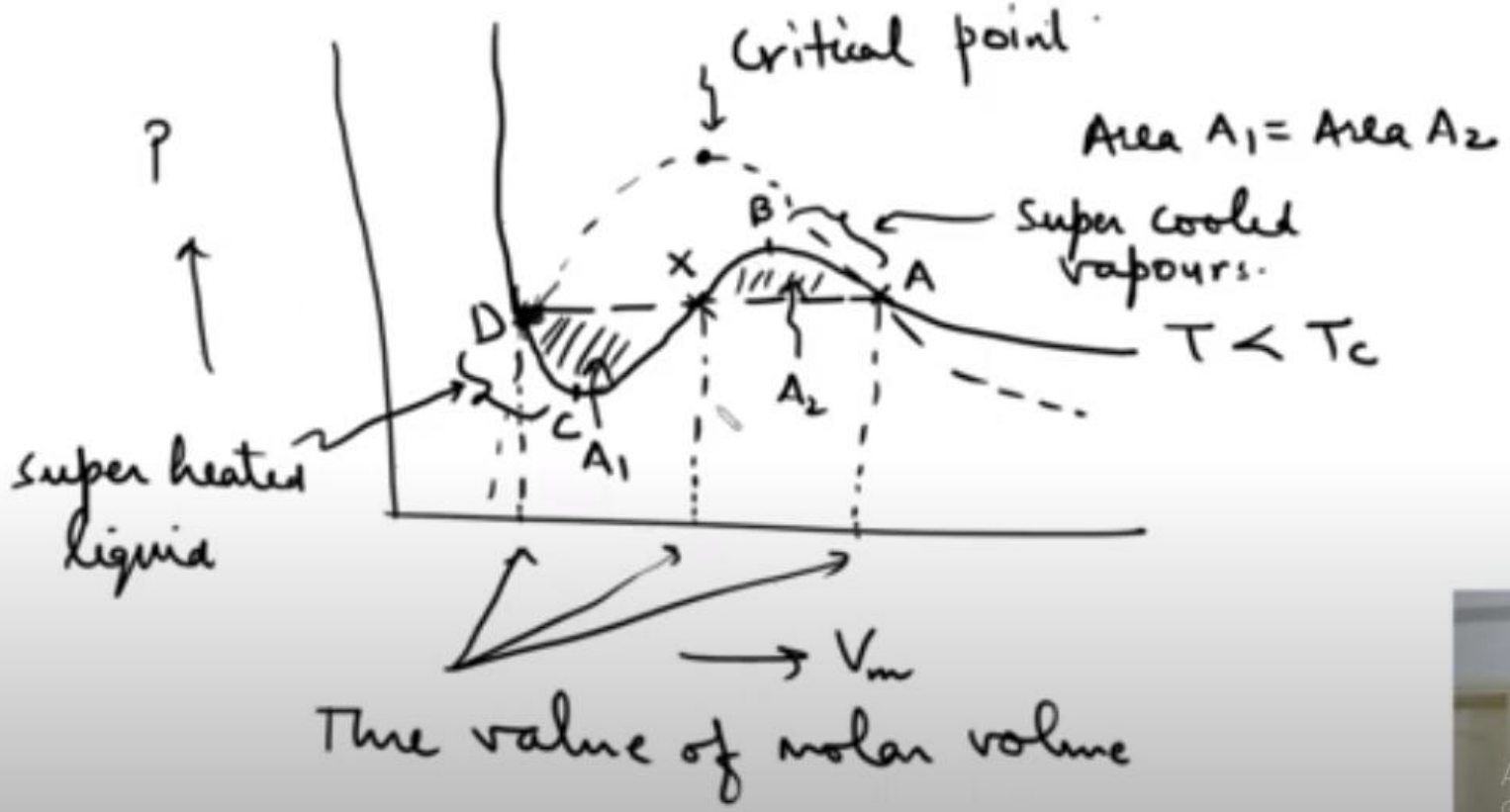
#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

distinct.



The value of molar volume

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The value of molar volume

* Region BC does not have any physical significance and not observed.

because, $\left(\frac{dP}{dV_m}\right)_T$ in BC region = +ve (Not possible)

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#23-ALTERNATE METHOD TO DETERMINE Pc,Vc,Tc|IIT ADVANCED|JEE MAIN|CHEMISTRY CLAS...

* Region BC does not have any physical significance and not observed because, $\left(\frac{dP}{dV_m}\right)_T$ in BC region = +ve (Not possible)

→ Vander Waal's eqⁿ is not applicable for meta stable region (or liq + vap) region. But applies well for pure vapour form and pure liquid form.

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14:45 / 15:20

#24-LAW OF CORRESPONDING STATES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|...

Law of corresponding states :

Reduced parameters :

- Reduced pr, $P_r = \frac{P}{P_c}$
- Reduced vol, $V_r = \frac{V_m}{V_c}$
- Reduced temp, $T_r = \frac{T}{T_c}$

→ dimensionless terms

1:38 / 14:29

#24-LAW OF CORRESPONDING STATES|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY|...

van der Waal's equation of state

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$$\left(P_r \times P_c + \frac{a}{V_r^2 \times V_c^2}\right)\left(V_r \times V_c - b\right) = R T_r \times T_c$$

$$\left(P_r \times \frac{a}{27b^2} + \frac{a}{V_r^2 \times 9b^2}\right)\left(V_r \times 3b - b\right) = RT_r \times \frac{8a}{27Rb}$$

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3:46 / 14:29

$$\left(P_r \times \frac{}{27b^2}\right)\left(V_r^2 \times 9b^2\right)\left(\quad\right) \qquad 27RT$$

$$\Rightarrow \left(\frac{P_r}{27} + \frac{1}{9V_r^2}\right)\left(3V_r - 1\right) = \frac{8T_r}{27}$$

$$\boxed{\left(P_r + \frac{3}{V_r^2}\right)\left(3V_r - 1\right) = 8T_r}$$

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$$\left(P_r \times \frac{}{27 b^2} \right) \left(V_r^2 \times 9b^2 \right) \left(\qquad \right) \qquad 27RT$$

$$\Rightarrow \left(\frac{P_r}{27} + \frac{1}{9V_r^2}\right)(3V_r - 1) = \frac{8T_r}{27}$$

$$\boxed{\left(P_r + \frac{3}{V_r^2}\right)(3V_r - 1) = 8T_r}$$

→ Reduced equation of state (as it involves reduced parameters)

no gas dependent constant.

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no gas dependent constant.

(as it involves reduced parameters)

$$V_r = f(T_r, P_r)$$

'Law of corresponding states' state that 'if two gases have same reduced Pr and reduced temp Tr, then their reduced volume will also be same'.

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reduced temp T_R, then ...
will also be same'.

Q → Calculate the volume occupied by 2 mol of N_2 (follows vanderWaal's eq^n. of state) at 200K and 10.1325 MPa pressure if

$$\frac{P_R V_R}{T_R} = 2.2$$

[Take $R = \frac{1}{12}$ lit atm mol^{-1} K^{-1}]

$$\frac{P_r V_r}{T_r} = 2.2 \quad \left[\text{Take } R = \frac{1}{12} \text{ lit atm mol}^{-1} K^{-1}\right]$$

Soln:

$$\frac{P_r \times V_r}{T_r} = 2.2 \Rightarrow \frac{\frac{P}{P_c} \times \frac{V_m}{V_c}}{\frac{T}{T_c}} = 2.2$$

$$\frac{P \times V}{n \times T} = 2.2 \times \frac{P_c V_c}{T_c}$$

For vanderWaal's gas

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T_c

$$\frac{P \times V}{n \times T} = 2.2 \times \frac{P_c V_c}{T_c}$$

For vanderWaal's gas

$$Z_c = \frac{3}{8}$$

compressibility factor at critical point

Proof: $Z_c = \frac{P_c \times V_c}{R T_c}$

$$= \frac{\frac{a}{27b^2} \times 3b}{R \times \frac{8a}{27Rb}} = \frac{3}{8}$$

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$$\frac{P \times V}{n \times T} = 2.2 \times \frac{P_c V_c}{T_c}$$

Since $\frac{P_c V_c}{RT_c} = Z_c = \frac{3}{8}$

$$\Rightarrow \frac{P_c V_c}{T_c} = \frac{3}{8} \times R$$

$$\frac{100 \times V}{2 \times 200} = 2.2 \times \frac{3}{8} \times \frac{1}{1}$$

For vanderWaal's gas

$$Z_c = \frac{3}{8}$$

Compressibility factor at critical point

Proof: $Z_c = \frac{P_c \times V_c}{RT_c}$

$$= \frac{\frac{a}{27b^2} \times 3b}{R \times \frac{8a}{27Rb}} = \frac{3}{8}$$

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$$\frac{P_c V_c}{T_c} = \frac{3}{8} \times R$$

at critical point

Proof: $Z_c = \frac{P_c \times V_c}{RT_c}$

$$= \frac{\frac{a}{27b^2} \times 3b}{R \times \frac{8a}{27Rb}} = \frac{3}{8}$$

$$\frac{100 \times V}{2 \times 200} = 2.2 \times \frac{3}{8} \times \frac{1}{12}$$

$$V = \frac{1.1}{12} \times 3 = \frac{1.1}{4}$$

$$= 0.275 \text{ lit} \quad \text{Ans}$$

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#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

Questions on Real gas :

Q → what mass of O_2 gas can be filled in 1 lit vessel at 1 atm and 300K if compressibility factor is 0.8 under these conditions.

Sol : $PV = Z \times nRT$

1:52 / 18:29

Gaseous state of Real gas :

Q → what mass of O_2 gas can be filled in 1 lit vessel at 1 atm and 300K if compressibility factor is 0.8 under these conditions.

Sol : $PV = Z \times nRT$

$$1 \times 1 = 0.8 \times \frac{W_{O_2}}{32} \times \frac{1}{1} \times 300$$

$$W_{O_2} = \frac{40}{25}\,g = \frac{8}{5} = 1.6\,g$$

$$1 \times 1 = 0.8 \times \frac{W_{O_2}}{40} \times \frac{1}{32} \times \frac{25}{300}$$

$$W_{O_2} = \frac{40}{25}\,g = \frac{8}{5} = 1.6\ g\ O_2$$

Imp

Q → For a fixed amount of a real gas when a graph of Z vs P was plotted then at very high pr Slope was observed to be $0.01\ atm^{-1}$. At the same condition if a graph is plotted between PV vs P, then for 2 mol of gas, 'Y' intercept is found to be 40 lit atm. Find excluded volume of

5 seconds

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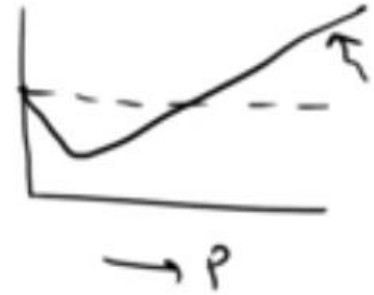
Q→ For a fixed amount of a real gas when a graph of Z vs P was plotted then at very high pr slope was observed to be 0.01 atm^{-1}. At the same condition if a graph is plotted between PV vs P, then for 2 mol of gas, 'Y' intercept is found to be 40 lit atm. Find excluded volume of gas per mol and also size (radius) of gas molecule.

Sol^{n} :



slope $= \frac{b}{RT}$

$Z = 1 + \frac{Pb}{RT}$ for high pr.

$\Rightarrow \frac{b}{RT} = 0.01$

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#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

PV vs P graph, Y-intercept means

$P \to 0$, behaviour is ideal, $Z = 1$

$PV = \underbrace{ZnRT} + 0 \times P$

intercept $= ZnRT = 1 \times 2 \times RT = 40$

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#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

$P \to 0$, behaviour is ideal, $Z = 1$

$PV = \underbrace{ZnRT} + 0 \times P$

intercept $= ZnRT = 1 \times 2 \times RT = 40$

$RT = 20$

$\frac{b}{RT} = 0.01 \Rightarrow$ $b = 0.2$ lit/mol

↑ excluded volume per mol

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#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

$\frac{b}{RT} = 0.01 \Rightarrow b = 0.2$ lit/mol

↑ excluded volume per mol.

$b = 4 \times N_A \times \frac{4}{3}\pi r^3$

$\Rightarrow r =$ ___ Å

10:37 / 18:29

$\frac{b}{RT} = 0.01 \Rightarrow$ $b = 0.2$ lit/mol

↑ excluded volume per mol.

$b = 4 \times N_A \times \frac{4}{3}\pi r^3 \Rightarrow 0.2 \times 1000 = 4 \times 6 \times 10^{23} \times \frac{4}{3}\pi r^3$

(0.2 × 1000 ↑ cm^3)

$\Rightarrow r = ?$ cm

#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

$\Rightarrow$ $\ell = \frac{}{}$ θ_n $\times \frac{1}{3}\pi$ (R)

cm^3

Q → At low pressure if $RT = 2\sqrt{aP}$, then molar volume occupied by vanderWaal's gas is

(A) $\frac{2RT}{P}$ (B) $\frac{2P}{RT}$ (C) $\frac{RT}{2P}$ (D) $\frac{2RT}{a}$

Sol^n: At low pr, $\left(P + \frac{a}{V_m^2}\right)(V_m - 0) = RT$

$$PV_m + \frac{a}{V_m} - RT = 0$$

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#25-QUESTIONS ON REAL GAS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KVPY GASEOUS STATE

$$P V_m + \frac{a}{V_m} - RT = 0$$

$$P V_m^2 - RT V_m + a = 0$$

$$V_m = \frac{RT \pm \sqrt{(RT)^2 - 4 \times P \times a}}{2P}$$

Since $RT = 2\sqrt{a \times P}$

$(RT)^2 = 4aP$

$$V_m = \frac{RT}{2P}$$

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Q → A gas at 250 K and 15 atm pr. has molar volume 12% smaller than that calculated from ideal gas law.

Find (i) Compressibility factor

(ii) Molar vol. of gas

(iii) whether attractive or repulsive forces are dominant.

Soln: $T = 250\,K$, $P = 15$ atm, $(V_m)_{real} = 0.88 \times (V_m)_{ideal}$

$\Rightarrow Z = 0.88$ (attractive forces)

$PV_m = Z \times RT$

$$V_m = \frac{0.88 \times \frac{1}{12} \times 250}{15} = \frac{6}{3} = \frac{0.88 \times 25}{6 \times 3 \times 4} = \frac{11}{9}$$

Eudiometry (volume measurement)

→ An experimental method to determine composition of gas mixture or to find molecular formula of a compound (e.g organic compounds) is called eudiometry.

#26-EUDIOMETRY VOLUME MEASUREMENT|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|K...

Eudiometry.

OR

or Eudiometry is gas mixture analysis.

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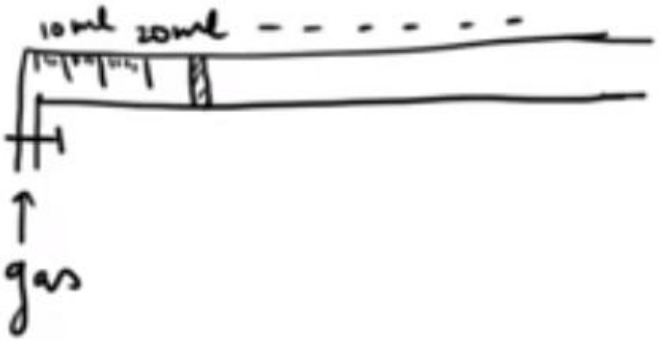
#26-EUDIOMETRY VOLUME MEASUREMENT|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|K...

Eudiometry is gas mixture analysis.



→ volumes of gas(es) are measured at contant pr. and temp (by using ~~eudio~~ eudiometer tube)

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4:11 / 23:34

pr. and temp (by using ~~eudio~~ eudiometer tube)

→ Another form of mole concept where we read stoichiometry of a gaseous reaction in terms of volumes rather than no. of moles.

For gas, $PV = nRT$

at same P & T, $V \propto n$

For gas, $PV = nRT$

at same P & T, $V \propto n$

e.g. $2A_{(g)} + 3B_{(g)} \longrightarrow C_{(g)} + 2D_{(g)}$ (balanced rex)

2 mol 3 mol 1 mol 2 mol ≡

2 lit 3 lit 1 lit 2 lit ≡

For gas, $PV = nRT$

at same P & T, $V \propto n$

e.g. $2A_{(g)} + 3B_{(g)} \longrightarrow C_{(g)} + 2D_{(g)}$ (balanced eqn)

2 mol 3 mol 1 mol 2 mol

2 lit 3 lit 1 lit 2 lit

2 ml 3 ml 1 ml 2 ml

Note: Some solvents (or reagents) are used

Imp to absorb gas(es).

gas	solvent
CO_2, Cl_2, SO_2	aq. KOH
CO	ammonical cuprous chloride (CuCl)

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GASEOUS STATE

CO_2, Cl_2, $SO_{2(g)}$	aq. KOH
$CO_{(g)}$	ammonical cuprous chloride (CuCl)
$O_{2(g)}$	alkaline pyrogallol
$O_{3(g)}$	Turpentine oil
NH_3, $HCl_{(g)}$	water, $H_2O_{(l)}$

$O_2(g)$	alkaline pyrogallol
$O_3(g)$	Turpentine oil
NH_3, $HCl(g)$	water, $H_2O(l)$
water vapour, $H_2O(vap)$	Anhy $CuSO_4$ / Anhy $CaCl_2$ / Conc H_2SO_4

#26-EUDIOMETRY VOLUME MEASUREMENT|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|K...

Assumptions taken in Eudiometry :

(1) Most common products of a reaction are assumed to be produced if not mentioned in question.

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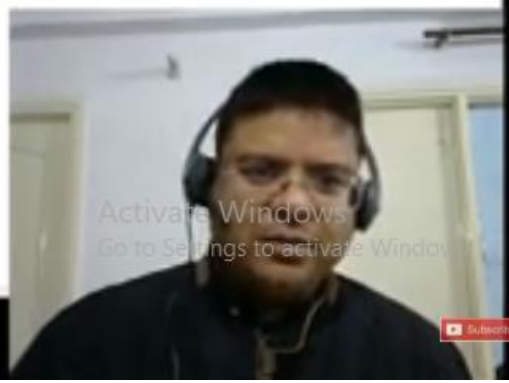
① Most common products of a reaction are assumed to be produced if not mentioned in question.

Product

→ Combustion of hydrocarbon ⟶ $CO_{2(g)} + H_2O_{(l)}$

→ $NH_{3(g)} \xrightarrow{\text{spark}} N_{2(g)} + H_{2(g)}$



① Most common products of a reaction are assumed to be produced if not mentioned in question.

Product

→ Combustion of hydrocarbon ⟶ $CO_{2(g)} + H_2O_{(l)}$

→ $NH_{3(g)} \xrightarrow{\text{spark}} N_{2(g)} + H_{2(g)}$

or $N_{2(g)} + H_{2(g)} \xrightarrow{\text{spark}} NH_{3(g)}$

Product

→ Combustion of hydrocarbon ⟶ $CO_{2(g)} + H_2O_{(l)}$

→ $NH_{3(g)} \xrightarrow{spark} N_{2(g)} + H_{2(g)}$

or $N_{2(g)} + H_{2(g)} \xrightarrow{spark} NH_{3(g)}$

→ Organic compound $(C_xH_yO_zS_mP_qN_r) \xrightarrow{combustion} CO_2 + H_2O + SO_2 + P_4O_{10} + N_2\uparrow$

* However if organic compound to nitro compound, then NO_2 will be formed)

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(2) N_2 gas present in a gas mixture on combustion will not form any oxide.

e.g $(H_2 + O_2 + N_2 + CH_4)$

$\downarrow + O_2$ (combustion)

$H_2O + N_2\uparrow + CO_2 + H_2O$

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$\downarrow +O_2$ (combustion)

$H_2O + (N_2\uparrow) + CO_2 + H_2O$

→ $N_{2(g)}$ & $O_{2(g)}$ react only under extreme conditions and in presence of catalyst.

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→ $N_{2(g)}$ & $O_{2(g)}$ react only under extreme conditions and in presence of catalyst.

(3) On combustion of hydrocarbon or an organic compound, H_2O will be considered liquid if temperature of volume measurement is not mentioned in the question.

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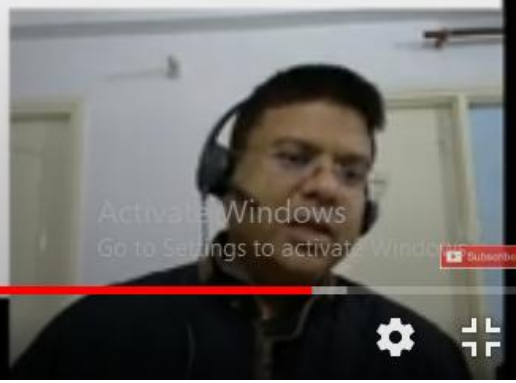
→ $N_{2(g)}$ & $O_{2(g)}$ react only under extreme conditions and in presence of catalyst.

(3) On combustion of hydrocarbon or an organic compound, H_2O will be considered liquid if temperature of volume measurement is not mentioned in the question.

(Assume all volume measurements are at room temp.)



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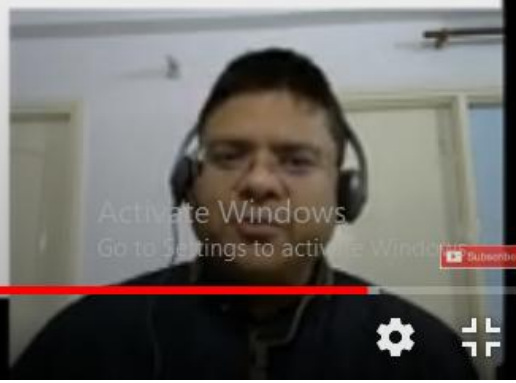
#26-EUDIOMETRY VOLUME MEASUREMENT|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|K...

organic compound, H_2O will be considered liquid if temperature of volume measurement is not mentioned in the question.

(Assume all volume measurements are at room temp.)

* In such case, vol. of $H_2O_{(l)}$ droplets is ignored wrt the other gases present.



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21:48 / 23:34

→ Alkane, C_nH_{2n+2}

Alkene, C_nH_{2n}

Alkyne, C_nH_{2n-2}

→ Hydrocarbon, C_xH_y

Question on EUDIOMETRY

Q→ 10 ml of hydrocarbon$_{(g)}$ requires 40 ml of O_2 gas for complete combustion and 30 ml of CO_2 gas is produced. Find molecular formula of hydrocarbon.

Sol^n:: C_xH_y

for complete combustion and 30 ml of CO_2 gas is produced. Find molecular formula of hydrocarbon.

Soln:

$$\underset{10\,ml}{C_xH_y(g)} + \underset{40\,ml}{\left(x+\frac{y}{4}\right)O_2(g)} \longrightarrow \underset{30\,ml}{xCO_2(g)} + \frac{y}{2}\underset{X}{H_2O(l)}$$

$$10\left(x+\frac{y}{4}\right) = 40 \qquad 10x = 30$$

$$x = 3 \checkmark$$

$$3 + \frac{y}{4} = 4$$

$$y = 4$$

#27-EUDIOMETRY NUMERICAL PROBLEMS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KV...

Sol :: $C_xH_y(g) + \left(x+\frac{y}{4}\right)O_2(g) \longrightarrow xCO_2(g) + \frac{y}{2}H_2O(l)$

10 ml, 40 ml, 30 ml, x

$10\left(x+\frac{y}{4}\right) = 40$

$3 + \frac{y}{4} = 4$

$y = 4$

$10x = 30$

$x = 3$

$\Rightarrow \boxed{C_3H_4}$ Ans

4:09 / 25:35

#27-EUDIOMETRY NUMERICAL PROBLEMS|IIT ADVANCED|JEE MAIN|CHEMISTRY CLASS 12|KV...

Q → 1 lit of O_2 gas is passed through an ozonizer and the final volume of gas mixture is found to be 820ml. If this gas mixture is passed through turpentine oil, find vol of gas remaining.

Soln:

$3O_2 \longrightarrow 2O_3$

V ml $\quad \frac{2}{3}V$ ml $\Rightarrow$

$(1000 - V) + \frac{2}{3}V = 820$,

6:55 / 25:35

mixture is passed through turpentine oil, find vol of gas remaining.

Soln:

$3O_2 \longrightarrow 2O_3$

V ml $\quad \frac{2}{3}V$ ml

$(1000-V)+\frac{2}{3}V = 820$

$\Rightarrow 1000 - \frac{V}{3} = 820$

$\frac{V}{3} = 180$

$V = 540$ ml

V_{O_2} remaing $= 1000 - 540$

$= \underline{460 \text{ ml}}$ Ans.

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$= \underline{460\ ml}$ Ans.

Q → A sample of water gas has composition by volume of 50% H_2, 45% CO and 5% CO_2 gases. Find vol. in lit of water gas which on treatment with excess of steam will produce 10 lit H_2 under same P&T Condition.

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with excess of steam will produce 10 lit H_2 under same P&T Condition.

Sol^n :

$H_2 + CO + CO_2$ — V lit

0.5V, 0.45V, 0.05V lit

$CO_{(g)} + H_2O_{(g)} \longrightarrow CO_{2(g)} + H_{2(g)}$

0.45V (excess steam)

0.45 V lit

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$$CO_{(g)} + H_2O_{(g)} \longrightarrow CO_{2(g)} + H_{2(g)}$$

0.5V (excess steam)

0.45V lit

$$(V_{H_2})_{total} = (0.5V + 0.45V) = 10 \text{ lit}$$

$$V = \frac{10}{0.95} \text{ lit}$$

11:43 / 25:35

(If)

Q → A 20 ml sample of hydrocarbon on treatment with excess of $O_{2(g)}$ observed a volume contraction of 60 ml. On passing through KOH solution, another contraction of 80 ml is observed. Find molecular formula of H.C.

Sol^n:

$$\underset{20\ ml}{C_xH_y(g)} + \left(x+\frac{y}{4}\right)O_2(g) \rightarrow x\,CO_2(g) + \frac{y}{2}H_2O(l)$$

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observed. Find molecular formula of H.C.

Sol^n:

$$C_xH_y(g) + \left(x+\frac{y}{4}\right)O_2(g) \rightarrow \underset{20x}{xCO_2(g)} + \frac{y}{2}\underset{\times}{H_2O(l)}$$

$2\times$ ml (excess)

$\Rightarrow$ volume contraction = volume spent − vol. obtained

$\Rightarrow$ vol. expansion = vol. obtained − vol. spent

vol. contraction $= 60 = \left\{20 + 20\left(x+\frac{y}{4}\right)\right\} - 20x$

$\Rightarrow 20 + 5y = 60$

$y = 8$

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$\Rightarrow$ vol. expansion = vol. obtained - vol. spent

vol. contraction $= 60 = \{20 + 20(x + \frac{y}{4})\} - 20x$

$\Rightarrow$ $20 + 5y = 60$

$y = 8$

$20x = 80$

$x = 4$

$\Rightarrow$ C_4H_8

17:13 / 25:35

Q → 15 ml of a gaseous oxide of nitrogen was mixed with hydrogen in an Eudiometer tube till the volume increased to 42 ml. On sparking, the volume reduced to 27 ml. The resulting mixture was mixed with 10 ml of O_2 and on further explosion volume reduced to 19 ml. Find molecular formula of oxide of nitrogen. Assume no formation of ammonia.

Sol^n: $N_xO_y(g) + yH_2(g) \longrightarrow \frac{x}{2}N_2 + yH_2O(l)$

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no formation of ammonia.

Sol^{n}:

$$N_xO_y(g) + yH_2(g) \longrightarrow \frac{x}{2}N_2 + yH_2O_x(l)$$

N_xO_y: ~~15~~ ml (L.R)

H_2: 27 ml, $(27 - 15y)$

N_2: $7.5x$ ml

$$27 - 15y + 7.5x = 27$$

$$x = 2y$$

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$x = 2y$ —— (1)

$$H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow \underset{x}{X_2O_{(l)}}$$

$(27 - 15y)$ 10ml

$$10 - \left(\frac{27-15y}{2}\right)$$

$$10 - \frac{27-15y}{2} + 7.5$$

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow \underset{x}{X_2O}(l)$$

$(27 - 15y)$ 10ml

$$10 - \left(\frac{27-15y}{2}\right)$$

$$10 - \frac{27-15y}{2} + 7.5x = 19 \text{ ml}$$

$$-13.5 + 7.5y + 7.5x = 9$$

M_2O

$$x + y = 3$$

$$y = 1 \Rightarrow x = 2$$